Introduction to de Rham Theory

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These notes are from the January 2024 Intersession at IMSA. They cover differential forms, the basics of de Rham cohomology, and some applications in electromagnetism. Prerequisites are single-variable differential and integral calculus.

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1 Differential Vector Calculus

1.1 Vector Fields

Vector Calculus is the study of differentiation and integration of *vector fields*. Vector Calculus many applications in physics.

Definition 1.1 (Vector field). A *vector field* is a map $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$, which can be though of as an assignment of a vector $\mathbf{v}(x, y, z)$ to each point (x, y, z) in $\mathbb{R}^3$:

$$\mathbf{F}(x, y, z) = P(x, y, z)\hat{\mathbf{i}} + Q(x, y, z)\hat{\mathbf{j}} + R(x, y, z)\hat{\mathbf{k}}$$

where $\hat{\mathbf{i}} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$, $\hat{\mathbf{j}} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$, and $\hat{\mathbf{k}} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$ are the *standard unit vectors* for $\mathbb{R}^3$.

Example 1.2 (Gravitational Field). *Newton's Law of Gravitation* states that the magnitude of the gravitational force between two objects with masses m and M is

$$||\mathbf{F}|| = \frac{GMm}{r^2}$$

where r is the distance between the objects and G is the gravitational constant. Let's assume that the object with mass M is located at the origin in $\mathbb{R}^3$. (For instance, M could be the mass of the Earth and the origin would be at its center.) Let the position vector of the object with mass m be $\mathbf{x} = (x, y, z)$. Then $r = ||\mathbf{x}||$, so $r^2 = ||\mathbf{x}||^2$. The gravitational force exerted on this second object acts toward the origin, and the unit vector in this direction is $-\frac{\mathbf{x}}{||\mathbf{x}||}$. Therefore the gravitational force acting on the object at $\mathbf{x} = (x, y, z)$ is

$$\mathbf{F}(\mathbf{x}) = -\frac{GMm}{||\mathbf{x}||^3}\mathbf{x}. \tag{1.1}$$

The function given by Equation (1.1) is called the *gravitational field*. This is a vector field since it associates a vector (the force $F(x)$) with every point x in space. Equation (1.1) can also be written in terms of its *component functions* as

$$\mathbf{F}(\mathbf{x}) = -\frac{GMm}{||\mathbf{x}||^3}(x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) = -\frac{GMm}{(x^2 + y^2 + z^2)^{3/2}}(x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}})$$

The gravitational field F is pictured in Figure 1 below.

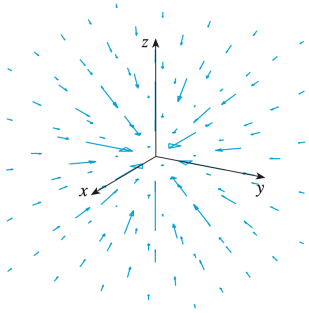


Figure 1: Gravitational force field.

1.2 Limits and Continuity

Definition 1.3 (Limits). Let $f(x,y,z)$ be a real-valued function of three variables, defined on an open subset U of $\mathbb{R}^3$ but perhaps not at the point $(x_0,y_0,z_0) \in U$ and let L be a real number. Then

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = L$$

means that, for every $\epsilon > 0$ there exist $\delta > 0$ such that

$$0 < \sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2} < \delta$$

implies $|f(x,y,z) - L| < \epsilon$. If $\mathbf{F}(x,y,z) = P(x,y,z)\hat{\mathbf{i}} + Q(x,y,z)\hat{\mathbf{j}} + R(x,y,z)\hat{\mathbf{k}}$ is a vector field defined on U (but perhaps not at $(x_0,y_0,z_0) \in U$) and $\mathbf{L} = L_1\hat{\mathbf{i}} + L_2\hat{\mathbf{j}} + L_3\hat{\mathbf{k}}$ is a vector in $\mathbb{R}^3$, then

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} \mathbf{F}(x,y,z) = \mathbf{L}$$

if and only if

$$\begin{aligned}
\lim_{(x,y,z)\to(x_0,y_0,z_0)} P(x,y,z) &= L_1, \\
\lim_{(x,y,z)\to(x_0,y_0,z_0)} Q(x,y,z) &= L_2, \text{ and} \\
\lim_{(x,y,z)\to(x_0,y_0,z_0)} R(x,y,z) &= L_3.
\end{aligned}$$

All the familiar limit laws hold for functions of three variables (sum and difference, constant multiple, product, quotient, composition, squeeze theorem, etc.), and the proofs are essentially the same as in the 1-variable case.

Definition 1.4 (Continuity). If f is a real-valued function defined on an open subset U of $\mathbb{R}^3$, then we say that f is *continuous* at (x_0,y_0,z_0) if

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = f(x_0,y_0,z_0).$$

If $\mathbf{F}(x,y,z) = P(x,y,z)\hat{\mathbf{i}} + Q(x,y,z)\hat{\mathbf{j}} + R(x,y,z)\hat{\mathbf{k}}$ is a vector field defined on an open subset U of $\mathbb{R}^3$, then we say that f is *continuous* at (x_0,y_0,z_0) if

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} \mathbf{F}(x,y,z) = \mathbf{F}(x_0,y_0,z_0).$$

which means that

$$\begin{aligned}
\lim_{(x,y,z)\to(x_0,y_0,z_0)} P(x,y,z) &= P(x_0,y_0,z_0),\\
\lim_{(x,y,z)\to(x_0,y_0,z_0)} Q(x,y,z) &= Q(x_0,y_0,z_0), \text{ and}\\
\lim_{(x,y,z)\to(x_0,y_0,z_0)} R(x,y,z) &= R(x_0,y_0,z_0).
\end{aligned}$$

Example 1.5. The function $f(x,y,z) = xy\sin(y^2z)$ is continuous for all $x,y,z \in \mathbb{R}^3$ since it is a product and composition of continuous functions.

1.3 Differential Calculus

1.3.1 Partial Derivatives

Recall that the derivative

$$\frac{df}{dx}(x_0) := \lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0}$$

of a real-valued function f at a point x is the instantaneous rate of change of f at x.

Definition 1.6 (Partial Derivative). Let $f : \mathbb{R}^3 \to \mathbb{R}$ is a real-valued function of three variables. Then the *partial derivative* of f at (x,y,z) with respect to x is the instantaneous rate of change in the x-direction:

$$\frac{\partial f}{\partial x}(x_0,y_0,z_0) := \lim_{x\to x_0} \frac{f(x,y_0,z_0)-f(x_0,y_0,z_0)}{x-x_0}.$$

Similarly, one defines the partial derivatives with respect to y and z by

$$\begin{aligned}
\frac{\partial f}{\partial y}(x_0,y_0,z_0) &:= \lim_{y\to y_0} \frac{f(x_0,y,z_0)-f(x_0,y_0,z_0)}{y-y_0},\\
\frac{\partial f}{\partial z}(x_0,y_0,z_0) &:= \lim_{z\to z_0} \frac{f(x_0,y_0,z)-f(x_0,y_0,z_0)}{z-z_0},
\end{aligned}$$

which give the instantaneous rate of change of f at (x_0,y_0,z_0) in the y- and z-directions, respectively.

We can interpret the partial derivatives of a function geometrically as follows. If $z = f(x,y)$ is a function of two variables, then its graph is a surface in $\mathbb{R}^3$. If we fix $y = b$, then $z = f(x,b)$ is a curve on this surface, and $\frac{\partial f}{\partial x}(a,b)$ is the slope of the tangent line to this curve at $(a,b,f(a,b))$. Similarly, $\frac{\partial f}{\partial y}(a,b)$ is the slope of the tangent line to the curve $z = f(a,y)$ at the point $(a,b,f(a,b))$.

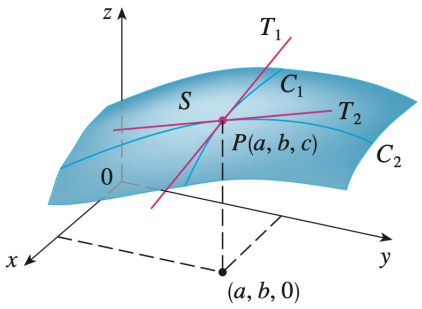


Figure 2: Geometric interpretation of partial derivatives.

Example 1.7. To calculate the partial derivatives of the function $f(x,y,z) = xy\sin(y^2z)$, simply treat two variables constant and differentiate as usual with respect to the other:

$$\frac{\partial f}{\partial x} = y\sin(y^2z), \quad \frac{\partial f}{\partial y} = x\sin(y^2z) + 2xy^2z\cos(y^2z), \quad \frac{\partial f}{\partial z} = xy^3\cos(y^2z).$$

1.3.2 Differentiable Functions

In one-variable calculus, if a function f is differentiable at x_0, then the tangent line to the graph of f at $(x_0, f(x_0))$ given by

$$y = f(x_0) + f'(x_0)(x - x_0)$$

is a good approximation to the graph of f near $(x_0, f(x_0))$.

If $z = f(x,y)$ is the graph of a function of two variables x, y, then the graph is a surface in $\mathbb{R}^3$. If the partial derivatives of f with respect to x and y are continuous at (x_0, y_0), then the tangent plane to the surface at $(x_0, y_0, f(x_0, y_0))$ given by

$$z = f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - x_0) \tag{1.2}$$

is a good approximation to the graph near $(x_0, y_0, f(x_0, y_0))$. If the partial derivatives are not continuous, then this tangent plane may not be a good approximation to the graph of the function for all values (x,y) arbitrarily close to (x_0, y_0). For example, consider the function

$$f(x,y) = \begin{cases} 1, & xy = 0, \\ 0, & xy \neq 0. \end{cases}$$

Note that $\frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0$, so Equation (1.2) says the tangent plane to the graph at the point $(0,0,1)$ is just the plane $z = 1$. However, this is not a good approximation for any point in $\mathbb{R}^2 \setminus \{\text{the coordinate axes}\}$.

Definition 1.8 (Differentiable function). We say that a function $f : U \to \mathbb{R}$, where U is an open subset of $\mathbb{R}^2$, is *differentiable* at (x_0, y_0) if the surface $z = f(x_0, y_0)$ has a tangent plane at $(x_0, y_0, f(x_0, y_0))$ which is a good approximation to the graph for nearby points. That is, f is *differentiable* if the partial derivatives exist **and are continuous** at (x_0, y_0, z_0).

For a surface in $\mathbb{R}^3$ defined by a relation of the form $F(x, y, z) = 0$ (such as the sphere $x^2 + y^2 + z^2 = 1$) which is not the graph of a function, then we can use multivariable implicit differentiation to find that the equation of the tangent plane to the surface at the point (x_0, y_0, z_0) is given by

$$\frac{\partial f}{\partial x}(x_0, y_0, z_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0, z_0)(y - y_0) + \frac{\partial f}{\partial z}(x_0, y_0, z_0)(z - z_0) = 0. \tag{1.3}$$

Again, if the partial derivatives are all continuous at (x_0, y_0, z_0), then this is a good approximation to the surface near this point. Note that in the case where $z = f(x, y)$ is a function of x and y, then $\frac{\partial f}{\partial z} = \frac{\partial z}{\partial z} = 1$, in which case Equation (1.3) agrees with Equation (1.2).

Example 1.9 (Tangent plane to the sphere). The tangent plane to the sphere $x^2 + y^2 + z^2 = 1$ at the point (x_0, y_0, z_0) is

$$2x_0(x - x_0) + 2y_0(y - y_0) + 2z_0(x - z_0) = 0.$$

This is pictured in Figure 3 below.

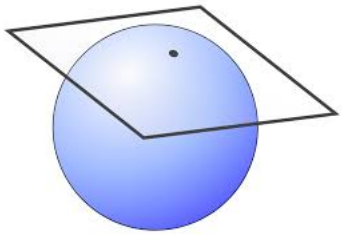


Figure 3: Tangent plane to a sphere.

1.3.3 Higher-order Partial Derivatives

Given the partial derivative $\frac{\partial f}{\partial x}$, we can form three second-order partial derivatives by differentiating $\frac{\partial f}{\partial x}$ with respect to x, y, and z:

$$\frac{\partial^2 f}{\partial x^2} := \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$
$$\frac{\partial^2 f}{\partial y \partial x} := \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$
$$\frac{\partial^2 f}{\partial z \partial x} := \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial x}\right)$$

Similarly, we can form

$$\begin{aligned}
\frac{\partial^2 f}{\partial x \partial y} &:= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \\
\frac{\partial^2 f}{\partial y^2} &:= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) \\
\frac{\partial^2 f}{\partial z \partial y} &:= \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial y}\right) \\
\frac{\partial^2 f}{\partial x \partial z} &:= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial z}\right) \\
\frac{\partial^2 f}{\partial y \partial z} &:= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial z}\right) \\
\frac{\partial^2 f}{\partial z^2} &:= \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial z}\right).
\end{aligned}$$

We can form higher-order partial derivatives in a similar fashion.

Example 1.10. For the function $f(x, y, z) = \sin(xyz)$, we have

$$\frac{\partial f}{\partial x} = yz\cos(xyz), \quad \frac{\partial f}{\partial y} = xz\cos(xyz), \quad \frac{\partial f}{\partial z} = xy\cos(xyz),$$

so the second partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = -y^2z^2\sin(xyz), \quad \frac{\partial^2 f}{\partial y \partial x} = z\cos(xyz) - xyz^2\sin(xyz), \quad \frac{\partial^2 f}{\partial z \partial x} = y\cos(xyz) - xy^2z\sin(xyz)$$

$$\frac{\partial^2 f}{\partial x \partial y} = z\cos(xyz) - xyz^2\sin(xyz), \quad \frac{\partial^2 f}{\partial y^2} = -x^2z^2\sin(xyz), \quad \frac{\partial^2 f}{\partial z \partial y} = x\cos(xyz) - x^2yz\sin(xyz)$$

$$\frac{\partial^2 f}{\partial x \partial z} = y\cos(xyz) - xy^2z\sin(xyz), \quad \frac{\partial^2 f}{\partial y \partial z} = x\cos(xyz) - x^2yz\sin(xyz), \quad \frac{\partial^2 f}{\partial z^2} = -x^2y^2\sin(xyz)$$

Notice that in Example 1.10 we have

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial z}, \quad \frac{\partial^2 f}{\partial x \partial z} = \frac{\partial^2 f}{\partial z \partial x}, \quad \frac{\partial^2 f}{\partial y \partial z} = \frac{\partial^2 f}{\partial z \partial y}.$$

The next theorem says that when the mixed second partial derivatives are continuous, they are always equal.

Theorem 1.11 (Clairaut's Theorem). Suppose f is defined on a disk D that contains the point (a, b). If the functions $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ are both continuous on D, then $\frac{\partial^2 f}{\partial x \partial y}(a, b) = \frac{\partial^2 f}{\partial y \partial x}(a, b)$.

Proof. For sufficiently small, non-zero values of h, consider the difference

$$\Delta(h) := [f(a + h, b + h) - f(a + h, b)] - [f(a, b + h) - f(a, b)].$$

If we let $g(x) := f(x, b+h) - f(x, b)$, then

$$\Delta(h) = g(a+h) - g(a).$$

By the Mean Value Theorem, there exists a number c between a and $a+h$ such that

$$\begin{aligned} g(a+h) - g(a) &= g'(c)h \\ &= h\left(\frac{\partial f}{\partial x}(c, b+h) - \frac{\partial f}{\partial x}(c, b)\right). \end{aligned}$$

Applying the Mean Value Theorem again, this time to $\frac{\partial f}{\partial x}$, we get a number d between b and $b+h$ such that

$$\begin{aligned} \frac{\partial f}{\partial x}(c, b+h) - \frac{\partial f}{\partial x}(c, b) &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)(c, d)h \\ &= \frac{\partial^2 f}{\partial y \partial x}(c, d)h. \end{aligned}$$

Combining these equations, we obtain

$$\Delta(h) = h^2 \frac{\partial^2 f}{\partial y \partial x}(c, d).$$

As $h \to 0$, then $(c, d) \to (a, b)$, so by continuity of $\frac{\partial^2 f}{\partial y \partial x}$ at (a, b) we have

$$\lim_{h \to 0} \frac{\Delta(h)}{h^2} = \lim_{(c,d) \to (a,b)} \frac{\partial^2 f}{\partial y \partial x}(c, d) = \frac{\partial^2 f}{\partial y \partial x}(a, b).$$

Similarly, by writing instead

$$\Delta(h) := [f(a+h, b+h) - f(a, b+h)] - [f(a+h, b) - f(a, b)]$$

and using the Mean Value Theorem twice and the continuity of $\frac{\partial^2 f}{\partial x \partial y}$ at (a, b), we obtain

$$\lim_{h \to 0} \frac{\Delta(h)}{h^2} = \frac{\partial^2 f}{\partial x \partial y}(a, b).$$

By uniqueness of limits, we conclude that

$$\frac{\partial^2 f}{\partial x \partial y}(a, b) = \frac{\partial^2 f}{\partial y \partial x}(a, b).$$

□

Definition 1.12 (C^k functions).

(a) Let k be a nonnegative integer. A real-valued function $f : U \to \mathbb{R}$, where U is an open subset of $\mathbb{R}^3$, is said to be *of class* C^k at a point $(x_0, y_0, z_0) \in U$ it its partial derivatives of all orders $j \leq k$ exist and are continuous at (x_0, y_0, z_0).

(b) The function f is said to be C^∞ or *smooth* at (x_0, y_0, z_0) if it is C^k for all $k \geq 0$; in other words, if its partial derivatives of all orders exist and are continuous at (x_0, y_0, z_0).

(c) A vector field $\mathbf{F} = P\hat{\mathbf{i}} + Q\hat{\mathbf{j}} + R\hat{\mathbf{k}}$ is C^k at (x_0, y_0, z_0) if its component functions P, Q, and R are C^k at (x_0, y_0, z_0), and $\mathbf{F}$ is C^∞ at (x_0, y_0, z_0) if P, Q and R are C^∞ at (x_0, y_0, z_0).

Example 1.13. A C^0 function is a continuous function. Polynomials, rational functions, sine and cosine are C^∞ everywhere they are defined. The function $f(x) = x^{1/3}$ is C^0 but not C^1 at $x = 0$. $g(x) = \int_0^x t^{1/3} dt$ has $g'(x) = x^{1/3}$, so g is C^1 but not C^2 at $x = 0$. In a similar way, one can construct a function which is C^k but not C^{k+1} at a given point.

It follows by repeatedly applying Theorem 1.11 that if f is a C^k function, then the individual partial derivatives appearing in any jth order partial derivative ($2 \leq j \leq k$) can be taken in any order.

Example 1.14. Consider the function $f(x, y, z) = xy^2z^3 + \arcsin(x\sqrt{z})$. To calculate the third-order partial derivative $\frac{\partial^3 f}{\partial x \partial y \partial z}$, we may use any of the 6 orderings of the individual partial derivatives. Since the second term is independent of y, the easiest way to get the answer is to first differentiate with respect to y:

$$\frac{\partial f}{\partial y} = 2xyz^3.$$

We then take the x and z derivatives to find

$$\frac{\partial^3 f}{\partial x \partial y \partial z} = \frac{\partial^3 f}{\partial z \partial x \partial y} = \frac{\partial}{\partial z}\left(\frac{\partial}{\partial x}\left(2xyz^3\right)\right) = \frac{\partial}{\partial z}\left(2yz^3\right) = 6yz^2.$$

1.3.4 Directional Derivatives

More generally, one can compute the instantaneous rate of change of f in *any* direction at (x_0, y_0, z_0). Specifying the direction by a unit vector $\hat{\mathbf{v}} = (v_1, v_2, v_3)$, we define the *directional derivative* of f at (x, y, z) in the direction $\hat{\mathbf{v}}$ by

$$D_{\hat{\mathbf{v}}} f(x_0, y_0, z_0) = \lim_{t \to 0} \frac{f(x_0 + tv_1, y_0 + tv_2, z_0 + tv_3) - f(x_0, y_0, z_0)}{t}.$$

Letting $(x, y, z) = (x(t), y(t), z(t)) = (x_0 + tv_1, y_0 + tv_2, z_0 + tv_3)$, we can write this as

$$\begin{aligned} D_{\hat{\mathbf{v}}} f(x_0, y_0, z_0) &= \lim_{t \to 0} \frac{f(x, y, z) - f(x_0, y_0, z_0)}{t} \\ &= \lim_{t \to 0} \frac{f(x, y, z) - f(x_0, y, z) + f(x_0, y, z) - f(x_0, y_0, z) + f(x_0, y_0, z) - f(x_0, y_0, z_0)}{t}. \end{aligned}$$

By the Mean Value Theorem, there exists a number a between x_0 and x, a number b between y_0 and y, and a number c between z_0 and z such that

$$\begin{aligned} f(x,y,z) - f(x_0,y,z) &= \frac{\partial f}{\partial x}(a,y,z)(x-x_0) \\ f(x_0,y,z) - f(x_0,y_0,z) &= \frac{\partial f}{\partial y}(x_0,b,z)(y-y_0) \\ f(x_0,y_0,z) - f(x_0,y_0,z_0) &= \frac{\partial f}{\partial z}(x_0,y_0,c)(z-z_0). \end{aligned}$$

Therefore

$$\begin{aligned} &D_{\hat{\mathbf{v}}} f(x_0,y_0,z_0) \\ &= \lim_{t\to 0} \frac{f(x,y,z) - f(x_0,y,z)}{t} + \lim_{t\to 0} \frac{f(x_0,y,z) - f(x_0,y_0,z)}{t} + \lim_{t\to 0} \frac{f(x_0,y_0,z) - f(x_0,y_0,z_0)}{t} \\ &= \lim_{t\to 0} \frac{\partial f}{\partial x}(a,y,z)\frac{x-x_0}{t} + \lim_{t\to 0} \frac{\partial f}{\partial y}(x_0,b,z)\frac{y-y_0}{t} + \lim_{t\to 0} \frac{\partial f}{\partial z}(x_0,y_0,c)\frac{z-z_0}{t} \\ &= \lim_{t\to 0} \frac{\partial f}{\partial x}(a,y,z) \lim_{t\to 0} \frac{x-x_0}{t} + \lim_{t\to 0} \frac{\partial f}{\partial y}(x_0,b,z) \lim_{t\to 0} \frac{y-y_0}{t} + \lim_{t\to 0} \frac{\partial f}{\partial z}(x_0,y_0,c) \lim_{t\to 0} \frac{z-z_0}{t} \\ &= \lim_{t\to 0} \frac{\partial f}{\partial x}(a,y,z)\frac{dx}{dt}\bigg|_{t=0} + \lim_{t\to 0} \frac{\partial f}{\partial y}(x_0,b,z)\frac{dy}{dt}\bigg|_{t=0} + \lim_{t\to 0} \frac{\partial f}{\partial z}(x_0,y_0,c) \lim_{t\to 0} \frac{dz}{dt}\bigg|_{t=0} \\ &= v_1 \lim_{t\to 0} \frac{\partial f}{\partial x}(a,y,z) + v_2 \lim_{t\to 0} \frac{\partial f}{\partial y}(x_0,b,z) + v_3 \lim_{t\to 0} \frac{\partial f}{\partial z}(x_0,y_0,c) \end{aligned}$$

If the partial derivatives of f are continuous at (x_0,y_0,z_0), then

$$\begin{aligned} \lim_{t\to 0} \frac{\partial f}{\partial x}(a,y,z) &= \frac{\partial f}{\partial x}(x_0,y_0,z_0) \\ \lim_{t\to 0} \frac{\partial f}{\partial y}(x_0,b,z) &= \frac{\partial f}{\partial y}(x_0,y_0,z_0) \\ \lim_{t\to 0} \frac{\partial f}{\partial z}(x_0,y_0,c) &= \frac{\partial f}{\partial z}(x_0,y_0,z_0), \end{aligned}$$

which shows that

$$D_{\hat{\mathbf{v}}} f(x_0,y_0,z_0) = v_1 \frac{\partial f}{\partial x}(x_0,y_0,z_0) + v_2 \frac{\partial f}{\partial y}(x_0,y_0,z_0) + v_3 \frac{\partial f}{\partial z}(x_0,y_0,z_0).$$

Defining the *gradient* of f at (x_0,y_0,z_0) to be the vector of partial derivatives

$$\nabla f(x_0,y_0,z_0) = \begin{pmatrix} \frac{\partial f}{\partial x}(x_0,y_0,z_0) & \frac{\partial f}{\partial y}(x_0,y_0,z_0) & \frac{\partial f}{\partial z}(x_0,y_0,z_0), \end{pmatrix}$$

we can write this as

$$D_{\hat{\mathbf{v}}} f = \hat{\mathbf{v}} \cdot \nabla f,$$

where both sides are evaluated at (x_0, y_0, z_0). Writing

$$D_{\hat{\mathbf{v}}} f = \hat{\mathbf{v}} \cdot \nabla f = \underbrace{||\hat{\mathbf{v}}||}_{=1} \, ||\nabla f|| \cos(\theta) = ||\nabla f|| \cos(\theta),$$

where θ is the angle between $\hat{\mathbf{v}}$ and ∇f, we see that the directional derivative is maximized when $\theta = 0$. That is, *the gradient points in the direction in which f is increasing most rapidly at* (x_0, y_0, z_0), and the magnitude $||\nabla f||$ gives this maximal rate of increase.

Example 1.15. The rate of change of the function $f(x, y) = xe^y$ at $(1, 2)$ in the direction $\hat{\mathbf{v}} = -\frac{\sqrt{3}}{2}\hat{\mathbf{i}} + \frac{1}{2}\hat{\mathbf{j}}$ is

$$D_{\hat{\mathbf{v}}}(1, 2) = \nabla f(1, 2) \cdot \hat{\mathbf{v}} = (e^2\hat{\mathbf{i}} + e^2\hat{\mathbf{j}}) \cdot (-\frac{\sqrt{3}}{2}\hat{\mathbf{i}} + \frac{1}{2}\hat{\mathbf{j}}) = e^2 \left(\frac{1 - \sqrt{3}}{2} \right).$$

The direction in which f has the maximal rate of increase at (x, y) is in the direction of $\nabla f = e^y\hat{\mathbf{i}} + xe^y\hat{\mathbf{j}}$, and the maximal rate of increase is $||\nabla f|| = \sqrt{(e^y)^2 + (xe^y)^2} = e^y\sqrt{1 + x^2}$.

1.3.5 Div, Grad, and Curl

Denote by $\nabla = \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}$ the mapping that takes f to ∇f. The symbol ∇ is called "nabla" or "del". Note that ∇ formally resembles a vector. Now, an ordinary vector $\mathbf{A}$ can be used to form three products:

(i) Scalar multiple: $\mathbf{A}c$

(ii) Dot product: $\mathbf{A} \cdot \mathbf{B}$

(iii) Cross product: $\mathbf{A} \times \mathbf{B}$.

In analogy with these products, one uses ∇ to define the following three operations:

(i) The *gradient* of a scalar function: $\nabla f = \frac{\partial f}{\partial x}\hat{\mathbf{i}} + \frac{\partial f}{\partial y}\hat{\mathbf{j}} + \frac{\partial f}{\partial z}\hat{\mathbf{k}}$

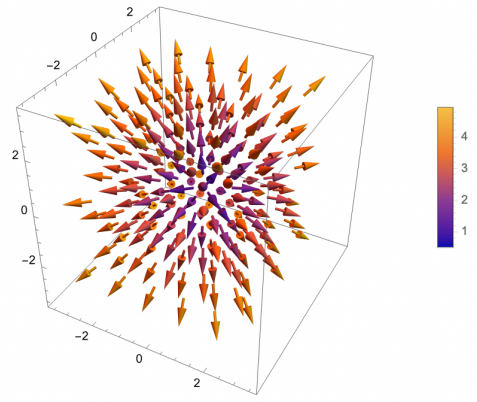
(ii) The *divergence* of a vector field: $\nabla \cdot \mathbf{F}$.

(iii) The *curl* of a vector field: $\nabla \times \mathbf{F}$

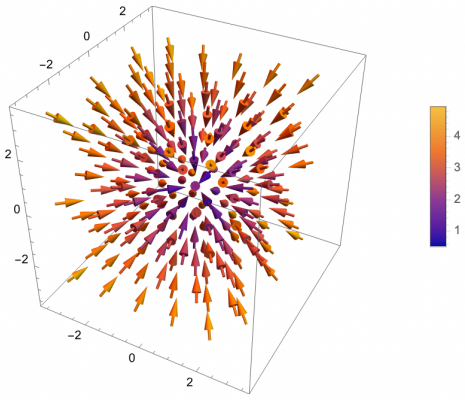
The divergence

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} P & Q & R \end{pmatrix} \\ &= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \end{aligned}$$

is a measure of how much the vector field $\mathbf{F}$ spreads out (diverges) at each point (x, y, z). The figure below shows an example of a vector field with positive divergence at every point and a vector field with negative divergence at every point. A point of positive divergence is called a *source* and a point of negative divergence is called a *sink*.



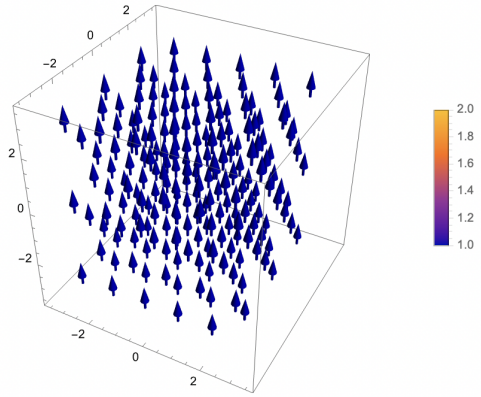
(a) The vector field $\mathbf{v} = (x, y, z)$ has a constant positive divergence.



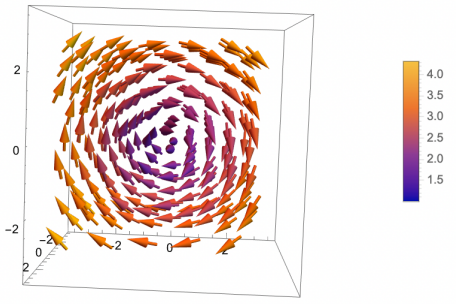
(b) The vector field $\mathbf{v} = (x, y, z)$ has a constant negative divergence.

Figure 4: Examples of vector fields with positive and negative divergence

The vector fields in the figure below have zero divergence.



(a) The vector field $\mathbf{v} = (0, 0, 1)$.



(b) The vector field $\mathbf{v} = (y, -x, 1)$

Figure 5: Examples of vector fields with zero divergence.

The curl of a vector field

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\hat{\mathbf{i}} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\hat{\mathbf{j}} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{\mathbf{k}}\end{aligned}$$

is a measure of how much the vector $\mathbf{v}$ "curls around" the point in question. The vector fields in Figure 4 both have zero curl, whereas the function in Figure 5(b) has a constant curl pointing in the negative z-direction.

1.3.6 Derivative Rules

The calculation of derivatives in one-variable calculus is facilitated by a number of general rules, such as the sum rule

$$\frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx},$$

the constant multiple rule

$$\frac{d}{dx}(cf) = c\frac{df}{dx},$$

the product rule

$$\frac{d}{dx}(fg) = f\frac{dg}{x} + g\frac{df}{dx},$$

and the quotient rule

$$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\frac{df}{dx} - f\frac{dg}{dx}}{g^2}.$$

Similar rules hold for the vector derivatives. We have the sum rules

$$\begin{aligned}\nabla(f+g) &= \nabla f + \nabla g\\ \nabla\cdot(\mathbf{v}+\mathbf{w}) &= \nabla\cdot\mathbf{v} + \nabla\cdot\mathbf{w}\\ \nabla\times(\mathbf{v}+\mathbf{w}) &= \nabla\times\mathbf{v} + \nabla\times\mathbf{w}\end{aligned}$$

and the constant multiple rules

$$\begin{aligned}\nabla(cf) &= c\nabla f\\ \nabla\cdot(c\mathbf{v}) &= c(\nabla\cdot\mathbf{v})\\ \nabla\times(c\mathbf{v}) &= c(\nabla\times\mathbf{v}),\end{aligned}$$

which are all straightforward to check.

The product rules are not quite so simple. There are two ways to construct a scalar as the product of two functions:

$$\begin{aligned}fg &\quad \text{(product of two scalar functions)},\\ \mathbf{v}\cdot\mathbf{w} &\quad \text{(dot product of two vector fields)},\end{aligned}$$

and two ways to make a vector:

$$\begin{aligned}f\mathbf{v} &\quad \text{(scalar multiple of a vector)},\\ \mathbf{v}\times\mathbf{w} &\quad \text{(cross product of two vector fields)}.\end{aligned}$$

Accordingly, there are *six* product rules, two for gradients:

$$\nabla(fg) = f\nabla g + g\nabla f, \tag{1.4}$$
$$\nabla(\mathbf{v}\cdot\mathbf{w}) = \mathbf{v}\times(\nabla\times\mathbf{w}) + \mathbf{w}\times(\nabla\times\mathbf{v}) + (\mathbf{v}\cdot\nabla)\mathbf{w} + (\mathbf{w}\cdot\nabla)\mathbf{v}, \tag{1.5}$$

two for divergences:

$$\nabla\cdot(f\mathbf{v}) = f(\nabla\cdot\mathbf{v}) + \mathbf{v}\cdot\nabla f, \tag{1.6}$$
$$\nabla\cdot(\mathbf{v}\times\mathbf{w}) = \mathbf{w}\cdot(\nabla\times\mathbf{v}) - \mathbf{v}\cdot(\nabla\times\mathbf{w}), \tag{1.7}$$

and two for curls:

$$\nabla\times(f\mathbf{v}) = f(\nabla\times\mathbf{v}) - \mathbf{v}\times(\nabla f), \tag{1.8}$$
$$\nabla\times(\mathbf{v}\times\mathbf{w}) = (\mathbf{w}\cdot\nabla)\mathbf{v} - (\mathbf{v}\cdot\nabla)\mathbf{w} + \mathbf{v}(\nabla\cdot\mathbf{w}) - \mathbf{w}(\nabla\cdot\mathbf{v}). \tag{1.9}$$

The proofs follow from the ordinary product rule for partial derivatives. For instance,

$$\begin{aligned}
\nabla\cdot(f\mathbf{v}) &= \begin{pmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\end{pmatrix}\cdot\begin{pmatrix}fv_1 & fv_2 & fv_3\end{pmatrix}\\
&= \frac{\partial}{\partial x}(fv_1) + \frac{\partial}{\partial y}(fv_2) + \frac{\partial}{\partial z}(fv_3)\\
&= \frac{\partial f}{\partial x}v_1 + f\frac{\partial v_1}{\partial x} + \frac{\partial f}{\partial y}v_2 + f\frac{\partial v_2}{\partial y} + \frac{\partial f}{\partial z}v_3 + f\frac{\partial v_3}{\partial z}\\
&= \left(\frac{\partial f}{\partial x}v_1 + \frac{\partial f}{\partial y}v_2 + \frac{\partial f}{\partial z}v_3\right) + \left(f\frac{\partial v_1}{\partial x} + f\frac{\partial v_2}{\partial y} + f\frac{\partial v_3}{\partial z}\right)\\
&= (\nabla f)\cdot\mathbf{v} + f(\nabla\cdot\mathbf{v}).
\end{aligned}$$

It is also possible to form three quotient rules

$$\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2},$$
$$\nabla\cdot\left(\frac{\mathbf{v}}{g}\right) = \frac{g(\nabla\cdot\mathbf{v}) - \mathbf{v}\cdot(\nabla g)}{g^2},$$
$$\nabla\times\left(\frac{\mathbf{v}}{g}\right) = \frac{g(\nabla\times\mathbf{v}) + \mathbf{v}\times(\nabla g)}{g^2},$$

however, these can be obtained quickly from the corresponding product rules by writing $\frac{f}{g} = fg^{-1}$ and $\frac{\mathbf{v}}{g} = \mathbf{v}g^{-1}$ so there is no need to memorize these.

1.3.7 Second Derivatives

We have just seen three different types of first derivative that can be formed with ∇: the gradient, divergence, and curl. By applying ∇ twice we can construct five different second derivatives. The gradient ∇f is a vector, so we can take its divergence and curl:

(1) Divergence of a gradient: $\nabla\cdot(\nabla f)$.

(2) Curl of a gradient: $\nabla \times (\nabla f)$.

The divergence $\nabla \cdot \mathbf{v}$ is a scalar - all we can do is take its gradient:

(3) Gradient of a divergence: $\nabla(\nabla \cdot \mathbf{v})$.

The curl $\nabla \times \mathbf{v}$ is a vector, so we can take its divergence and curl:

(4) Divergence of a curl: $\nabla \cdot (\nabla \times \mathbf{v})$.

(5) Curl of a curl: $\nabla \times (\nabla \times \mathbf{v})$.

Let's consider these one at a time:

(1) Divergence of a gradient:

$$\begin{aligned}\nabla \cdot (\nabla f) &= \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{pmatrix} \\ &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.\end{aligned}$$

This object, which we write as $\nabla^2 f$ for short, is called the *Laplacian* of f. Note that the Laplacian of a scalar f is a scalar. We will also talk about the Laplacian of a vector, $\nabla^2 \mathbf{v}$. By this we mean a vector quantity whose x-component is the Laplacian of v_1, and so on:

$$\nabla^2 \mathbf{v} := \begin{pmatrix} \nabla^2 v_1 & \nabla^2 v_2 & \nabla^2 v_3 \end{pmatrix}.$$

(2) Curl of a gradient:

$$\begin{aligned}\nabla \times (\nabla f) &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \begin{pmatrix} \frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \end{pmatrix}\end{aligned}$$

If f is of class C^1, then the mixed partial derivatives are equal by Theorem 1.11, so this is the zero vector. This shows that the curl of a gradient of a C^1 function is always zero:

$$\nabla \times (\nabla f) = 0.$$

This fact will play an important role later.

(3) Gradient of a divergence: $\nabla(\nabla \cdot \mathbf{v})$ does not have a special name. You should note though that $\nabla(\nabla \cdot \mathbf{v})$ is not the same as the Laplacian of a vector $\nabla^2 \mathbf{v}$:

$$\nabla^2 \mathbf{v} = (\nabla \cdot \nabla)\mathbf{v} \neq \nabla(\nabla \cdot \mathbf{v}). \tag{1.10}$$

(4) Divergence of a curl: Like the curl of a gradient, this is always zero for a C^1 vector field, since

$$\begin{aligned}
\nabla \cdot (\nabla \times \mathbf{v}) &= \left(\tfrac{\partial}{\partial x} \quad \tfrac{\partial}{\partial y} \quad \tfrac{\partial}{\partial z}\right) \cdot \left(\tfrac{\partial v_3}{\partial y} - \tfrac{\partial v_2}{\partial z} \quad \tfrac{\partial v_1}{\partial z} - \tfrac{\partial v_3}{\partial x} \quad \tfrac{\partial v_2}{\partial x} - \tfrac{\partial v_1}{\partial y}\right) \\
&= \frac{\partial}{\partial x}\left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y}\right) \\
&= \frac{\partial^2 v_3}{\partial x \partial y} - \frac{\partial^2 v_2}{\partial x \partial z} + \frac{\partial^2 v_1}{\partial y \partial z} - \frac{\partial^2 v_3}{\partial y \partial x} + \frac{\partial^2 v_2}{\partial z \partial x} - \frac{\partial^2 v_1}{\partial z \partial y} \\
&= \left(\frac{\partial^2 v_1}{\partial y \partial z} - \frac{\partial^2 v_1}{\partial z \partial y}\right) + \left(\frac{\partial^2 v_2}{\partial z \partial x} - \frac{\partial^2 v_2}{\partial x \partial z}\right) + \left(\frac{\partial^2 v_3}{\partial x \partial y} - \frac{\partial^2 v_3}{\partial y \partial x}\right) \\
&= 0 + 0 + 0 \\
&= 0.
\end{aligned}$$

This will also play an important role later.

(5) Curl of a curl: It is straightforward, though tedious, to check that

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}, \tag{1.11}$$

Thus, the curl of a curl gives seems to give nothing new, since the first term is just number (3) and the second is the Laplacian of the vector field $\mathbf{v}$. It should be noted, however, that Equation (1.11) is often used to *define* the Laplacian of a vector field, in preference to Equation (1.10), since the latter only holds in Cartesian coordinates. In curvilinear coordinates, the unit vectors themselves depend on position, so they too must be differentiated, leading to more complicated expressions for the Laplacian of a vector field.

Really, then, there are just two kinds of second derivatives (at least for C^1 functions): the Laplacian and the gradient of divergence. We could similarly work out third derivatives, and so on, in a similar fashion. Note also that everything we did above followed from the definition of the differential operator ∇. Even if you remembered *only* the definition of ∇, you should be able, in principle, to reconstruct all the rest.

2 Integral Vector Calculus

2.1 Multiple Integrals

Recall that the definite integral $\int_a^b f(x)dx$ in one-variable calculus is defined by partitioning the interval $[a,b]$ into n subintervals $[x_{i-1},x_i]$ of equal width $\Delta x = \frac{b-a}{n}$, choosing a sample point x_i^* in each subinterval, forming the Riemann sum

$$\sum_{i=1}^{n} f(x_i^*)\Delta x$$

and taking the limit of such sums as $n \to \infty$:

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x.$$

This limit exists if f is continuous or has at most finitely-many jump discontinuities on $[a,b]$. In the special case where $f(x) \geq 0$, $\int_a^b f(x)dx$ represents the area under the curve $y = f(x)$ from a to b.

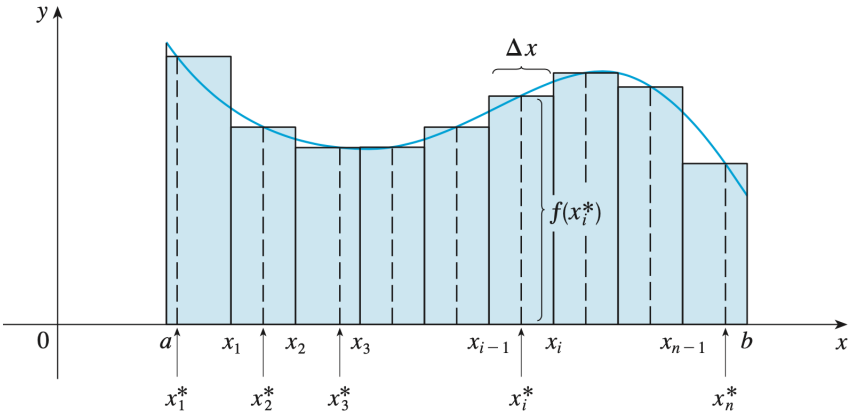


Figure 6: A Riemann sum.

In a similar manner we consider the integral of a function $f(x,y)$ of two variables defined on a closed rectangle

$$R = [a,b] \times [c,d] = \{(x,y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}.$$

Begin by partitioning $[a,b]$ into m subintervals $[x_{i-1},x_i]$ of equal width $\Delta x = \frac{b-a}{m}$ and $[c,d]$ into n subintervals $[y_{j-1},y_j]$ of equal width $\Delta y = \frac{d-c}{n}$. This partitions the rectangle R into mn subrectangles

$$R_{ij} = [x_{i-1},x_i] \times [y_{j-1},y_j] = \{(x,y) : x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j\}$$

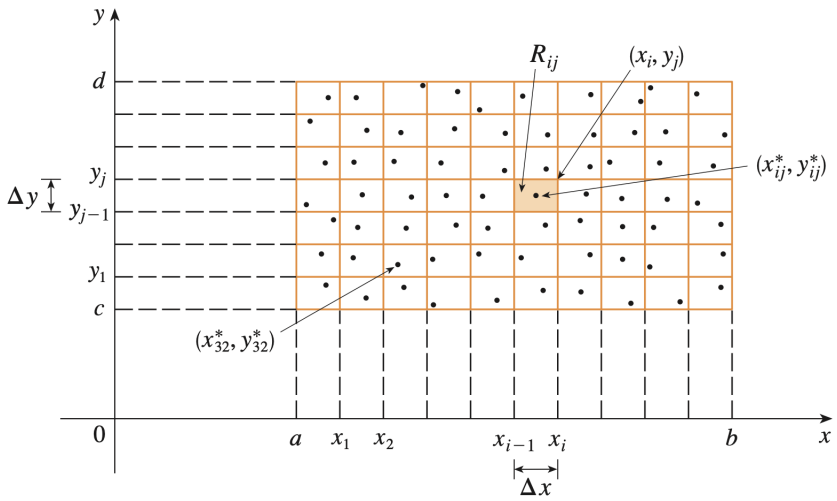


Figure 7: Partition of the rectangle R into mn subrectangles R_{ij} and a choice of sample points (x^*_{ij}, y^*_{ij}).

each with area $\Delta A = \Delta x \Delta y$. Now choose a sample point (x^*_{ij}, y^*_{ij}) in each R_{ij}, as shown in Figure 7 below.

Finally, form the Riemann sum

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij}) \Delta A$$

and define the integral by taking the limit as $m, n \to \infty$:

$$\iint_R f(x,y) dA := \lim_{n \to \infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij}) \Delta A.$$

Continuity of f is sufficient for the limit above to exist. This is called the *double integral* of f over R. Similar to before, in the special case where $f(x,y) \geq 0$, the $\iint_R f(x,y) dA$ is the volume of the solid that lies above R and below the surface $z = f(x,y)$.

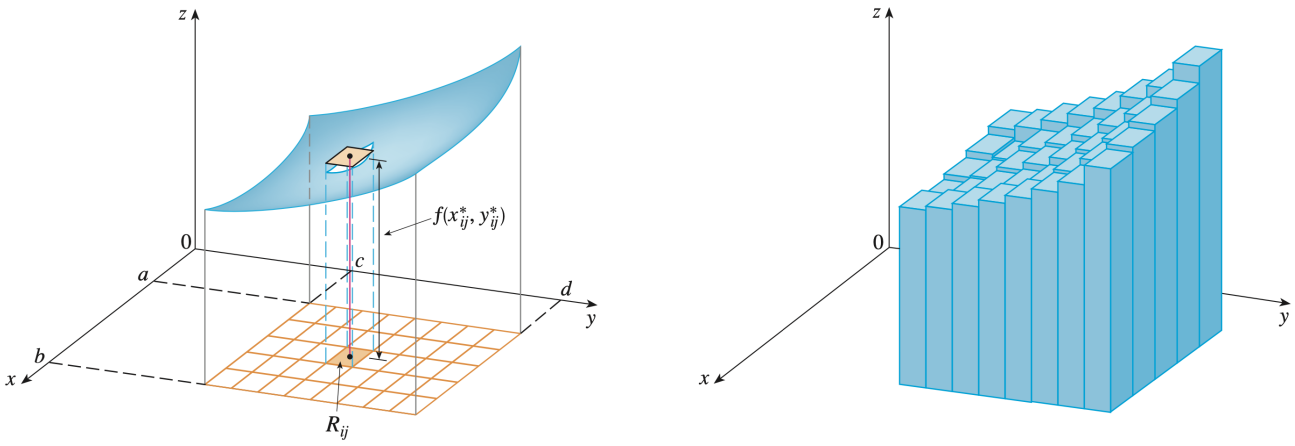


Figure 8: A Riemann sum for a positive function $f(x,y)$ over a rectangle.

It follows from the definition of the double integral that it has the familiar properties

$$\iint_R [f(x,y) + g(x,y)]dA = \iint_R f(x,y)dA + \iint_R g(x,y)dA$$
$$\iint_R cf(x,y)dA = c\iint_R f(x,y)dA$$

The following theorem says that double integrals can be computed as *iterated integrals*; that is by holding one variable constant at a time and integrating with respect to the other.

Theorem 2.1 (Fubini's Theorem). If f is continuous on the rectangle $R = [a,b] \times [c,d]$, then

$$\iint_R f(x,y)dA = \int_a^b \int_c^d f(x,y)dydx = \int_c^d \int_a^b f(x,y)dxdy.$$

More generally, this is true if f is bounded on R, f is discontinuous on only a finite number of smooth curves, and the iterated integrals exist.

To see why Fubini's theorem makes intuitive sense, consider the case where $f(x,y) > 0$. Then $\iint_R f(x,y)dA$ is the volume of V of the solid S that lies above R and under the surface $z = f(x,y)$. By the slicing method of calculating volumes, this volume is also given by

$$V = \int_a^b A(x)dx,$$

where $A(x)$ is the area of a cross-section of S in the plane through x perpendicular to the x-axis. Now $A(x)$ is the area under the curve C whose equation is $z = f(x,y)$, where x is held constant and $c \leq y \leq d$ (Figure 9a below), so

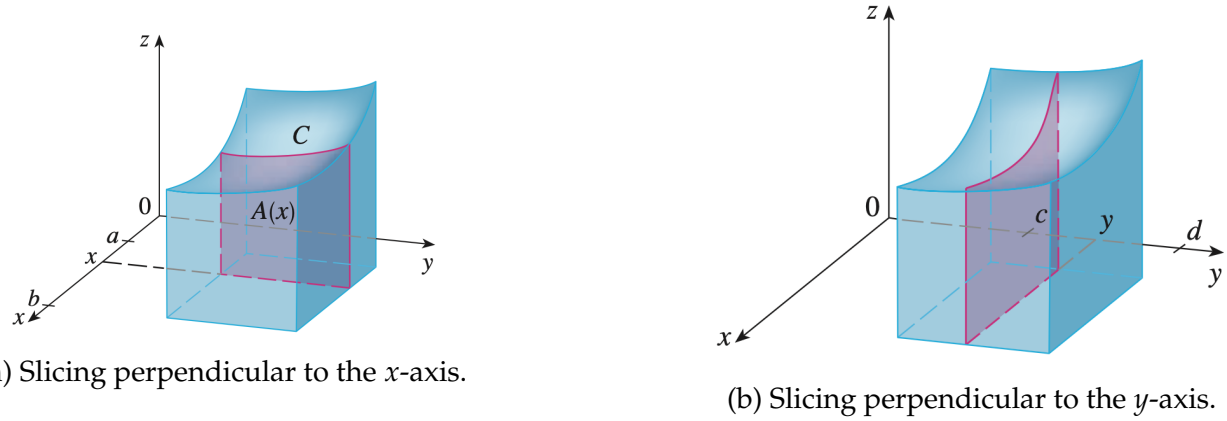
$$A(x) = \int_c^d f(x,y)dy$$

and therefore

$$\iint_R f(x,y)dA = V = \int_a^b A(x)dx = \int_a^b \int_c^d f(x,y)dydx.$$

A similar argument, using cross-sections perpendicular to the y-axis (See Figure 9b), shows that

$$\iint_R f(x,y)dA = \int_c^d \int_a^b f(x,y)dxdy.$$



(a) Slicing perpendicular to the x-axis.

(b) Slicing perpendicular to the y-axis.

Figure 9: Intuition behind Fubini's Theorem.

Example 2.2. Evaluate $\iint_R (x - 3y^2)dA$, where $R = [0,2] \times [1,2]$.
Solution: Fubini's Theorem gives

$$\begin{aligned}\iint_R (3 - 3y^2)dA &= \int_0^2 \int_1^2 (x - 3y^2)dydx \\ &= \int_0^2 (xy - y^3)\Big|_{y=1}^{y=2} dx \\ &= \int_0^2 (x - 7)dx = \left[\frac{x^2}{2} - 7x\right]_0^2 \\ &= -12.\end{aligned}$$

What if we want to integrate a function $f(x,y)$ over a bounded domain D which is not a rectangle? Since D is bounded, we can find a rectangle R which encloses D (see Figure 10) and extend f to a function $\widetilde{f}$ with domain R by setting

$$\widetilde{f}(x,y) = \begin{cases} f(x,y) & \text{if } (x,y) \in D, \\ 0 & \text{if } (x,y) \in R \setminus D. \end{cases}$$

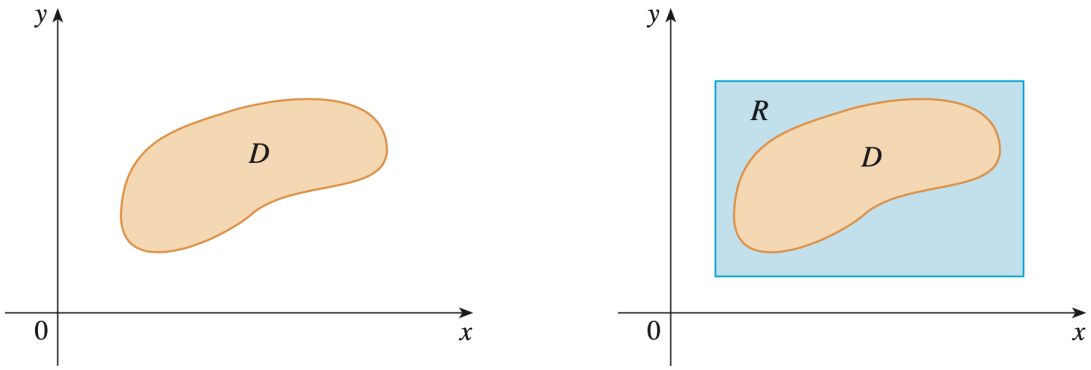


Figure 10: Domain of the function $\widetilde{f}$.

If $\widetilde{f}$ is integrable over R, then we define

$$\iint_D f(x,y)dA = \iint_R \widetilde{f}dA.$$

This definition makes sense since $\widetilde{f} = f$ on D and makes no contribution to the integral outside of D.

One similarly defines the *triple integral*

$$\iiint_B f(x,y,z)dV$$

of a function $f(x,y,z)$ over a box $B = [a,b] \times [c,d] \times [r,s]$ and a version of Fubini's Theorem for three variables says that it can be computed as an iterated integral, and that all 6 orderings of the iterated integrals are equal. If the domain D of f is a bounded region, then one defines $\widetilde{f}$ and the integral $\iiint_D f(x,y,z)dV$ similar to before.

2.2 Line and Surface Integrals

2.2.1 Line Integrals

For a one-dimensional force, such as the force $f(x) = kx$ required to stretch a spring with spring constant k by a distance x from its equilibrium length, the definite integral $W = \int_a^b f(x)dx$ represents the work exerted by the force in moving the particle from point a to point b. For example, the work done to stretch a spring $\sqrt{2}$ units from its equilibrium length is $W = \int_0^{\sqrt{2}} kxdx = k$.

More generally, the work done by a vector field

$$\mathbf{F}(x,y,z) = P(x,y,z)\hat{\mathbf{i}} + Q(x,y,z)\hat{\mathbf{j}} + R(x,y,z)\hat{\mathbf{k}}$$

in moving a particle along a curve C in $\mathbb{R}^3$ defined by a function $\mathbf{r}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} +$

$z(t)\hat{\mathbf{k}}$, $a \le t \le b$ is given by the *line integral*

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} := \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ &= \int_a^b (P(x(t), y(t), z(t))\hat{\mathbf{i}} + Q(x(t), y(t), z(t))\hat{\mathbf{j}} + R(x(t), y(t), z(t))\hat{\mathbf{k}}) \cdot (x'(t)\hat{\mathbf{i}} + y'(t)\hat{\mathbf{j}} + z'(t)\hat{\mathbf{k}}) dt \\ &= \int_a^b \left[P(x(t), y(t), z(t))x'(t) + Q(x(t), y(t), z(t))y'(t) + R(x(t), y(t), z(t))z'(t) \right] dt. \end{aligned}$$

Since $dx = x'(t)dt$, $dy = y'(t)dt$, and $dz = z'(t)dt$, we often write this as

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz$$

for short.

Example 2.3. To calculate the work done by the force field $\mathbf{F}(x, y) = x^2\hat{\mathbf{i}} - xy\hat{\mathbf{j}}$ in moving a particle along the quarter circle $\mathbf{r}(t) = \cos(t)\hat{\mathbf{i}} + \sin(t)\hat{\mathbf{j}}$, $0 \le t \le \frac{\pi}{2}$, just write everything in terms of t:

$$\begin{aligned} \mathbf{F}(\mathbf{r}(t)) &= \cos^2(t)\hat{\mathbf{i}} - \sin(t)\cos(t)\hat{\mathbf{j}} \\ \mathbf{r}'(t) &= -\sin(t)\hat{\mathbf{i}} + \cos(t)\hat{\mathbf{j}} \end{aligned}$$

form the dot product:

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -2\cos^2(t)\sin(t)$$

and integrate the result with respect to t over the interval $[0, \frac{\pi}{2}]$:

$$W = \int_0^{\pi/2} (-2\cos^2(t)\sin(t)) dt = 2\frac{\cos^3(t)}{3}\Big|_0^{\pi/2} = -\frac{2}{3}.$$

The definition of the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is given in terms of a parametrization $\mathbf{r}(t)$ of the curve C. It can be shown that the line integral is independent of the choice of parametrization, as long as the curve is traced out exactly once as the parameter t increases from a to b. If we let $-C$ denote the curve defined by the same points as C, but traversed with opposite orientation (see Figure 11 below), then

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}.$$

If C is a closed loop (that is, $\mathbf{r}(a) = \mathbf{r}(b)$), then we indicate this by writing a circle on the integral sign as in

$$\oint_C \mathbf{F} \cdot d\mathbf{r}.$$

The work done by a force field $\mathbf{F}$ over a closed loop C is called the *circulation* of the vector field $\mathbf{F}$ around C.

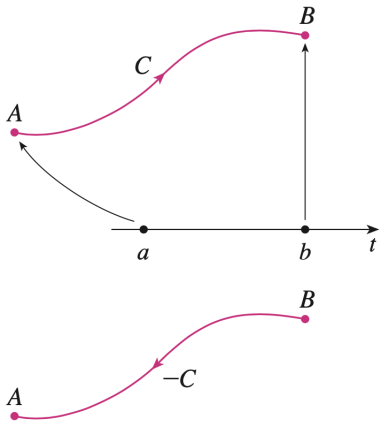


Figure 11: The two orientations of a curve C.

2.2.2 Surface Integrals

In much the same way that we describe a space curve by a vector-valued function $\mathbf{r}(t)$ of a single parameter t, we can describe a surface in $\mathbb{R}^3$ by a vector-valued function $\mathbf{r}(u,v)$ of two parameters u and v:

$$\mathbf{r}(u,v) = x(u,v)\hat{\mathbf{i}} + y(u,v)\hat{\mathbf{j}} + z(u,v)\hat{\mathbf{k}}$$

defined on a region D in the uv-plane. The set S of all points $(x,y,z) \in \mathbb{R}^3$ such that

$$x = x(u,v) \quad y = y(u,v) \quad z = z(u,v)$$

as (u,v) varies throughout D is called a *parametric surface*.

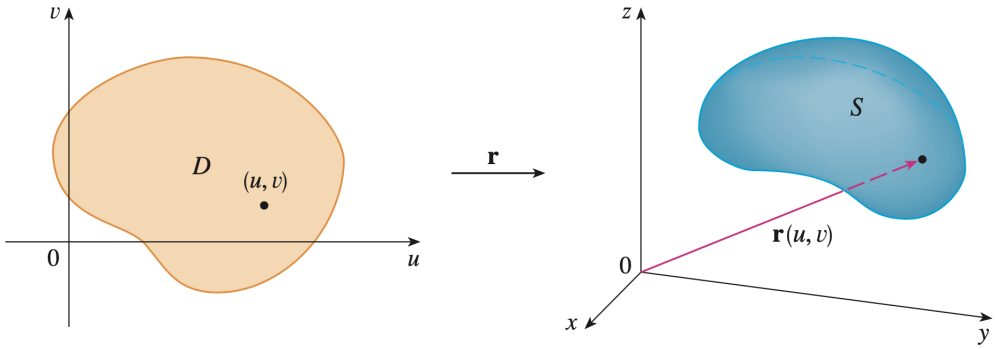


Figure 12: A parametric surface.

Example 2.4. Consider the vector-valued function $\mathbf{r}(u,v) = 2\cos(u)\hat{\mathbf{i}} + v\hat{\mathbf{j}} + 2\sin(u)\hat{\mathbf{k}}$, where $(u,v) \in [0,2\pi] \times (-\infty,\infty)$. For each fixed value of v, the curve $\mathbf{r}(u,v)$ is a circle of radius 2 in the xz-plane. Thus, the parametric surface traced out by the function $\mathbf{r}(u,v)$ is an infinite-cylinder of radius 2 whose axis is the y-axis. If $v \in [0,h]$ for some positive real number h, then the cylinder has finite height h and has a *boundary* at $v = 0$ and $v = h$, which consists of two disjoint circles.

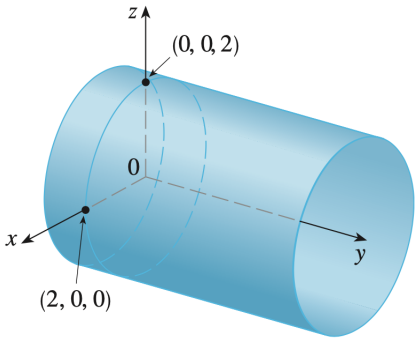


Figure 13: The cylinder traced out by $\mathbf{r}(u,v) = 2\cos(u)\hat{\mathbf{i}} + v\hat{\mathbf{j}} + 2\sin(u)\hat{\mathbf{k}}$.

Let's now find the tangent plane to a parametric surface S traced out by a vector function

$$\mathbf{r}(u,v) = x(u,v)\hat{\mathbf{i}} + y(u,v)\hat{\mathbf{j}} + z(u,v)\hat{\mathbf{k}}$$

at a point $P_0 = \mathbf{r}(u_0, v_0)$. If we fix $u = u_0$, then $\mathbf{r}(u_0, v)$ defines a curve C_1 on S through P_0. The tangent vector to C_1 at P_0 is given by the partial derivative of $\mathbf{r}$ with respect to v:

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0,v_0)\hat{\mathbf{i}} + \frac{\partial y}{\partial v}(u_0,v_0)\hat{\mathbf{j}} + \frac{\partial z}{\partial v}(u_0,v_0)\hat{\mathbf{k}}.$$

Similarly, if we fix $v = v_0$, then $\mathbf{r}(u, v_0)$ defines a curve C_2 on S through P_0, and its tangent vector at P_0 is

$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0,v_0)\hat{\mathbf{i}} + \frac{\partial y}{\partial u}(u_0,v_0)\hat{\mathbf{j}} + \frac{\partial z}{\partial u}(u_0,v_0)\hat{\mathbf{k}}.$$

If $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$, then the surface S is called *regular* (such a surface has no "corners"). For a regular surface, the *tangent plane* is the plane that contains the tangent vectors $\mathbf{r}_u, \mathbf{r}_v$, and which has the normal vector $\mathbf{r}_u \times \mathbf{r}_v$. (See Figure 14.)

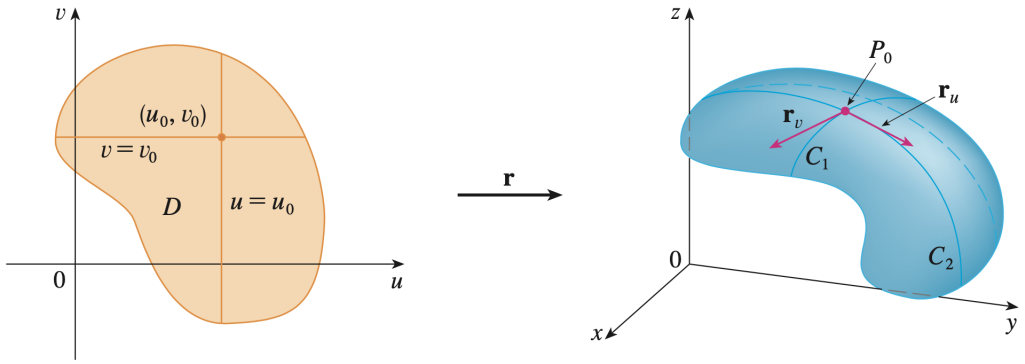


Figure 14: Tangent plane to a parametric surface at P_0.

Consider a surface S that has a tangent plane at every point (except at any boundary point). There are two unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2 = -\mathbf{n}_1$ at (x,y,z). (See Figure 15.)

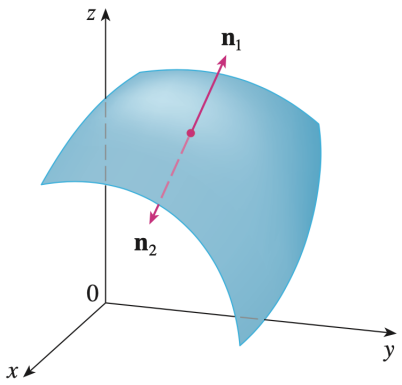


Figure 15: Two normal vectors at a point on a surface.

If it is possible to choose a unit normal vector $\mathbf{n}$ at every such point (x, y, z) so that $\mathbf{n}$ varies continuously over S, then S is called an *orientable surface* and the given choice of $\mathbf{n}$ provides S with an *orientation*. There are two possible orientations for any orientable surface. (See Figure 16.)

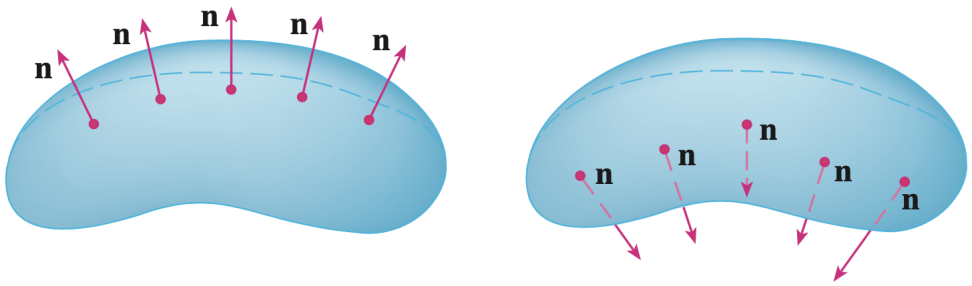


Figure 16: Two possible orientations of a surface.

If S is a regular orientable surface given in parametric form by a vector function $\mathbf{r}(u, v)$, then it is automatically supplied with an orientation induced unit normal vector

$$\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{||\mathbf{r}_u \times \mathbf{r}_v||}$$

and the opposite orientation is given by $-\mathbf{n}$.

Example 2.5. Consider again the cylinder traced out by the function

$$\mathbf{r}(u,v) = 2\cos(u)\hat{\mathbf{i}} + v\hat{\blacksquare} + 2\sin(u)\hat{\mathbf{k}}.$$

Since

$$\begin{aligned}\mathbf{r}_u &= -2\sin(u)\hat{\mathbf{i}} + 0\hat{\blacksquare} + 2\cos(u)\hat{\mathbf{k}}\\ \mathbf{r}_v &= 0\hat{\mathbf{i}} + 1\hat{\blacksquare} + 0\hat{\mathbf{k}},\end{aligned}$$

we have

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\blacksquare} & \hat{\mathbf{k}} \\ -2\sin(u) & 0 & 2\cos(u) \\ 0 & 1 & 0 \end{vmatrix} = -2\cos(u)\hat{\mathbf{i}} + 0\hat{\blacksquare} - 2\sin(u)\hat{\mathbf{k}},$$

(♣ this points inward; I probably switched u and v) and therefore the outward unit normal vector at the point $\mathbf{r}(u,v)$ is

$$\mathbf{n} = -\frac{\mathbf{r}_u \times \mathbf{r}_v}{||\mathbf{r}_u \times \mathbf{r}_v||} = \cos(u)\hat{\mathbf{i}} + 0\hat{\blacksquare} + \sin(u)\hat{\mathbf{k}}.$$

Since this is a continuous function of (u,v), this shows that the cylinder is orientable.

If $\mathbf{F}$ is a continuous vector field defined on an oriented parametric surface S with unit normal vector $\mathbf{n}$, then the *surface integral of* $\mathbf{F}$ *over* S (also called the *flux* of $\mathbf{F}$ through S) is defined as

$$\begin{aligned}\iint_S \mathbf{F}\cdot d\mathbf{S} &= \iint_S \mathbf{F}\cdot\mathbf{n}\, dS\\ &= \iint_D \mathbf{F}(\mathbf{r}(u,v))\cdot(\mathbf{r}_u\times\mathbf{r}_v)dudv,\end{aligned}$$

where D is the parameter domain. If a fluid with density $\rho(x,y,z)$ and velocity field $\mathbf{v}(x,y,z)$ is flowing in a region of space, then the rate of flow (mass per unit time) per unit area is $\rho\mathbf{v}$. Taking $\mathbf{F} = \rho\mathbf{v}$, then the flux integral above gives the rate of flow of the fluid across S. (Think of S as an imaginary surface that doesn't impede the fluid flow, like a fishing net across a stream.)

The integral is independent of the parametrization $\mathbf{r}(u,v)$ of the surface, up to changing orientation, in which case

$$\iint_{-S} \mathbf{F}\cdot d\mathbf{S} = -\iint_S \mathbf{F}\cdot d\mathbf{S}.$$

For a *closed surface* (i.e., a surface with empty boundary), that is, a surface that is the boundary of a solid region E, we denote this with a circle on the integral sign, as in the integral below:

$$\oiint_S \mathbf{F}\cdot d\mathbf{S}.$$

For a closed surface, the convention is that the *positive orientation* is the one for which the normal vectors point outward form E, and inward-pointing normals give the negative orientation.

Example 2.6. Let's calculate the flux of the vector field $\mathbf{F}(x,y,z) = z\hat{\mathbf{i}} + y\hat{\mathbf{j}} + x\hat{\mathbf{k}}$ across the unit sphere $x^2 + y^2 + z^2 = 1$. We can write the sphere as a parametrized surface by using *spherical coordinates* (ρ, θ, ϕ), which are shown in Figure 17 below.

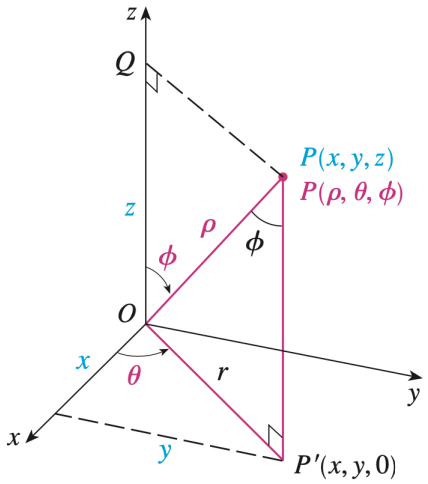


Figure 17: Definition of spherical coordinates.

The spherical coordinates (ρ, θ, ϕ) of a point P in $\mathbb{R}^3$ can be defined in terms of the vector $\overrightarrow{OP}$ as follows:

- $\rho = |\overrightarrow{OP}|$; that is, ρ is the distance from the origin O to P.
- θ is the angle between the positive x-axis and the projection of $\overrightarrow{OP}$ onto the xy-plane.
- ϕ is the angle between $\overrightarrow{OP}$ and the positive z-axis.

From Figure 17, we see that the rectangular coordinates (x, y, z) of P are given in terms of spherical coordinates (ρ, θ, ϕ) by

$$x = \rho\cos(\theta)\sin(\phi), \quad y = \rho\sin(\theta)\cos(\phi), \quad z = \rho\cos(\phi).$$

Using spherical coordinates, we can parametrize a point on the unit sphere by the function

$$\mathbf{r}(\phi, \theta) = \sin(\phi)\cos(\theta)\hat{\mathbf{i}} + \sin(\phi)\sin(\theta)\hat{\mathbf{j}} + \cos(\phi)\hat{\mathbf{k}}$$

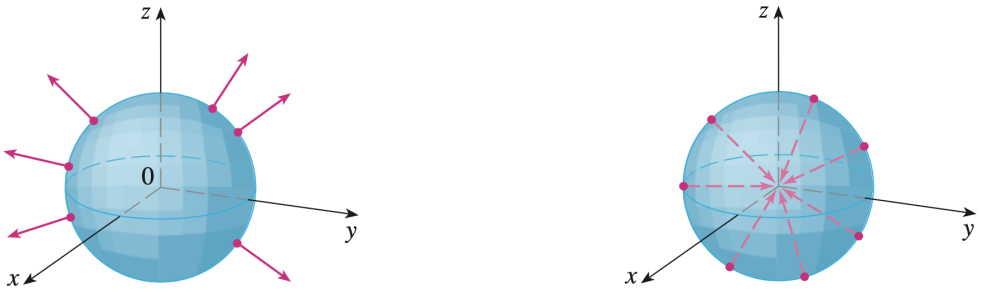
where the parameter domain D is the rectangle $D = \{(\phi, \theta) : 0 \leq \phi \leq \pi,\ 0 \leq \theta \leq 2\pi\}$. The partial derivatives of $\mathbf{r}$ with respect to ϕ and θ are

$$\mathbf{r}_\phi = \cos(\phi)\cos(\theta)\hat{\mathbf{i}} + \cos(\phi)\sin(\theta)\hat{\mathbf{j}} + -\sin(\phi)\hat{\mathbf{k}},$$
$$\mathbf{r}_\theta = -\sin(\phi)\sin(\theta)\hat{\mathbf{i}} + \sin(\phi)\cos(\theta)\hat{\mathbf{j}} + 0\hat{\mathbf{k}}$$

so the unit normal vector is

$$\begin{aligned}\mathbf{n} &= \mathbf{r}_\phi \times \mathbf{r}_\theta \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \cos(\phi)\cos(\theta) & \cos(\phi)\sin(\theta) & -\sin(\phi) \\ -\sin(\phi)\sin(\theta) & \sin(\phi)\cos(\theta) & 0 \end{vmatrix} \\ &= \sin^2(\phi)\cos(\theta)\hat{\mathbf{i}} + \sin^2(\phi)\sin(\theta)\hat{\mathbf{j}} + \sin(\phi)\cos(\phi)\hat{\mathbf{k}}.\end{aligned}$$

Note that $\mathbf{n} = \sin(\phi)(\sin(\phi)\cos(\theta)\hat{\mathbf{i}} + \sin(\phi)\sin(\theta)\hat{\mathbf{j}} + \cos(\phi)\hat{\mathbf{k}}) = \sin(\phi)\mathbf{r}(\phi,\theta)$. Since $0 \le \phi \le \pi$, $\sin(\phi) \ge 0$, so $\mathbf{n}$ points in the same direction as the position vector, that is, outward from the sphere. (♣ $\mathbf{n} = \mathbf{0}$ at the north and south poles. Explain why this is OK.)



(a) Positive orientation. (b) Negative orientation.

Figure 18: Positive and negative orientations of a closed surface.

The integrand is then

$$\begin{aligned}\mathbf{F}(\mathbf{r}(\phi,\theta)) \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) &= \cos(\phi)\sin^2(\phi)\cos(\theta) + \sin^3(\phi)\sin^2(\theta) + \sin^2(\phi)\cos(\phi)\cos(\theta) \\ &= 2\sin^2(\phi)\cos(\phi)\cos(\theta) + \sin^3(\phi)\sin^2(\theta)\end{aligned}$$

and therefore the flux is

$$\begin{aligned}\oiint_S \mathbf{F}\cdot d\mathbf{S} &= \iint_D \mathbf{F}(\mathbf{r}(\phi,\theta)) \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta)\, dA \\ &= \int_0^{2\pi}\int_0^{\pi} (2\sin^2(\phi)\cos(\phi)\cos(\theta) + \sin^3(\phi)\sin^2(\theta))\, d\phi d\theta \\ &= 2\int_0^{\pi}\sin^2(\phi)\cos(\phi)d\phi \int_0^{2\pi}\cos(\theta)d\theta + \int_0^{\pi}\sin^3(\phi)d\phi\int_0^{2\pi}\sin^2(\theta)d\theta \\ &= 0 + \int_0^{\pi}\sin^3(\phi)d\phi\int_0^{2\pi}\sin^2(\theta)d\theta \\ &= \int_0^{\pi}(\sin(\phi) - \sin(\phi)\cos^2(\phi))d\phi \int_0^{2\pi}\frac{1}{2}(1+\cos(2\theta))d\theta \\ &= (-\cos(\phi) + \frac{1}{3}\cos^3(\phi))\Big|_0^{\pi} \cdot \frac{1}{2}(\theta + \frac{1}{2}\sin(2\theta))\Big|_0^{2\pi} \\ &= \frac{4\pi}{3}.\end{aligned}$$

2.3 The Fundamental Theorems

An essential theorem for computing integrals in one-variable calculus is the Fundamental Theorem of Calculus.

Theorem 2.7 (Fundamental Theorem of Calculus).

(i) If f is continuous on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t)dt, \quad a \leq x \leq b$$

is continuous on $[a, b]$ and differentiable on (a, b), and $g'(x) = f(x)$.

(ii) If f is continuous on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is any antiderivative of f, that is, a function such that $F' = f$.

Part (i) of the FTC says that every continuous function has an antiderivative, and part (ii) says that the integral of a continuous function f over an interval $[a, b]$ is given in terms of an antiderivative, and only depends on its values at the boundary of the interval. One encounters several generalizations of the Fundamental Theorem of Calculus in Vector Calculus which have this same structure

2.3.1 Fundamental Theorem for Line Integrals

If we think of the gradient vector ∇f of a function f of two or three variables as a sort of derivative of f, then the following theorem can be regarded as a version of the Fundamental Theorem for line integrals.

Theorem 2.8 (Fundamental Theorem for Line Integrals). Let C be a regular curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be function of two or three variables which is C^1 on C. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(b)).$$

A vector field $\mathbf{F}$ of the form $\mathbf{F} = \nabla f$ for some function f is said to be *conservative*, and the function f is called a *potential* function for $\mathbf{v}$. Theorem 2.8 says that the integral of a conservative vector field is given by the net change in the potential at the endpoints. For non-conservative vector fields, the value of the line integral from A to B in general depends on the path taken from A to B.

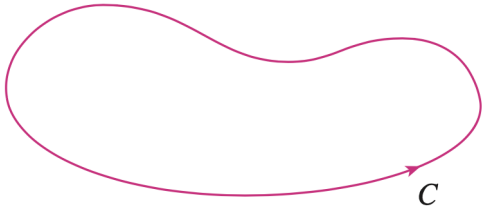
Example 2.9. The gravitational field $\mathbf{F}(\mathbf{x}) = -\frac{GMm}{||\mathbf{r}||^3}\mathbf{r}$ is conservative with potential $f = \frac{GMm}{\sqrt{x^2+y^2+z^2}}$. By Theorem 2.8, the work done in moving a particle with mass m from

$(3,4,12)$ to the point $(2,2,0)$ is

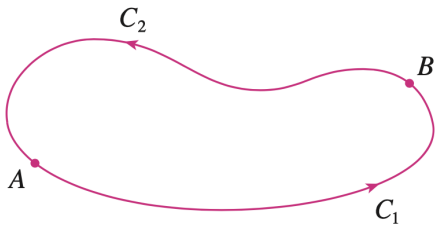
$$\begin{aligned} W &= \int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} \\ &= f(2,2,0) - f(3,4,12) \\ &= \frac{GMm}{\sqrt{2^2+2^2}} - \frac{GMm}{\sqrt{3^2+4^2+12^2}} \\ &= GMm\left(\frac{13-2\sqrt{2}}{16\sqrt{2}}\right). \end{aligned}$$

Corollary 2.10. If $\mathbf{F}$ is conservative, then $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed loop C.

Proof. Choose any two points A and B on C and regard C as being composed of the path C_1 from A to B followed by the path C_2 from B to A.



(a) A closed curve C.



(b) Choose $A \neq B$ so that $C = C_1 \cup C_2$.

Figure 19

We then have

$$\begin{aligned} \int_C \mathbf{F}\cdot d\mathbf{r} &= \int_{C_1} \mathbf{F}\cdot d\mathbf{r} + \int_{C_2} \mathbf{F}\cdot d\mathbf{r} \\ &= \int_{C_1} \mathbf{F}\cdot d\mathbf{r} - \int_{-C_2} \mathbf{F}\cdot d\mathbf{r}. \end{aligned}$$

Since $\mathbf{F}$ is conservative, $\int_C \mathbf{F}\cdot d\mathbf{r}$ is path-independent. Since C_1 and $-C_2$ are both paths from A to B, $\int_{C_1} \mathbf{F}\cdot d\mathbf{r} = \int_{-C_2} \mathbf{F}\cdot d\mathbf{r}$, and therefore $\int_C \mathbf{F}\cdot d\mathbf{r} = 0$. □

2.3.2 Fundamental Theorem for Curls

Stokes' Theorem is a version of the Fundamental Theorem for the curl of a vector field.

Theorem 2.11 (Stokes' Theorem). Let S be an oriented piecewise-regular surface that is bounded by a simple, closed, piecewise-regular curve C with positive orientation. Let $\mathbf{v}$ be a vector field whose component functions have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S. Then

$$\iint_S \nabla \times \mathbf{v} \cdot d\mathbf{S} = \oint_C \mathbf{v} \cdot d\mathbf{r}$$

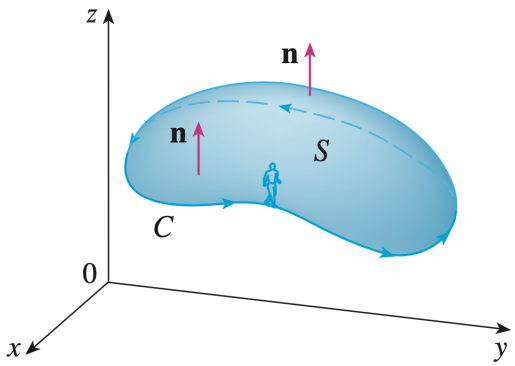


Figure 20: The surface S and its boundary curve C in Stokes' Theorem.

Stokes' Theorem says that if $\mathbf{F}$ is the curl of a vector field $\mathbf{v}$, then the flux of $\mathbf{F}$ through S is equal to the circulation of $\mathbf{v}$ around the boundary curve C.

Example 2.12. Suppose $\mathbf{F}(x,y,z) = z^2\hat{\mathbf{i}} - 3xy\hat{\mathbf{j}} + x^3y^3\hat{\mathbf{k}}$ and S is the part of the paraboloid $z = 5 - x^2 - y^2$ above the plane $z = 1$. The boundary curve C is the circle $x^2 + y^2 = 4$, which can be parametrized as $x(t) = 2\cos(t), y(t) = 2\sin(t), z(t) = 1\ 0 \leq t \leq 2\pi$. By Stokes' Theorem

$$\begin{aligned}
\iint_S \nabla \times \mathbf{F} \cdot d\mathbf{S} &= \oint_C \mathbf{F} \cdot d\mathbf{r} \\
&= \int_0^{2\pi} (\hat{\mathbf{i}} - 12\sin(t)\cos(t)\hat{\mathbf{j}} + 64\sin^3(t)\cos^3(t)\hat{\mathbf{k}}) \cdot (-2\sin(t)\hat{\mathbf{i}} + 2\cos(t)\hat{\mathbf{j}} + 0\hat{\mathbf{k}})dt \\
&= \int_0^{2\pi} (-2\sin(t) - 24\sin(t)\cos^2(t))dt \\
&= (2\cos(t) + 8\cos^3(t))\Big|_0^{2\pi} \\
&= 0.
\end{aligned}$$

2.3.3 Fundamental Theorem for Divergences

Gauss's Theorem (also known as the *Divergence Theorem*) gives a version of the Fundamental Theorem for divergences.

Theorem 2.13 (Divergence Theorem). Let E be a bounded region in $\mathbb{R}^3$ and let S be the boundary surface of E , given with positive (outward) orientation. Let $\mathbf{v}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E. Then

$$\iiint_E \nabla \cdot \mathbf{v}\, dV = \oiint_S \mathbf{v} \cdot d\mathbf{S}$$

Example 2.14. Let $\mathbf{F}(x, y, z) = z\hat{\mathbf{i}} + y\hat{\mathbf{j}} + x\hat{\mathbf{k}}$. By the Divergence Theorem, the flux of $\mathbf{F}$ through the unit sphere $x^2 + y^2 + z^2 = 1$ is

$$\begin{aligned}
\oiint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \nabla \cdot \mathbf{F}\, dV \\
&= \iiint_E \left(\frac{\partial z}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial x}{\partial z} \right) dV \\
&= \iiint_E (0 + 1 + 0) dV \\
&= \iiint_E dV \\
&= \text{vol}(\text{unit sphere}) \\
&= \frac{4\pi}{3}.
\end{aligned}$$

3 Electromagnetism

Vector calculus is used extensively in physics. As an application, we now discuss *electromagnetism*.

3.1 The Helmholtz Theorem

The laws of electricity and magnetism are expressed in terms of *electric* and *magnetic fields*, denoted by $\mathbf{E}$ and $\mathbf{B}$, respectively. Like many physical laws, these are most compactly expressed as differential equations. Since $\mathbf{E}$ and $\mathbf{B}$ are vectors, the differential equations involve vector derivatives: divergence and curl. This raises an important mathematical question: To what extent is a vector function determined by its divergence and curl? In other words, if the divergence of a vector field $\mathbf{F}$ is a specified (scalar) function D,

$$\nabla \cdot \mathbf{F} = D,$$

and the curl of $\mathbf{F}$ is a specified (vector) function $\mathbf{C}$,

$$\nabla \times \mathbf{F} = \mathbf{C},$$

(which, for consistency, must be divergenceless,

$$\nabla \cdot \mathbf{C} = 0,$$

since div curl $= 0$), does this uniquely determine the function $\mathbf{F}$?

In general, the answer is no. For example, it is straightforward to check that both $\mathbf{F} = \mathbf{0}$ and $\mathbf{F} = yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}$ have zero divergence and curl everywhere. However, to solve a differential equation uniquely, one must also supply appropriate *boundary conditions*. In electrodynamics, one typically requires that the fields go to zero "at infinity" (i.e., far away from all charges). In this case, the *Helmholtz Theorem* says that such a vector field *is* uniquely determined by its divergence and curl:

Theorem 3.1 (Helmholtz Theorem). Suppose we are given a scalar function D and a divergenceless vector function $\mathbf{C}$ such that $\lim_{||\mathbf{r}||\to\infty} \frac{D(\mathbf{r})}{||\mathbf{r}||^2} = 0$ and $\lim_{||\mathbf{r}||\to\infty} \frac{\mathbf{C}(\mathbf{r})}{||\mathbf{r}||^2} = 0$, then there is a unique vector field $\mathbf{F}$ satisfying $\nabla \cdot \mathbf{F} = D$, $\nabla \times \mathbf{F} = \mathbf{C}$, and $\lim_{||\mathbf{r}||\to\infty} \mathbf{F}(\mathbf{r}) = 0$.

A proof of this theorem can be found in texts on electromagnetism. In the following, we will discuss the differential equations which, by Theorem 3.1, determine the electric and magnetic fields.

3.2 Maxwell's Equations pre-Maxwell

3.2.1 Electric Fields

In 1785, French physicist Charles-Augustin de Coulomb (1736-1806) published three influential papers leading to the development of the theory of electromagnetism.

It one places a point particle of charge q Coulombs at position $\mathbf{r}'$ and define the *electric field*, $\mathbf{E}$, sourced by q, to be the force per unit charge at a point $\mathbf{r} \in \mathbb{R}^3 \backslash \{\mathbf{r}'\}$, Coulomb observed experimentally that

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r} - \mathbf{r}'}{||\mathbf{r} - \mathbf{r}'||^3},$$

where $\epsilon_0 \approx 8.85 \times 10^{-12}\ C^2 N^{-1} m^{-2}$ is a fundamental constant of nature, called the *permittivity of free space*.

For $n \geq 1$ point particles of charge q_i Coulombs located at position $\mathbf{r}_i$, $i = 1, \ldots, n$, the electric field at a point $\mathbf{r} \in \mathbb{R}^3 \backslash \{\mathbf{r}_1, \ldots, \mathbf{r}_n\}$ is the vector sum

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^{n} \frac{q_i(\mathbf{r} - \mathbf{r}_i)}{||\mathbf{r} - \mathbf{r}_i||^3}.$$

More generally, if we have a continuous charge distribution specified by a charge density $\rho(\mathbf{r}')$, then the total charge contained in a bounded region D is

$$q_{\text{enc}} = \iiint_D \rho(r') dV$$

and the total electric field at $\mathbf{r}$ is given by

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_D \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{||\mathbf{r} - \mathbf{r}'||^3} dV.$$

The flux, Φ_E, of $\mathbf{E}$ through any closed surface S is, by definition,

$$\Phi_E = \oiint_S \mathbf{E} \cdot d\mathbf{S}.$$

Gauss's Law says that

$$\Phi_E = \frac{q_{\text{enc}}}{\epsilon_0}, \tag{3.1}$$

that is, the flux of the electric field through the surface S is equal to the total charge enclosed by S divided by ϵ_0.

Example 3.2. Let's verify Gauss's Law for a point particle of charge q Coulombs located at the origin. Since the electric field

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0}\frac{\mathbf{r}}{||\mathbf{r}||^3}$$

is radially symmetric and the flux integral $\oiint_S \mathbf{E}\cdot d\mathbf{S}$ independent of the closed surface S enclosing the charge, the integral is most easily computed by taking S to be a sphere of radius R centered at the origin. The outward normal vector on the sphere is just $\frac{\mathbf{r}}{R}$, so

$$\begin{aligned}
\Phi_E &= \oiint_S \mathbf{E}\cdot d\mathbf{S} \\
&= \frac{q}{4\pi\epsilon_0}\oiint_S \frac{\mathbf{r}}{R^3}\cdot\frac{\mathbf{r}}{R}dA \\
&= \frac{q}{4\pi\epsilon_0}\frac{R^2}{R^4}\oiint_S dA \\
&= \frac{q}{4\pi\epsilon_0 R^2}\text{area(sphere)} \\
&= \frac{q}{4\pi\epsilon_0 R^2}4\pi R^2 \\
&= \frac{q}{\epsilon_0}.
\end{aligned}$$

We can convert Equation (3.1) to a differential equation using the following theorem:

Theorem 3.3. If $\iiint_D f_1(x,y,z)dV = \iiint_D f_2(x,y,z)dV$ for all regions D in $\mathbb{R}^3$, then $f_1 = f_2$.

By the Divergence Theorem,

$$\begin{aligned}
\iiint_D \nabla\cdot\mathbf{E}\,dV &= \oiint_S \mathbf{E}\cdot d\mathbf{S} \\
&= \frac{q_{\text{enc}}}{\epsilon_0} \\
&= \iiint_D \frac{\rho(\mathbf{r})}{\epsilon_0}\,dV,
\end{aligned}$$

so by Theorem 3.3

$$\nabla\cdot\mathbf{E} = \frac{\rho(\mathbf{r})}{\epsilon_0}.$$

This is the differential form of Gauss's Law.

3.2.2 Magnetic Fields

In 1820, Danish physicist Hans Christian Ørsted (1777-1851) published his discovery that a compass needle was deflected from magnetic north by a nearby electric current, confirming a direct relationship between electricity and magnetism.

The magnetic field, $\mathbf{B}$, at a point $\mathbf{r} \in \mathbb{R}^3$ sourced by a point charge of q Coulombs located at position $\mathbf{r}'$ and moving with velocity $\mathbf{v}$ is given by the *Biot-Savart Law*,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 q}{4\pi}\left(\frac{\mathbf{v}\times(\mathbf{r}-\mathbf{r}')}{||\mathbf{r}-\mathbf{r}'||^3}\right). \tag{3.2}$$

named after French physicists Jean-Baptiste Biot and Félix Savart, who discovered this relationship in 1820. The constant μ_0 is called the *permeability of free space*; in SI units, it is $4\pi \times 10^{-7}\frac{Ns^2}{C^2}$.

In the case of a magnetic field that arises from a continuous, charged medium (such as an electric current moving through a wire), rather than a single moving charge, we replace q with a suitable charge density function ρ (again measured in Coulombs per unit area) and the velocity of a single particle by the *velocity vector field* $\mathbf{v}$ of the charges. Then we define the *current density field* $\mathbf{J}$ by

$$\mathbf{J}(\mathbf{r}) = \rho(\mathbf{r})\mathbf{v}(\mathbf{r}),$$

which sources a magnetic field

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi}\iiint_D \mathbf{J}(\mathbf{r}')\times\frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3}\,dV \tag{3.3}$$

Taking the divergence of this equation (note that the divergence is taken with respect to the *unprimed* variables), we get

$$\begin{aligned}\nabla\cdot\mathbf{B} &= \nabla\cdot\frac{\mu_0}{4\pi}\iiint_D \mathbf{J}(\mathbf{r}')\times\frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3}\,dV \\ &= \frac{\mu_0}{4\pi}\iiint_D \nabla\cdot\left(\mathbf{J}(\mathbf{r}')\times\frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3}\right)\,dV.\end{aligned}$$

Invoking the product rule $\nabla\cdot(\mathbf{v}\times\mathbf{w}) = \mathbf{w}\cdot(\nabla\times\mathbf{v}) - \mathbf{v}\cdot(\nabla\times\mathbf{w})$, the integrand can be rewritten as

$$\nabla\cdot\left(\mathbf{J}(\mathbf{r}')\times\frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3}\right) = \frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3}\cdot(\nabla\times\mathbf{J}(\mathbf{r}')) - \mathbf{J}(\mathbf{r}')\cdot\left(\nabla\times\frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3}\right).$$

Now $\nabla\times\mathbf{J}(\mathbf{r}') = 0$ since $\mathbf{J}(\mathbf{r}')$ does not depend on the primed variables. It is also straightforward to check that $\nabla\times\frac{\mathbf{r}-\mathbf{r}'}{||\mathbf{r}-\mathbf{r}'||^3} = 0$. Hence, the integrand is zero, and therefore

$$\nabla\cdot\mathbf{B} = 0. \tag{3.4}$$

Equation (3.4) is *Gauss's Law for magnetic fields*.

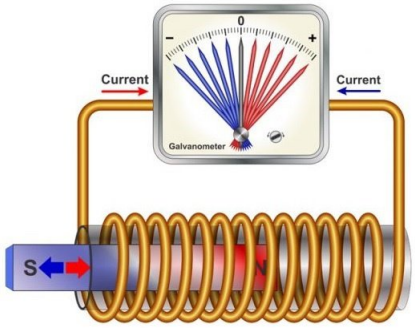
By the Divergence Theorem, if S is a closed surface bounding a region E in $\mathbb{R}^3$, then

$$\begin{aligned} 0 &= \iiint_E \nabla \cdot \mathbf{B}\, dV \\ &= \oiint_S \mathbf{B} \cdot d\mathbf{S}. \end{aligned}$$

This is interpreted as the statement that there are no magnetic monopoles.

3.2.3 Faraday's Law

English physicist Michael Faraday (1791-1867) observed that a change in magnetic flux creates an electric field. For example, pushing the north pole of a magnet in and out of a coil of wire creates a circular electric field which drives an alternating current through the wire.



Quantitatively, the change in magnetic flux across a surface S equals the circulation of $\mathbf{E}$ (called in physics *electromotive force*, which is actually work and not a force) around the boundary of C of the surface S. This relation can be written as

$$\frac{d\Phi_B}{dt} = -\oint_C \mathbf{E} \cdot d\mathbf{r}$$

where $\Phi_B(t) = \iint_S \mathbf{B} \cdot d\mathbf{S}$, and C and S are oriented consistently. We can rewrite $\oint_C \mathbf{E} \cdot d\mathbf{r}$ as a double integral over S using Stokes' Theorem to obtain the differential form of *Faraday's Law*:

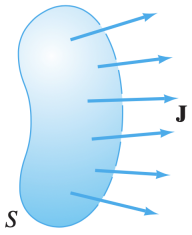
$$\begin{aligned} \frac{d\Phi_B}{dt} &= -\oint_C \mathbf{E} \cdot d\mathbf{r} \\ \frac{d}{dt}\iint_S \mathbf{B} \cdot d\mathbf{S} &= -\oint_C \mathbf{E} \cdot d\mathbf{r} \\ \frac{d}{dt}\iint_S \mathbf{B} \cdot d\mathbf{S} &= -\iint_S \nabla \times \mathbf{E} \cdot d\mathbf{S} \\ \iint_S \frac{d\mathbf{B}}{dt} \cdot d\mathbf{S} &= -\iint_S \nabla \times \mathbf{E} \cdot d\mathbf{S}. \end{aligned}$$

Since this is true for all surfaces S, by a theorem similar to Theorem 3.3 this implies which implies $\nabla \times \mathbf{E} = -\frac{d\mathbf{B}}{dt}$. This is the differential form of Faraday's Law.

3.2.4 Ampére's Law

The total *electric current*, I, through a surface S is the flux of the vector field $\mathbf{J} = \rho\mathbf{v}$ (where ρ is the charge density and $\mathbf{v}$ is the velocity field of the charges) through S

$$I = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$



If the boundary of S is a closed loop C, then *Ampére's Law* says that the circulation of $\mathbf{B}$ around C is proportional to the current passing through C:

$$\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I$$

We can use Stokes's Theorem to rewrite Ampére's Law as a differential equation:

$$\begin{aligned} \oint_C \mathbf{B} \cdot d\mathbf{r} &= \mu_0 I \\ \oint_C \mathbf{B} \cdot d\mathbf{r} &= \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{S} \\ \iint_S \nabla \times \mathbf{B} \cdot d\mathbf{S} &= \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{S}. \end{aligned}$$

Since this is true for all surfaces S, the integrands must be equal. Hence,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \tag{3.5}$$

Equation (3.5) is the differential form of Ampére's Law.

3.2.5 Summary so far

For electric and magnetic fields which go to zero far from all charges, the Helmholtz Theorem says that these are unique specified by their divergence and curl. We have found that these are given by

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \quad \text{(Gauss's Law for Electric Fields)} \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{dt} \quad \text{(Faraday's Law)} \\
\nabla \cdot \mathbf{B} &= 0 \quad \text{(Gauss's Law for Magnetic Fields)} \\
\nabla \times \mathbf{B} &= \mu_0 \mathbf{J} \quad \text{(Ampére's Law)}
\end{aligned}$$

In the next next section we will see that, for time-varying electric fields, these equations contain an inconsistency. This was corrected by Scottish physicist James Clerk Maxwell (1831-1879), and the resulting equations have since been called *Maxwell's Equations*. This correction was crucial to Maxwell's derivation in 1865 that electric and magnetic fields travel through space as waves moving at the speed of light, leading to his proposal that light is an electromagnetic wave. Maxwell's theory was confirmed experimentally by German physicist Heinrich Hertz (1857-1894), who was the first to conclusively prove the existence of electromagnetic waves in 1888.

3.3 Maxwell's Equations

The classical theory of electrodynamics, in its current form, was completed by James Clerk Maxwell (1831-1879) in his work *A Treatise on Electricity and Magnetism*, published in 1873. One of Maxwell's greatest achievements, and one of the greatest scientific discoveries of the 19th century, was to use his mathematical model to propose that light is an electromagnetic phenomenon in 1864. Remarkably, Maxwell's breakthrough was not the result of performing any experiments; it was purely through the use of mathematics, which we now discuss.

Before Maxwell, it was known that the electric and magnetic fields $\mathbf{E}(x,y,z,t)$ and $\mathbf{B}(x,y,z,t)$, sourced by a charge density $\rho(x,y,z,t)$ and current density $\mathbf{J}(x,y,z,t)$, are described by the following system of four coupled partial differential equations:

$$\text{(Gauss's law)} \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \text{(Magnetic Gauss law)} \quad \nabla \cdot \mathbf{B} = 0,$$
$$\text{(Faraday's law)} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \text{(Ampére's law)} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J},$$

where ϵ_0 and μ_0 are fundamental constants (similar to Newton's gravitational constant G), which can be measured in the laboratory.

In addition to Maxwell's Equations, on physical grounds we also require local conservation of charge. That is, not only is the total charge in the universe a constant (global conservation of charge), but charge can only move from one location to another by traveling through the intermediate space. Let's now see what condition this puts on ρ and $\mathbf{J}$.

The total charge q is a region D of space is $q = \iiint_D \rho(x,y,z,t)dV$. For charge to be conserved locally, the rate $\frac{dq}{dt}$ at which the charge enclosed in D changes must be equal to the rate at which charge is flowing out of D, which is the total current $I = \oiint_S \mathbf{J} \cdot d\mathbf{S}$ flowing though the boundary surface; that is $I = -\frac{dq}{dt}$. Differentiating under the integral and using the Divergence Theorem,

$$\begin{aligned} I &= -\frac{dq}{dt} \\ \oiint_S \mathbf{J} \cdot d\mathbf{S} &= -\iiint_D \frac{\partial \rho}{\partial t}\, dV \\ \iiint_D \nabla \cdot \mathbf{J} \cdot d\mathbf{S} &= -\iiint_D \frac{\partial \rho}{\partial t}\, dV, \end{aligned}$$

which implies

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}. \tag{3.6}$$

Equation (3.6) is called the *continuity equation*.

A *steady current* is one for which $\frac{d\rho}{dt} = 0$, which describes a flow of charge in which the charge density at each point never changes. Maxwell noticed that Ampère's Law is not consistent with local conservation of charge for non-steady currents, which we will now see.

We have seen that div curl = 0. If we take the divergence of Faraday's law, we get

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{E}) &= \nabla \cdot (-\frac{\partial \mathbf{B}}{\partial t}) \\ &= -\frac{\partial}{\partial t}(\nabla \cdot \mathbf{B}) \\ &= 0 \qquad\qquad \text{(by Gauss's Law for Magnetic Fields)}\end{aligned}$$

in accordance with the identity div curl $= 0$.

However, if we take the divergence of Ampére's law, we get

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{B}) &= \nabla \cdot (\mu_0 \mathbf{J}) \\ &= \mu_0 (\nabla \cdot \mathbf{J}) \\ &= -\mu_0 \frac{\partial \rho}{\partial t}.\end{aligned}$$

The right-hand side is non-zero for a non-steady current, in contradiction with the identity div curl $= 0$. This shows that Ampére's Law is not consistent with local charge conservation.

Maxwell noted that

$$\begin{aligned}\nabla \cdot \mathbf{J} &= -\frac{\partial \rho}{\partial t} \\ &= -\frac{\partial}{\partial t}(\epsilon_0 \nabla \cdot \mathbf{E}) \qquad \text{(by Gauss's Law for electric fields)} \\ &= -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right).\end{aligned}$$

Since $\nabla \cdot (\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}) = 0$, Maxwell proposed replacing $\mathbf{J}$ in Ampère's Law with $\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ in order to make it consistent with conservation of charge. The resulting equation is called the Ampère-Maxwell law:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Note that, for a steady current $\mathbf{J}$, the electric field $\mathbf{E}$ produced by $\mathbf{J}$ is constant, so $\frac{\partial \mathbf{E}}{\partial t} = 0$ and the Ampère-Maxwell law reduces to the original Ampère law:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + 0 = \mu_0 \mathbf{J}.$$

These equations therefore give identical predictions for magnetostatic phenomena, which must be the case, since we already knew Ampère's law is valid for steady currents. On the other hand, the additional term

$$\mathbf{J}_d := \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

which Maxwell called the *displacement current*, implies a new physical phenomenon: a time-varying electric field sources a magnetic field, just as Faraday's law says that a time varying magnetic field sources an electric field. This magnetic field however, turns out to be hard to detect in ordinary electromagnetic experiments, which is why Faraday and others never discovered it in the laboratory. It ultimately took 24 years for this effect to be measured by Heinrich Hertz (1857-1894) in 1888, confirming Maxwell's theory.

The four equations

$$\text{(Gauss's law)} \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \text{(Magnetic Gauss law)} \quad \nabla \cdot \mathbf{B} = 0,$$
$$\text{(Faraday's law)} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \text{(Ampère-Maxwell law)} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

including the corrected Ampére-Maxwell law, are now known collectively as *Maxwell's Equations*. [1]

[1] Maxwell also did important work in thermodynamics, and there is actually another set of four equations called the "Maxwell relations" in thermodynamics.

3.4 Electromagnetic Waves

The most striking prediction of Maxwell's theory is existence of electromagnetic waves, which implies that light is an electromagnetic phenomenon.

To see this, consider a region of empty space, with no charge or current. In such a region, $\rho = \mathbf{J} = 0$, so Maxwell's equations become

$$\text{(Gauss's law)} \quad \nabla \cdot \mathbf{E} = 0, \quad \text{(Magnetic Gauss law)} \quad \nabla \cdot \mathbf{B} = 0,$$
$$\text{(Faraday's law)} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \ \text{(Ampère-Maxwell law)} \ \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

A common technique in the theory of differential equations is to decouple a system of first-order equations by replacing them with an equivalent system of second-order equations.

Taking the curl of Faraday's Law, the left-hand-side can be written as

$$\begin{aligned} \nabla \times \nabla \times \mathbf{E} &= \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \ \text{(by the double curl identity)} \\ &= \nabla(0) - \nabla^2 \mathbf{E} \ \text{(by Gauss's Law for Electric Fields)} \\ &= -\nabla^2 \mathbf{E} \ \text{(the gradient of a constant function is zero)}, \end{aligned}$$

while the right hand side becomes

$$\begin{aligned} \nabla \times (-\frac{\partial \mathbf{B}}{\partial t}) &= -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) \\ &= -\frac{\partial}{\partial t}(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}) \ \text{(by the Ampère-Maxwell law)} \\ &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \ \text{(since } \mu_0, \epsilon_0 \text{ are constants)}, \end{aligned}$$

which shows that the electric field $\mathbf{E}$ satisfies the three-dimensional *wave equation*:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Similarly, taking the curl of the Ampère-Maxwell Law, the left-hand-side can be written as

$$\begin{aligned} \nabla \times \nabla \times \mathbf{B} &= \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} \\ &= \nabla(0) - \nabla^2 \mathbf{B} \ \text{(by the magnetic Gauss law)} \\ &= -\nabla^2 \mathbf{B} \ \text{(the gradient of a constant function is zero)}, \end{aligned}$$

while the right hand side becomes

$$\begin{aligned} \nabla \times (\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}) &= \mu_0 \epsilon_0 \nabla \times \frac{\partial \mathbf{E}}{\partial t} \ \text{(since } \mu_0, \epsilon_0 \text{ are constants)} \\ &= \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\nabla \times \mathbf{E}) \ \text{(by Exercise **??**(c))} \\ &= \mu_0 \epsilon_0 \frac{\partial}{\partial t}(-\frac{\partial \mathbf{B}}{\partial t}) \ \text{(by the Ampére-Maxwell law)} \\ &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}, \end{aligned}$$

hence

$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2},$$

which shows that the magnetic field $\mathbf{B}$ satisfies the same differential equation.

A pair of solutions $(\mathbf{E}, \mathbf{B})$ to the wave equations

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2},$$
$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2},$$

is called an *electromagnetic wave*. The general form of the wave equation is

$$\nabla^2 \mathbf{F} = \frac{1}{v^2} \frac{\partial^2 \mathbf{F}}{\partial t^2},$$

where v is the speed of the wave, so Maxwell's equations imply that empty space supports the propagation of electromagnetic waves, traveling at a speed

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \ m/s,$$

which happens to be precisely the speed of light, c. The values of μ_0, ϵ_0, and c had already been measured quite precisely in Maxwell's time, leading him to write "*We can scarcely avoid the conclusion that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena.*" After Newton's work, which unified mechanics, Maxwell's work in electromagnetism has been called the "second great unification in physics."

Note the crucial role played by Maxwell's displacement current in the derivation of the wave equation. Without it, $\mathbf{E}$ would merely satisfy Laplace's equation $\nabla^2 \mathbf{E} = 0$, as in electrostatics, and we could have no electromagnetic waves!

3.5 Unification

3.5.1 Electric-Magnetic Duality

Maxwell's equations have some intriguing properties. To avoid distractions, we will work in units where the speed of light c, the vacuum permittivity ϵ_0, the mass of the electron m_e, and the charge of the proton e are all equal to 1. (Note that in these units $\mu_0 = \frac{1}{c^2\epsilon_0}$ is also equal to 1.) This amounts to choosing the following units for length, mass, time, and electric charge:

Quantity	Expression	Metric Value
Length	$\epsilon_0 e^2/m_e c^2$	2.77×10^{-36} m
Mass	m_e	9.11×10^{-31} kg
Time	$\epsilon_0 e^2/m_e c^3$	9.23×10^{-45} s
Electric Charge	e	1.602×10^{-19} C

In these units, Maxwell's equations are

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0,$$
$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j}.$$

Notice that these naturally come in two pairs. The pair that does not involve the electric charge and current densities

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0,$$

looks very much like the pair that does

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j},$$

up to a minus sign in the second pair. This symmetry is clearest in the vacuum Maxwell equations, where the charge and current densities vanish:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0,$$
$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = 0.$$

Then the transformation $\mathbf{B} \mapsto \mathbf{E}$, $\mathbf{E} \mapsto -\mathbf{B}$ exchanges the two pairs of equations. This symmetry is called *electric-magnetic duality* and is a clue that the electric and magnetic fields are part of a unified whole, the *electromagnetic field*. This symmetry can be extended to the inhomogeneous case by introducing magnetic charge and current densities ρ_m and $\mathbf{j}_m$, respectively, so that Maxwell's Equations become

$$\nabla \cdot \mathbf{B} = \rho_m, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{j}_m,$$
$$\nabla \cdot \mathbf{E} = \rho_e, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j}_e,$$

and supplementing the above transformation with $(\rho_e, \mathbf{j}_e) \mapsto (-\rho_m, \mathbf{j}_m)$, $(\rho_m, \mathbf{j}_m) \mapsto (\rho_e, -\mathbf{j}_e)$. No magnetic monopoles have ever been observed, so experimentally it seems that $\rho_m = 0$ and $\mathbf{J}_m = 0$. However, we will have more to say about monopoles later.

3.5.2 Special Relativity

We found previously that Maxwell's Equations predict electromagnetic waves which propagate at the speed of light, c. However, an observer moving toward or away from the light source at a constant speed v should observe the light traveling at a speed $c + v$ or $c - v$, respectively. Thus, it would seem that, unlike Newton's Laws, Maxwell's Equations do not satisfy the *Principle of Galilean Relativity*, which says the laws of physics are the same in all inertial (i.e., unaccelerated) reference frames.

Example 3.4. Suppose $\mathscr{O}$ is an inertial reference frame and $\mathscr{O}'$ is another reference frame moving with constant velocity $\mathbf{V}$ relative to $\mathscr{O}$. If the coordinate axes of $\mathscr{O}$ and $\mathscr{O}'$ overlap at $t = 0$, then the position vectors measured with respect to these reference frames are related by

$$\mathbf{x}' = \mathbf{x} - \mathbf{V}t.$$

Let $\mathbf{F}$ be the net force on a particle. Then

$$\begin{aligned}
\mathbf{F} &= m\mathbf{a} \quad \text{(by Newton's Second Law, since } \mathscr{O} \text{ is inertial)} \\
&= m\frac{d^2\mathbf{x}}{dt^2} \\
&= m\frac{d^2}{dt^2}(\mathbf{x}' + \mathbf{V}t) \\
&= m\frac{d^2\mathbf{x}'}{dt^2} + m\frac{d^2}{dt^2}(\mathbf{V}t) \\
&= m\mathbf{a}' + m\frac{d}{dt}\mathbf{V} \\
&= m\mathbf{a}' + 0 \quad \text{(since } \mathbf{V} \text{ is constant)} \\
&= m\mathbf{a}'.
\end{aligned}$$

This shows that any reference frame undergoing constant relative motion with respect to an inertial frame is inertial. [2]

It would therefore seem that Maxwell's Equations only hold in one reference frame. Just as water wave must have a supporting substance, i.e., a "medium", to move across (in this case water), and audible sound requires a medium to transmit its wave motions (such as air or water), it was believed in the 19th century that light must also require a medium, the *luminiferous aether*", to transmit its wave motions. Because light can travel

[2]Note that the net force $\mathbf{F}$ is the same in all reference frames, since the sum of the forces on an object (which could be gravitational, air resistance, contact forces, etc.) are what they are independent of the motion of an observer relative to the object. Similarly, the mass m and time t are assumed to be the same in all reference frames in Newtonian mechanics.

through the vacuum of space, it was assumed that even the vacuum must be filled with aether.

Because the speed of light is so great, and because material bodies, such as planets, pass through the aether without obvious friction or drag, it was assumed to have a highly unusual combination of properties. Designing experiments to investigate these properties was a high priority of 19th century physics.

In 1887, Michelson and Morley performed a famous experiment using a high-precision interferometer to measure the Earth's velocity relative to the aether and found a null result. At the time, they thought this result proved that the Earth dragged the aether along with it as it moved through space. However, this explanation was contradicted by other experiments.

This puzzle was resolved by Albert Einstein in 1905 in his paper titled *On the Electrodynamics of Moving Bodies* in which he formulated the *Special Theory of Relativity*, which is based on two postulates:

1. (Relativity Principle) The laws of physics are the same in all inertial reference frames.
2. (Invariance of the Speed of Light) The speed of light in a vacuum is the same for all observers.

The second postulate was motivated by Maxwell's theory of electromagnetism and the lack of evidence for the existence of the aether. In Special Relativity, the aether plays no role. However, the theory implies modifications to long-held assumptions about the nature of space and time.

Consider two reference frames $\mathscr{O}$ and $\mathscr{O}'$, which overlap at $t = 0$ and with $\mathscr{O}'$ moving with constant speed v in the positive x-direction. In Newtonian Mechanics, the coordinates in the two reference frames are related by the *Galilean transformation*

$$\begin{aligned} t' &= t \\ x' &= x - vt \\ y' &= y \\ z' &= z. \end{aligned}$$

In particular, there is a universal time. However, according to Special Relativity, the coordinates are related by the *Lorentz transformation* (in units with $c = 1$)

$$\begin{aligned} t' &= \gamma(t - vx) \\ x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z. \end{aligned}$$

where $\gamma = \frac{1}{\sqrt{1-v^2}}$. Note that the Lorentz transformation mixes the time and space variables together, which is the origin of the length contraction, time dilation, and other predictions of Special Relativity. When v is small compared to the speed of light, $v \approx 0$ and $\gamma \approx 1$, so in this case the Lorentz transformation agrees with the Galilean transformation. Thus, the Galilean transformation above can be thought of as an approximation to the true transformation, which is valid at low speeds.

The Lorentz transformation above can be written in a way that resembles a rotation in 3d space by using the *hyperbolic functions*

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \quad \cosh(x) = \frac{e^x + e^{-x}}{2}, \quad \tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

If we define the *rapidity* ζ so that $\tanh(\zeta) = v$, then we have $\cosh(\zeta) = \gamma$ and $\sinh(\zeta) = \gamma v$, so we can write the Lorentz transformation above as

$$\begin{aligned} t' &= \cosh(\zeta)t - \sinh(\zeta)x, \\ x' &= -\sinh(\zeta)t + \cosh(\zeta)x, \\ y' &= y, \\ z' &= z, \end{aligned}$$

which resembles a rotation by an angle ζ in the tx-plane. This suggests that we should not think points $\mathbf{x} = (x, y, z)$ in space and time t separately, but as components of a *4-vector* (t, x, y, z), which labels a point in *spacetime*. It is conventional to denote a component of a 4-vector using Greek letters for spacetime indices and Latin letters for spatial indices; that is x^μ is an element of a 4-vector, where $\mu = 0, 1, 2, 3$, with $x^0 = t$, and x^i with $i = 1, 2, 3$ labeling the spatial components x, y, z.

It turns out that Maxwell's Equations are invariant under Lorentz transformations, so they are automatically compatible with Special Relativity. Indeed, this was the main fact that led Einstein to Special Relativity. Under the Lorentz transformation above, the charge and current densities transform as

$$\begin{aligned} \rho' &= \cosh(\zeta)\rho - \sinh(\zeta)j_x, \\ j'_x &= -\sinh(\zeta)\rho + \cosh(\zeta)j_x, \\ j'_y &= j_y, \\ j'_z &= j_z, \end{aligned}$$

which shows that we should think of the charge and current densities as a 4-vector $\mathbf{J}^\mu = (\rho, \mathbf{j}^i)$ called the *4-current*. The transformation of the electric and magnetic fields under the same Lorentz transformation is somewhat more complicated:

$$\begin{aligned} E'_x &= E_x, \\ E'_y &= \cosh(\zeta)E_y - \sinh(\zeta)B_z, \\ E'_z &= \sinh(\zeta)B_y + \cosh(\zeta)E_z, \end{aligned}$$

$$\begin{aligned} B'_x &= B_x, \\ B'_y &= \cosh(\zeta)B_y + \sinh(\zeta)E_z, \\ B'_z &= -\sinh(\zeta)E_y + \cosh(\zeta)B_z. \end{aligned}$$

We see that the components of the electric and magnetic fields get mixed together under a Lorentz transformation, so, by the philosophy above, we should think of the electric

and magnetic fields not as separate entities, but as components of the *electromagnetic field* on spacetime. However, the electromagnetic field is clearly not a 4-vector, as it has 6 independent components, which get mixed together in a rather subtle way.

What sort of object, then, is the electromagnetic field? It turns out that everything we have discussed so far becomes much clearer when rewritten in the language of *differential forms*. Differential forms were introduced around 1900 by Cartan and Poincaré, and are a powerful unifying notion:

- The operations gradient, curl, and divergence can all be seen as instances of a single operation on differential forms, called the *exterior derivative*. Moreover, this allows us to generalize vector calculus to higher dimensions.
- Differential forms are the natural objects to be integrated over higher-dimensional versions of curves and surfaces, called *manifolds*.
- The Fundamental Theorem of Calculus, Fundamental Theorem of Line Integrals, Stokes' Theorem, and the Divergence Theorem are all instances of a single theorem, which again holds in higher dimensions as well.

As we will see, the electromagnetic field is best understood as a *2-form* defined on spacetime. In addition to their application to physics, differential forms provide an important *functor* called the *de Rham complex*, which is used to construct an important set of invariants of a smooth manifold called the *de Rham cohomology* of the manifold. These invariants belong to a branch of math called *algebraic topology*, as they allow us to use algebra to study the topology of manifolds. After defining the de Rham cohomology and calculating some examples, we will explore some physical consequences.

3.6 Exercises

Monochromatic Plane Waves

1. Let $\mathbf{E}_0$ and $\mathbf{B}_0$ be constant vectors in $\mathbb{R}^3$, $\mathbf{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector, $\mathbf{n} = n_x\hat{i} + n_y\hat{j} + n_z\hat{k}$ a vector such that $||\mathbf{n}|| = \frac{\omega}{c}$, where ω is the angular frequency of the wave and c is the speed of light, and δ is a real constant, independent of x, y, z, and t.

   (a) Show that

   $$\mathbf{E} = \mathbf{E}_0 \cos(\mathbf{n} \cdot \mathbf{r} - \omega t),$$
   $$\mathbf{B} = \mathbf{B}_0 \cos(\mathbf{n} \cdot \mathbf{r} - \omega t + \delta)$$

   are solutions to the wave equations

   $$\nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \nabla^2 \mathbf{B} = \frac{1}{c^2}\frac{\partial^2 \mathbf{B}}{\partial t^2}.$$

   Solutions of this form are called *monochromatic plane waves*, and describe electromagnetic waves of frequency ω traveling in the direction $\mathbf{n}$ at speed c.

   (b) Show that the electric and magnetic Gauss laws imply that the vector $\mathbf{n}$ is orthogonal to both $\mathbf{E}$ and $\mathbf{B}$. Since $\mathbf{n}$ points in the direction in which the wave is propagating, this shows that the wave has no component along the direction of propagation. Such waves are said to be *transverse*.

   (c) Show that Faraday's law implies [3] $\frac{1}{c}(\hat{n} \times \mathbf{E}_0) = \mathbf{B}_0$, where $\hat{n} := \frac{\mathbf{n}}{||\mathbf{n}||}$. Show that this implies that $\mathbf{E}$ and $\mathbf{B}$ are in phase and mutually perpendicular.

   (d) Use the previous equation to show that the amplitudes of $\mathbf{E}$ and $\mathbf{B}$ are related by

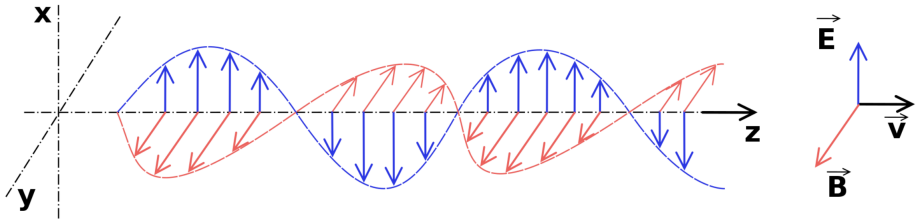
   $$||\mathbf{B}_0|| = \frac{1}{c}||\mathbf{E}_0||.$$

   This shows that the amplitude of the magnetic field component of the wave is 8 orders of magnitude smaller than that of the electric field component, [4] which explains why it was not noticed in experiments before Maxwell's theory.

By the results of this exercise, a picture of a monochromatic plane wave (with magnetic field amplitude greatly enlarged for visibility) is given by the following:

In this way, the changing electric field generates the magnetic field, and vice versa, so that this process sustains itself and we obtain a propagating wave. An animation of this picture can be found here.

[3] The Ampére-Maxwell law does not yield an independent condition; it simply reproduces this equation.
[4] Despite this fact, it turns out that the electric and magnetic fields of the wave have equal energy density.



Lorentz-Invariance of Maxwell's Equations

2. Show that Maxwell's Equations

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0,$$
$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j},$$

are invariant under the Lorentz Transformation

$$\begin{aligned}
t' &= \cosh(\zeta)t - \sinh(\zeta)x, \\
x' &= -\sinh(\zeta)t + \cosh(\zeta)x, \\
y' &= y, \\
z' &= z, \\
\rho' &= \cosh(\zeta)\rho - \sinh(\zeta)j_x, \\
j'_x &= -\sinh(\zeta)\rho + \cosh(\zeta)j_x, \\
j'_y &= j_y, \\
j'_z &= j_z, \\
E'_x &= E_x, \\
E'_y &= \cosh(\zeta)E_y - \sinh(\zeta)B_z, \\
E'_z &= \sinh(\zeta)B_y + \cosh(\zeta)E_z, \\
B'_x &= B_x, \\
B'_y &= \cosh(\zeta)B_y + \sinh(\zeta)E_z, \\
B'_z &= -\sinh(\zeta)E_y + \cosh(\zeta)B_z.
\end{aligned}$$

That is, show that

$$\nabla' \cdot \mathbf{B}' = 0, \quad \nabla' \times \mathbf{E}' + \frac{\partial \mathbf{B}'}{\partial t'} = 0,$$
$$\nabla' \cdot \mathbf{E}' = \rho', \quad \nabla' \times \mathbf{B}' - \frac{\partial \mathbf{E}'}{\partial t'} = \mathbf{j}'.$$

Note that the derivatives with respect to the transformed variables $t,', x', y', z'$ are given in terms of the original variables t, x, y, z as

$$\frac{\partial}{\partial t'} = \cosh(\zeta)\frac{\partial}{\partial t} + \sinh(\zeta)\frac{\partial}{\partial x}$$
$$\nabla' = \left(\sinh(\zeta)\frac{\partial}{\partial t} + \cosh(\zeta)\frac{\partial}{\partial x}\right)\hat{\mathbf{i}} + \frac{\partial}{\partial y}\hat{\mathbf{j}} + \frac{\partial}{\partial z}\hat{\mathbf{k}}.$$

4 Linear Algebra

Here we review some linear algebra we will need.

4.1 Vector Spaces

Definition 4.1 (Vector space). Let $\mathbb{R}$ denote the field of real numbers. A *vector space over* $\mathbb{R}$ (or *real vector space*) is a set V endowed with two operations

$$\begin{aligned} + : V \times V &\to V \\ (v, w) &\mapsto v + w, \end{aligned}$$

called *vector addition*, and

$$\begin{aligned} \cdot : \mathbb{R} \times V &\to V \\ (c, v) &\mapsto cv, \end{aligned}$$

called *scalar multiplication*, satisfying the following properties:

A1. (Associativity of addition) $u + (v + w) = (u + v) + w$ for all $u, v, w \in V$.
A2. (Commutativity of addition) $u + v = v + u$ for all $u, v \in V$.
A3. (Existence of an additive identity) There exists an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
A4. (Existence of additive inverses) For every $v \in V$ there exists a $w \in V$ such that $v + w = 0$.

S1. (Associativity of multiplication) $(xy)v = x(yv)$ for all $x, y \in \mathbb{R}, v \in V$.
S2. (Distributivity over scalar addition) $(x + y)v = xv + yv$ for all $x, y \in \mathbb{R}, v \in V$
S3. (Distributivity over vector addition) $x(v + w) = xv + xw$ for all $x \in \mathbb{R}, v, w \in V$.
S4. (Multiplication by 1 fixes each vector) $1v = v$ for all $v \in V$.

The elements of V are called *vectors* and the elements of $\mathbb{R}$ are called *scalars*.

Example 4.2 ($\mathbb{R}^n$). It is straightforward to check that $\mathbb{R}^n$ is a vector space under the operations

$$\begin{aligned} (x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) &= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) \\ c(x_1, x_2, \ldots, x_n) &= (cx_1, cx_2, \ldots, cx_n) \end{aligned}$$

Indeed, the definition of a vector space came from abstracting the properties of $\mathbb{R}^n$.

Example 4.3 (The empty set). The empty set $\emptyset$ cannot be given the structure of a vector space, since axiom A3 fails. Thus, every vector space is necessarily nonempty.

Example 4.4 (The zero vector space). Any singleton set $\{a\}$ can be given the structure of a vector space by defining $a + a = a$ and $ca = a$ for all $c \in \mathbb{R}$. Since a is the zero vector in axiom A3, we usually denote this by writing $a = 0$. This vector space is called the *zero vector space* or *trivial vector space*.

Example 4.5 (Matrices). The set of $m \times n$ matrices with real entries is a vector space under the entrywise operations

$$(A + B)_{ij} = A_{ij} + B_{ij}$$
$$(cA)_{ij} = cA_{ij}.$$

Example 4.6 (Functions). The set of all real-valued functions defined on a common domain A is a vector space under the pointwise operations

$$(f + g)(x) = f(x) + g(x)$$
$$(cf)(x) = cf(x).$$

4.2 Subspaces

Definition 4.7 (Subspace). Let V be a vector space. If W is a nonempty subset of V which is closed under vector addition and scalar multiplication, then it is straightforward to check that W is itself a vector space under the same operations of V. In this case, we say that W is a *subspace* of V.

The next proposition shows that we can check closure under addition and scalar multiplication all at once.

Proposition 4.8 (Subspace criterion). A subset W of a vector space V is a subspace if and only if $w, w' \in W$ and $c \in \mathbb{R}$ implies $cw + w' \in W$.

Proof. Suppose W is closed under addition and scalar multiplication and let $w, w' \in W$ and $c \in \mathbb{R}$. Then $cw \in W$ since W is closed under scalar multiplication and therefore $cw + w' \in W$ since W is closed under vector addition.

Now suppose W has the property that $w, w' \in W$ and $c \in \mathbb{R}$ implies $cw + w' \in W$. Taking $c = 1$, this implies $1 \cdot w + w' = w + w' \in W$, hence W is closed under vector addition. Taking $w' = 0$ (which is in W since W is a subspace), $cw + 0 = cw \in W$, which shows that W is closed under scalar multiplication. □

Example 4.9. Every vector space has at least two subspaces: $\{0\}$ and V itself.

Example 4.10. The set $C^\infty(\mathbb{R}^n)$ of smooth functions on $\mathbb{R}^n$ is a subspace of all real-valued functions on $\mathbb{R}^n$.

Definition 4.11 (Linear combination, span). Let V be a vector space. A finite sum of the form $\sum_{i=1}^k c^i v_i$, where c^i are scalars and $v_i \in V$, is called a *linear combination* of the vectors $v_1, \ldots, v_k$.

If S is an arbitrary subset of V, then the set of all linear combinations of elements of S is called the *span of* S and is denoted by $\operatorname{span}(S)$; it is the smallest subspace of V containing S.

If $V = \operatorname{span}(S)$, we say that S *spans* V. By convention, a linear combination of no elements is considered to sum to zero, and so that $\operatorname{span}(\emptyset) = \{0\}$.

If $V = \operatorname{span}(S)$ with S finite, then we say that V is *finite-dimensional*; otherwise we say that V is infinite-dimensional.

4.3 Bases and Dimension

Definition 4.12 (Linearly independent set). Let V be a vector space. A subset S pf V is said to be *linearly dependent* if there exists a linear relation of the form $\sum_{i=1}^{k} a^i v_i = 0$, where $v_1, \ldots, v_k$ are distinct elements of S and at lease one of the coefficients a^i is nonzero; S is said to be *linearly independent* otherwise. In other words, S is linearly independent if and only if the only linear combination of distinct elements of S that sums to zero is the one in which all the scalar coefficients are zero.

Remark 4.13. Note that every set containing the zero vector is linearly dependent. By convention, the empty set is considered to be linearly independent.

It is frequently important to work with ordered k-tuples of vectors in V; such a k-tuple is denoted by $(v_1, \ldots, v_k)$ or (v_i), with parentheses instead of braces to distinguish it from the (unordered) set of elements $\{v_1, \ldots, v_k\}$.

Definition 4.14 (Linearly independent k-tuple). We say that $(v_1, \ldots, v_k)$ is a *linearly dependent k-tuple* if there are scalars $(a^1, \ldots, a^k)$, not all zero, such that $\sum_{i=1}^{k} a^i v_i = 0$; it is a *linearly independent k-tuple* otherwise.

Remark 4.15. The only difference between a linearly independent set and a linearly independent k-tuple is that the latter cannot have repeated vectors. For example if $v \in V$ is a nonzero vector, the ordered pair (v, v) is linearly dependent, while the set $\{v, v\} = \{v\}$ is linearly independent. On the other hand, if $(v_1, \ldots, v_k)$ is any linearly independent k-tuple, then the set $\{v_1, \ldots, v_k\}$ is also linearly independent.

Proposition 4.16. Let V be a vector space.

(a) If $S \subseteq V$ is linearly independent, then every subset of S is linearly independent.

(b) If $S \subseteq V$ is linearly dependent or spans V, then every subset of V that properly contains S is linearly dependent.

(c) A subset $S \subseteq V$ containing more than one element is linearly dependent if and only if some element $v \in S$ can be expressed as a linear combination of elements $S \setminus \{v\}$.

(d) If $(v_1, \ldots, v_k)$ is a linearly dependent k-tuple in V with $v_1 \neq 0$, then some v_i can be expressed as a linear combination of the preceding vectors $(v_1, \ldots, v_{i-1})$.

Proof. (a) Suppose $S \subseteq V$ is linearly independent. Then $\sum_{i=1}^{k} a^i v_i = 0$ implies $a^i = 0$ for all $i = 1, \ldots, k$ whenever $v_1, \ldots, v_k$ are vectors in S. If W is a subset of S and $\sum_{i=1}^{k} a^i v_i = 0$, then since each $v_i \in W$ is also in S, this is a linear combination of vectors in S and therefore we have $a^i = 0$ for all $i = 1, \ldots, k$. Hence, W is linearly independent.

(b) Suppose $S \subseteq V$ is linearly dependent. Then there exist vectors $v_1, \ldots, v_k$ in S and scalars $a^1, \ldots, a^k$ with at least one $a^j \neq 0$, such that $\sum_{i=1}^{k} a^i v_i = 0$. If W is a subset that properly contains S, then for any $w \in W$ and $a^{k+1} = 0$ we have $\sum_{i=1}^{k} a^i v_i + a^{k+1} w = 0$ with $a^j \neq 0$, which shows that W is linearly dependent.

Suppose now that $\operatorname{span}(S) = V$ and let W be a subset of V that properly contains S. Note that V cannot be the zero vector space, since the only proper subset is $\emptyset$, which is linearly independent. Since S spans V, S must contain a non-zero vector. Since V is not the zero vector space and S spans V, S must contain a nonzero vector. If $W = S \cup \{0\}$, then W is linearly dependent because it contains the zero vector. Otherwise, W properly contains S and contains a nonzero vector w. Since S spans V, there exist scalars $a^1, \ldots, a^k$ and vectors $v_1, \ldots, v_k \in S$ such that $\sum_{i=1}^k a^i v_i = w$. Since $w \neq 0$, at least one term $a^j v_j \neq 0$, which means that $a^j \neq 0$ and $v_j \neq 0$. It then follows that $\sum_{i=1}^k (-a^i) v_i + w = 0$ with $a^j \neq 0$, hence W is linearly dependent.

(c) Suppose S is linearly dependent. Then there exists a linear relation of the form $\sum_{i=1}^k a^i v_i = 0$ with $a^j \neq 0$. It follows that $v_j = \sum_{i \neq j} \left(-\frac{a^i}{a^j} \right) v_i$. Now suppose $v \in S$ can be written as $v = \sum_{i=1}^k a^i v_i$ for $v_1, \ldots, v_k \in S \setminus \{v\}$. Then $v - \sum_{i=1}^k a^i v_i = 0$, where the coefficient of v is $1 \neq 0$, which shows that S is linearly dependent.

(d) If $(v_1, \ldots, v_k)$ is a linearly dependent k-tuple, then there exists a k-tuple of scalars $(a^1, \ldots, a^k)$, not all zero, such that $\sum_{i=1}^k a^i v_i = 0$. Choose j to be the largest index such that $a^j \neq 0$. Then

$$v_j = \sum_{i=1}^{j-1} \left(-\frac{a^i}{a^j} v_i \right).$$

If $j = 1$, then there are no preceding vectors, so the claim is vacuously true.

□

Definition 4.17 (Basis). A *basis for* V is a subset $S \subseteq V$ that is linearly independent and spans V. If S is a basis for V, then every element of V has a *unique* expression as a linear combination of elements of S.

If V has a finite basis, then V is said to be *finite-dimensional*, and otherwise it is *infinite-dimensional*. The trivial vector space $\{0\}$ is finite-dimensional, because it has the empty set as a basis.

Definition 4.18 (Ordered basis). If V is finite-dimensional, an *ordered basis for* V is a basis endowed with a specific ordering of the basis vectors, or equivalently a linearly independent n-tuple (E_i) that spans V.

For most purposes, ordered bases are more useful than unordered bases, so we always assume, often without comment, that each basis comes with a given ordering.

If $(E_1, \ldots, E_n)$ is an (ordered) basis for V, each vector $v \in V$ has a unique expression as a linear combination of basis vectors:

$$v = \sum_{i=1}^n v^i E_i.$$

The real numbers v^i are called the *components of* v with respect to this basis, and the ordered n-tuple $(v^1, \ldots, v^n)$ is called its *basis representation*. (Note that this definition requires an ordered basis.)

Lemma 4.19. Let V be a vector space. If V is spanned by a set of n vectors, then every subset of V containing more than n vectors is linearly dependent.

Proof. Suppose $\operatorname{span}(\{v_1, \ldots, v_n\}) = V$. To prove the lemma, it suffices to show that every subset of V containing exactly $n+1$ vectors is linearly dependent. Let $\{w_1, \ldots, w_{n+1}\}$ be such a set. If $w_i = 0$ for some i, then the set is linearly dependent, so assume $w_i \neq 0$ for all $i = 1, \ldots, n$. By Proposition 4.16(b), the set $\{w_1, v_1, \ldots, v_n\}$ is linearly dependent, and thus so is the ordered $(n+1)$-tuple $(w_1, v_1, \ldots, v_n)$. By 4.16(d), one of the vectors v_j can be written as a linear combination of $\{w_1, v_1, \ldots, v_j\}$, and thus the set $\{w_1, v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n\}$ still spans V. Renumbering the v_i's if necessary, we may assume that the set $\{w_1, v_2, \ldots, v_n\}$ spans V.

Now suppose by induction that $\{w_1, w_2, \ldots, w_{k-1}, v_k, \ldots, v_n\}$ spans V. As before, the $(n+1)$-tuple $(w_1, w_2, \ldots, w_{k-1}, w_k, v_k, \ldots, v_n)$ is linearly dependent, so one of the vectors in this list can be written as a linear combination of the preceding ones. If one of the w_i's can be so written, then the set $\{w_1, \ldots, w_{n+1}\}$ is linearly dependent and we are done. Otherwise, one of the v_j's can be so written, and after reordering we may assume that $\{w_1, w_2, \ldots, w_k, v_{k+1}, \ldots, v_n\}$ still spans V. Continuing by induction, by the time we get to $k = n$, if we have not already shown that the w_i's are dependent, we conclude that the set $\{w_1, \ldots, w_n\}$ spans V. But this means that the set $\{w_1, \ldots, w_{n+1}\}$ is linearly dependent by Proposition 4.16(b). □

Proposition 4.20. If V is a finite-dimensional, all bases for V contain the same number of elements.

Proof. If $\{E_1, \ldots, E_n\}$ is a basis for V with n elements, then Lemma 4.19 implies that every set containing more than n elements is linearly dependent, so no basis can have more than n elements. On the other hand, if there were a basis containing fewer than n elements, then Lemma 4.19 would imply that $\{E_1, \ldots, E_n\}$ is linearly dependent, which is a contradiction. □

Because of the preceding proposition, if V is a finite-dimensional vector space, it makes sense to define the *dimension of* V, denoted by $\dim V$, to be the number of elements in any basis.

Proposition 4.21. Let V be a finite-dimensional vector space.

(a) Every set that spans V contains a basis, and every linearly independent subset of V is contained in a basis.

(b) Every subspace $S \subseteq V$ is finite-dimensional and satisfies $\dim S \leq \dim V$, with equality if and only if $S = V$.

(c) Show that $\dim V = 0$ if and only if $V = \{0\}$.

Example 4.22. The *standard basis* for $\mathbb{R}^n$ consists of the n vectors $e_1, \ldots, e_n$, where $e_i = (0, \ldots, 1, \ldots, 0)$ is the vector with a 1 in the ith place and zeros elsewhere. This shows that $\mathbb{R}^n$ has dimension n. Any element $x \in \mathbb{R}^n$ can be written $(x^1, \ldots, x^n) = \sum_{i=1}^n x^i e_i$, so its components with respect to the standard basis are just its coordinates $(x^1, \ldots, x^n)$.

4.4 Quotient vector spaces

Definition 4.23 (Cosets). If V is a vector space and W is a subspace of V, a *coset* of W in V is a subset of the form

$$v + W = \{v + w : w \in W\}$$

for some $v \in V$.

Two cosets $v + W$ and $v' + W$ are equal if and only if $v' = v + w$ for some $w \in W$, or equivalently, if and only if $v' - v \in W$. This introduces an equivalence relation on V:

$$v \sim v' \iff v' - v \in W \iff v + W = v' + W.$$

A coset of W in V is simply an equivalence class under this equivalence relation. Any element of $v + W$ is called a *representative* of the coset $v + W$.

Definition 4.24 (Quotient vector space). The set V/W of all cosets in W in V is again a vector space, with addition and scalar multiplication defined by

$$(u + W) + (v + W) = (u + v) + W$$
$$r(v + W) = rv + W$$

for $u, v \in V$ and $r \in \mathbb{R}$.

We call V/W the *quotient vector space* or the *quotient space of V by W*.

Note that $(0 + W) + (v + W) = (0 + v) + W = v + W$, so the coset W is the zero vector in V/W.

Example 4.25. For $V = \mathbb{R}^2$ and W a line through the origin in $\mathbb{R}^2$, a coset of W in $\mathbb{R}^2$ is a line parallel to W. The quotient space $\mathbb{R}^2/W$ is the collection of lines in $\mathbb{R}^2$ parallel to W.

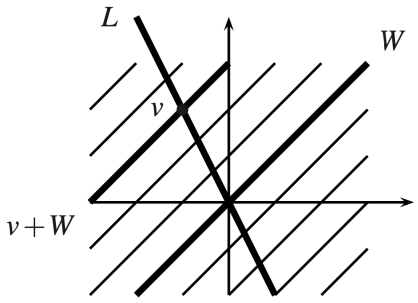


Figure 21: Quotient vector space of $\mathbb{R}^2$ by W.

Proposition 4.26. If V be a finite-dimensional vector space and W a subspace of V, then

$$\dim V/W = \dim V - \dim W.$$

Proof. Let $\{w_1, \ldots, w_m\}$ be a basis for W. Since $\{w_1, \ldots, w_m\}$ is a linearly independent subset of V, it can be extended to a basis $\{w_1, \ldots, w_m, v_1, \ldots, v_n\}$ for V. We will show that the set $\{v_1 + W, \ldots, v_n + W\}$ is a basis for V/W.

First,

$$\begin{aligned} 0_{V/W} = W &= \sum_{i=1}^{n} c^i (v_i + W) \\ &= \left(\sum_{i=1}^{n} c^i v_i \right) + W \end{aligned}$$

implies $\sum_{i=1}^{n} c^i v_i \in W$. But $\sum_{i=1}^{n} c^i v_i \in V \setminus W$ by definition, so $\sum_{i=1}^{n} c^i v_i \in W \cap (V \setminus W) = \{0\}$, which implies $\sum_{i=1}^{n} c^i v_i = 0$. Since any subset of a linearly independent set is linearly independent, $\{v_1, \ldots, v_n\}$ is linearly independent, so $\sum_{i=1}^{n} c^i v_i = 0$ implies $c^i = 0$ for all i. This proves that $\{v_1 + W, \ldots, v_n + W\}$ is linearly independent.

Now let $v + W \in V/W$. Since $\{w_1, \ldots, w_m, v_1, \ldots, v_n\}$ is a basis for V, there exist scalars $a^1, \ldots, a^m, b^1, \ldots, b^n$ such that $v = \sum_{i=1}^{m} a^i w_i + \sum_{j=1}^{n} b^j v_j$. Since $\sum_{i=1}^{m} a^i w_i \in W$,

$$\begin{aligned} v + W &= \sum_{i=1}^{m} a^i w_i + \sum_{j=1}^{n} b^j v_j + W \\ &= \sum_{j=1}^{n} b^j v_j + W \\ &= \sum_{j=1}^{n} b^j (v_j + W) \end{aligned}$$

which shows that $\operatorname{span}(\{v_1 + W, \ldots, v_n + W\}) = V/W$, and therefore $\{v_1 + W, \ldots, v_n + W\}$ is a basis for V/W. This proves that

$$\dim V/W = n = (n + m) - m = \dim V - \dim W.$$

□

4.5 Linear Transformations

Definition 4.27 (Linear transformation, isomorphism). Let V and W be vector spaces over $\mathbb{R}$. A map $f : V \to W$ is called a *linear transformation*, a *vector space homomorphism*, a *linear operator*, or a *linear map* over $\mathbb{R}$ if for all $u, v \in V$ and $c \in \mathbb{R}$,

$$\begin{aligned} f(u + v) &= f(u) + f(v) \\ f(cv) &= cf(v). \end{aligned}$$

If, additionally, f is a bijection, then it is called a *vector space isomorphism*, and we say that V and W are isomorphic, which we denote by $V \cong W$.

Proposition 4.28.

(a) If $f : V \to W$ is an isomorphism, then so is $f^{-1} : W \to V$.

(b) If $f : V \to U$ and $g : U \to W$ is an isomorphism, then so is $g \circ f : V \to W$

Proof.

(a) Since the inverse of a bijection is a bijection, all we need to show is that f^{-1} is linear. Since f is surjective, given $w, w' \in W$, there exist $v, v', v'' \in V$ such that $w = f(v)$ and $w' = f(v')$, $cw + w' = f(v'')$. It follows that

$$\begin{aligned} cw + w' &= f(v'') \\ cf(v) + f(v') &= f(v'') \\ f(cv + v') &= f(v'') \quad \text{(since } f \text{ is linear)} \\ cv + v' &= v'' \quad \text{(since } f \text{ is injective)} \\ cf^{-1}(w) + f^{-1}(w') &= f^{-1}(cw + w'), \end{aligned}$$

which proves that f^{-1} is linear, and therefore an isomorphism.

(b) Since the composition of a bijection is a bijection, all we need to show is that the composition of linear maps is linear. Let $v, v' \in V$ and $c \in \mathbb{R}$. Then

$$\begin{aligned} (g \circ f)(cv + v') &= g(f(cv + v')) \\ &= g(cf(v) + f(v')) \quad \text{(since } f \text{ is linear)} \\ &= cg(f(v)) + g(f(v')) \quad \text{(since } g \text{ is linear)} \\ &= c(g \circ f)(v) + (g \circ f)(v'), \end{aligned}$$

which shows that $g \circ f$ is linear, and hence an isomorphism.

□

Definition 4.29 (Kernel and image). If $f : V \to W$ is a linear transformation, the *kernel* of f is the set

$$\ker f = \{v \in V : f(v) = 0\}$$

and the *image* of f is the set

$$\operatorname{im} f = \{w \in : w = f(v) \text{ for some } v \in V\}.$$

Proposition 4.30. If $f : V \to W$ is a linear map, then $\ker f$ is a subspace of V and $\operatorname{im} f$ is a subspace of W.

Proof. Let $v, w \in \ker f$. Then $f(cv + w) = cf(v) + f(w) = c0 + 0 = 0 + 0 = 0$, so $cv + w \in \ker f$. This shows that $\ker f$ is a subspace of V.

Let $w, w' \in \operatorname{im} f$. Then there exist $v, v' \in V$ such that $w = f(v)$ and $w' = f(v')$. It follows that $cw + w' = cf(v) + f(v') = f(cv + v')$, which shows that $cw + w' \in \operatorname{im} f$ and therefore $\operatorname{im} f$ is a subspace of W. □

Proposition 4.31. Let $f : V \to W$ be a linear map. Then $\ker f = \{0\}$ if and only if f is injective.

Proof. For any linear map $f(0) = f(0+0) = f(0) + f(0)$. Subtracting $f(0)$ from both sides gives $f(0) = 0$. Hence $0 \in \ker f$ (or, $\{0\} \subseteq \ker f$) for any linear map.

Suppose f is injective. Then $f(v) = f(v') \implies v = v'$. If $v \in \ker f$, then $f(v) = 0 = f(0)$, so $v = 0$. This shows that $\ker f \subseteq \{0\}$, and therefore $\ker f = \{0\}$.

Now suppose $\ker f = \{0\}$. If $f(v) = f(v')$, then $0 = f(v) - f(v') = f(v - v')$, which says that $v - v' \in \ker f$, and therefore $v - v' = 0$ which implies $v = v'$. Hence, f is injective. □

Proposition 4.32. Let V be an n-dimensional vector space and let $(v_1, \ldots, v_n)$ be an ordered basis for V. Then the map

$$f : \mathbb{R}^n \to V$$
$$f(x^1, \ldots, x^n) = \sum_{i=1}^{n} x^i v_i$$

is an isomorphism.

Proof.

$$\begin{aligned} f(c(x^1, \ldots, x^n) + (y^1, \ldots, y^n)) &= f(cx^1 + y^1, \ldots, cx^n + y^n) \\ &= \sum_{i=1}^{n} (cx^i + y^i) v_i \\ &= c \sum_{i=1}^{n} x^i + \sum_{i=1}^{n} y^i \\ &= cf(x^1, \ldots, x^n) + f(y^1, \ldots, y^n) \end{aligned}$$

which shows that f is linear.

Since $(v_1, \ldots, v_n)$ is linearly independent,

$$f(x^1, \ldots, x^n) = \sum_{i=1}^{n} x^i v_i = 0$$

implies $x^i = 0$ for all i. This proves that $\ker f = \{0\}$, hence f is injective by Proposition 4.31.

Since $(v_1, \ldots, v_n)$ spans V, given any vector $v \in V$, there exist scalars $c^1, \ldots, c^n$ such that

$$v = \sum_{i=1}^{n} c^i v_i = f(c^1, \ldots, c^n),$$

which shows that f is surjective. This proves that f is a bijection, and hence an isomorphism. □

Proof. (Alternative proof that f is bijective.) Since $(v_1, \ldots, v_n)$ is an ordered basis for V, ever $v \in V$ can be written uniquely as

$$v = \sum_{i=1}^{n} c^i v_i$$

for some scalars $c^1, \ldots, c^n$. The map

$$\begin{aligned} g : V &\to \mathbb{R}^n \\ g(v) &= (c^1, \ldots, c^n) \end{aligned}$$

is easily seen to be a 2-sided inverse for f, hence f is a bijection. □

Proposition 4.33. Two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension.

Proof. Suppose $\dim V = \dim W = n$, then by Proposition 4.32 there exist isomorphisms $f : V \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to W$, and therefore $g \circ f : V \to W$ is an isomorphism.

Now suppose that $V \cong W$. Let $f : V \to W$ be an isomorphism, and let $\{v_1, \ldots, v_n\}$ be a basis for V. We will show that $\{f(v_1), \ldots, f(v_n)\}$ is a basis for W.

First, note that

$$\begin{aligned} 0 &= \sum_{i=1}^{n} c^i f(v_i) \\ &= f\left(\sum_{i=1}^{n} c^i v_i \right) \end{aligned}$$

implies $\sum_{i=1}^{n} c^i v_i \in \ker f$. Since f is injective, by Proposition **??** $\ker f = \{0\}$, and therefore

$$\sum_{i=1}^{n} c^i v_i = 0.$$

Since $\{v_1, \ldots, v_n\}$ is linearly independent, this implies $c^i = 0$ for all $i = 1, \ldots, n$. Hence, $\{f(v_1), \ldots, f(v_n)\}$ is linearly independent.

Now let $w \in W$. Since f is surjective, there exists $v \in V$ such that $w = f(v)$. Since $\{v_1, \ldots, v_n\}$ is a basis for V, there exist scalars $c^1, \ldots, c^n$ such that $v = \sum_{i=1}^{n} c^i v_i$. Since f is linear, it follows that

$$\begin{aligned} w &= f(v) \\ &= f\left(\sum_{i=1}^{n} c^i v_i \right) \\ &= \sum_{i=1}^{n} c^i f(v_i), \end{aligned}$$

which shows that $\{f(v_1), \ldots, f(v_n)\}$ spans W and is therefore a basis for W, proving that $\dim W = \dim V$. □

4.6 First Isomorphism Theorem

By Proposition 4.30, if $f : V \to W$ is a linear map then $\ker f$ is a subspace of V and $\operatorname{im} f$ is a subspace of W. We can therefore form the quotient spaces $V/\ker f$ and $W/\operatorname{im} f$. The latter space, $W/\operatorname{im} f$, is called the *cokernel* of f, denoted by $\operatorname{coker} f$.

Theorem 4.34 (First isomorphism theorem). Let $f : V \to W$ be a linear map. Then f induces an isomorphism $\bar{f} : V/\ker f \cong \operatorname{im} f$, where

$$\bar{f} : V/\ker f \to \operatorname{im} f$$
$$\bar{f}(v + \ker f) = f(v).$$

Proof. To see that this map is well-defined, suppose $v + \ker f = v' + \ker f$. Then $v' - v \in \ker f$, and therefore

$$\bar{f}(v' + \ker f) = f(v') = f(v + (v' - v)) = f(v) + f(v' - v) = f(v) + 0 = \bar{f}(v + \ker f).$$

Hence, $\bar{f}$ is well-defined.

Let us now check that $\bar{f}$ is linear. Since

$$c(v + \ker f) + (v' + \ker f) = (cv + v') + \ker f,$$

we have

$$\begin{aligned}\bar{f}(c(v + \ker f) + (v' + \ker f)) &= \bar{f}((cv + v') + \ker f)\\ &= f(cv + v')\\ &= cf(v) + f(v')\\ &= c\bar{f}(v + \ker f) + \bar{f}(v' + \ker f),\end{aligned}$$

which proves that $\bar{f}$ is linear.

To see that $\bar{f}$ is injective, note that

$$0 = \bar{f}(v + \ker f) = f(v)$$

which implies $v \in \ker f$, hence $v + \ker f = \ker f = 0_{V/\ker f}$. This shows that $\ker \bar{f} \subseteq \{0\}$ and therefore $\ker \bar{f} = \{0\}$. Hence, $\bar{f}$ is injective by Proposition 4.31.

Finally, to show that $\bar{f}$ is surjective, let $w \in \operatorname{im} f$. Then there exists $v \in V$ such that $w = f(v) = \bar{f}(v + \ker f)$. This proves that $\bar{f}$ is a bijection, and therefore an isomorphism. □

Corollary 4.35 (Rank-nullity law). If V is a finite-dimensional vector space and $f : V \to W$ a linear transformation, then

$$\dim V = \dim \ker f + \dim \operatorname{im} f.$$

Proof. By the First Isomorphism Theorem and Proposition 4.26.

$$\dim \operatorname{im} \mathrm{f} = \dim V/\ker f = \dim V - \dim \ker f.$$

Adding $\dim \ker f$ to both sides gives

$$\dim V = \dim \ker f + \dim \operatorname{im} f.$$

□

4.7 Direct sums

The *sum* of two subspaces A and B of a vector space V is the subspace

$$A + B = \{a + b \in V : a \in A, b \in B\}.$$

Since $c(a+b) + (a'+b') = (ca+a') + (cb+b')$ with $ca + a' \in A$ and $cb + b' \in B$, this is indeed a subspace of V.

If $A \cap B = \{0\}$, this sum is called an *internal direct sum* and written $A \oplus_i B$. In this case, every $v \in V$ has a representation as $v = a + b$ for a unique $a \in A$ and $b \in B$. Indeed, if $a + b = a' + b' \in A \oplus_i B$, then

$$a - a' = b' - b \in A \cap B = \{0\},$$

and therefore $a = a'$ and $b = b'$.

In contrast to the internal direct sum, if A and B are any two vector spaces, then we can form their *direct sum*

$$A \oplus B = \{(a, b) : a \in A, b \in B\},$$

which is the same thing as their Cartesian product. [5]

In fact, these two notions are isomorphic: if A and B are both subspaces of V with $A \cap B = \{0\}$, then

$$\begin{aligned} \varphi : A \oplus B &\to A \oplus_i B \\ (a, b) &\mapsto a + b \end{aligned}$$

is an isomorphism. For this reason, in the literature the internal direct sum is normally denoted by $A \oplus B$, just like the external direct sum.

If $V = A \oplus_i B$, then A is called a *complimentary subspace* to B in V.

Proposition 4.36. If W is a subspace of V and W' is a complementary subspace to W, then the map

$$\begin{aligned} \varphi : W' &\to V/W \\ w' &\mapsto w' + W. \end{aligned}$$

is an isomorphism.

Proof. To see that φ is linear,

$$\begin{aligned} \varphi(cw_1' + w_2') &= cw_1' + w_2' + W \\ &= c(w_1' + W) + (w_2' + W) \\ &= c\varphi(w_1') + \varphi(w_2'), \end{aligned}$$

hence φ is linear.

[5] The direct sum and direct product differ for infinite-dimensional vector spaces.

Now suppose $\varphi(w') = w' + W = W = O_{V/W}$. Then $w' \in W$ and therefore $w' \in W \cap W' = \{0\}$, hence $w' = 0$. This shows that $\ker \varphi = \{0\}$, hence φ is injective.

Let $v + W \in V/W$. Since $v = w + w'$ for some $w \in W$ and $w' \in W'$,

$$v + W = w + w' + W = w' + W = \varphi(w'),$$

which shows that φ is surjective, and hence a bijection, completing the proof that φ is an isomorphism. □

4.8 Algebra over $\mathbb{R}$

An *algebra* over the field $\mathbb{R}$ is a vector space A over $\mathbb{R}$ with a multiplication map

$$\cdot : A \times A \to A$$

which satisfies the following properties for all $a, b, c \in A$ and $r \in \mathbb{R}$:

(i) (Associativity) $(ab)c = a(bc)$,

(ii) (Left- and right-distributivity) $(a + b)c = ac + bc$ and $a(b + c) = ab + ac$,

(iii) (Homogeneity) $r(ab) = (ra)b = a(rb)$.

If A and A' are algebras over $\mathbb{R}$, then an *algebra homomorphism* is a linear map $L : A \to A'$ that preserves the algebra multiplication: $L(ab) = L(a)L(b)$ for all $a, b \in A$.

Example 4.37 ($C^\infty(\mathbb{R}^n)$). The vector space $C^\infty(\mathbb{R}^n)$ of all smooth, real-valued functions on $\mathbb{R}^n$ is an algebra with respect to the product

$$(fg)(p) := f(p)g(p).$$

4.9 Exercises

Vector spaces

1. If you haven't already, check that the sets in Examples 1.2, 1.3, 1.5, and 1.6 satisfy the axioms in Definition 1.1. What is the zero vector in each case? What is the inverse of a non-zero element?

2. Let $\mathbb{R}^\omega$ denote the set of infinite sequences of real numbers. An element of $\mathbb{R}^\omega$ is therefore of the form

$$v = (v_1, v_2, \dots, v_n, \dots)$$

with $v_i \in \mathbb{R}$ for all $i \in \mathbb{N}$. Show that $\mathbb{R}^\omega$ is a vector space with respect to the operations

$$\begin{aligned} v + w &:= (v_1 + w_1, v_2 + w_2, + \dots, v_n + w_n, \dots) \\ cv &:= (cv_1, cv_2, \dots, cv_n, \dots). \end{aligned}$$

3. Show that the set of polynomials of degree $\leq n$, which we denote by P_n, is a vector space.

4. Let V be the set of positive real numbers, and define vector addition to be ordinary multiplication and scalar multiplication to be exponentiation:

$$\begin{aligned} u + v &:= uv \\ cu &:= u^c. \end{aligned}$$

Is this vector space isomorphic to a more familiar vector space?

5. There are additional properties of any vector space which follow immediately from Definition 1.1. Use Definition 1.1 to prove that if V is a vector space, then the following statements are true:

   (a) The zero vector $0 \in V$ in axiom A3 is unique.

   (b) $c0 = 0$ for all $c \in \mathbb{R}$.

   (c) $0v = 0$ for all $v \in V$.

   (d) For each $v \in V$, the additive inverse w in axiom A4 is unique. It therefore makes sense to write $w = -v$.

   (e) $-v = (-1)v$ for all $v \in V$.

   (f) If $cv = 0$, then either $x = 0$ or $v = 0$.

Subspaces

5. Show that a line through the origin is a subspace of $\mathbb{R}^n$.
6. Show that the following are subspaces of the vector space of $M^n(\mathbb{R})$ of $n \times n$ matrices with real entries:
   (a) Symmetric matrices
   (b) Upper triangular matrices
   (c) Lower triangular matrices
   (d) Diagonal matrices
7. Show that the set of all continuously differentiable functions on (a, b) is a subspace of the set of all functions on (a, b).

Bases

8. A 2×2 matrix A with *complex* entries is called *Hermitian* if it is equal to its conjugate transpose: $\overline{A}^T = A$. Show that the matrices
$$\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$$
form a basis for the *real* vector space of Hermitian matrices.
9. Prove Proposition 4.21.

Linear transformations / isomorphisms

10. Show that the vector space in problem 4 is isomorphic to a more familiar vector space.
11. Show that $V/\{0\} \cong V$.
12. Given a linear map $T : V \to V$ and a subspace W of V, define a linear map $\overline{T} : V/W \to V/W$ by
$$\overline{T}(v + W) = \overline{T}v + W.$$
Show that $\overline{T}$ is well-defined if and only if $\overline{T}w \in W$ for all $w \in W$.

5 Vector Fields and Differential Forms

5.1 Manifolds

An *n-dimensional manifold* is a space that looks like $\mathbb{R}^n$, but perhaps not globally.
For example, the sphere

$$x^2 + y^2 + z^2 = 1$$

looks locally like the plane $\mathbb{R}^2$, which is why some people though the Earth was flat. We call this a *2-sphere*, S^2, to indicate that it is a 2-dimensional manifold.

- Similarly, while the space we live in looks locally like $\mathbb{R}^3$, we have no way yet of ruling out the possibility that it is really S^3, the *3-sphere*:

  $$w^2 + x^2 + y^2 + z^2 = 1$$

  and in fact in many models of cosmology space is a 3-sphere.

- String Theory is a theory of physics which unifies General Relativity and Quantum Mechanics. A surprising prediction of String Theory is the spacetime is 10-dimensional. In order to describe our universe, the 6 extra dimensions are assumed to be a compact 6-dimensional manifold, so that the spacetime looks approximately 4-dimensional on large distance scales. The geometry of this manifold affects the spectrum of particles, as well as their masses and interactions. There are a huge number of choices for this 6-manifold, and while many "toy models" have been studied since the mid-1980s, a 6-manifold has not yet been identified which reproduces our universe, and this remains an area of intense research.

- The Hamiltonian formulation of mechanics uses *phase spaces*, which could be very high-dimensional. For a single particle moving in $\mathbb{R}^3$, the coordinates of a point in phase space are given by the three components of the particle's position together with the three components of the particle's momentum, so the phase space is $\mathbb{R}^6$. For N particles moving in $\mathbb{R}^3$, phase space is $\mathbb{R}^{6N}$. If the system of particles is a gas, then N is on the order of Avogadro's number. If the particle motion is constrained, then the topology of the phase space can be non-trivial (i.e., a manifold globally different from $\mathbb{R}^n$). For example, for a mass at the end of a pendulum swinging in a plane, the phase space is S^1. The phase space of a double pendulum is the 2-torus $T^2 \cong S^1 \times S^1$.

This hopefully gives an idea of why it is necessary to learn about manifolds to do modern physics. Since every manifold looks locally like $\mathbb{R}^n$, a good understanding of $\mathbb{R}^n$ is essential in generalizing differential and integral calculus to a manifold. One big difference between $\mathbb{R}^n$ and a general n-manifold is that $\mathbb{R}^n$ is special in having a set of standard global coordinates. This allows one to define all constructions on $\mathbb{R}^n$ in terms of the standard coordinates and to carry out all computations explicitly.

Since a manifold does not, in general, have standard coordinates, only-coordinate independent concepts will make sense on a manifold. For example, on an n-manifold it is

not possible to integrate functions, because the integral of a function depends on a set of coordinates. For example, if we integrate the constant function $f = 1$ over a unit ball, then $\int f dV$ is the volume of the ball, which is not invariant under changes of coordinates.

As we will see, it is only because the existence of global coordinates permits an identification of functions with differential n-forms on $\mathbb{R}^n$ that integration of functions becomes possible on $\mathbb{R}^n$. While the study of manifolds is fascinating in its own right, since our goal is to do physics, we will instead turn our attention to the basic types of fields that are defined on manifolds: vector fields and differential forms.

5.2 Tangent Vectors

Recall that we can view a vector at a point $p \in \mathbb{R}^n$ either geometrically as an arrow emanating from p

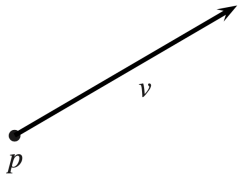


Figure 22: A vector v at p.

or algebraically as a column of numbers

$$v = \begin{bmatrix} v^1 \\ v^2 \\ v^3 \end{bmatrix}.$$

Define the *tangent space to* $\mathbb{R}^n$ as the vector space of all arrows emanating from p, which we denote by $T_p\mathbb{R}^n$. By the correspondence above, $T_p\mathbb{R}^n \cong \mathbb{R}^n$, so we can identify the tangent space to $\mathbb{R}^n$ at p with $\mathbb{R}^n$ itself.

A vector field on a manifold M can similarly be pictured geometrically as an arrow attached to each point of M.

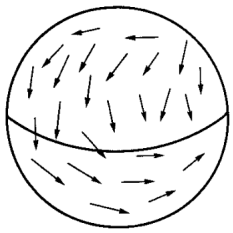


Figure 23: A vector field on S^2.

As before, we would like to define the *tangent space to M at a point p* as the vector space of all arrows emanating from p. For example, when $M = S^2$, this is just the tangent plane to S^2 at p. While this works just fine for S^2, there are other manifolds, such as projective spaces, which do not sit inside $\mathbb{R}^n$ in any natural way. Thus, for manifolds, **it is much more useful to describe tangent vectors algebraically**.

The key is to identify a tangent vector v at p with the operator $D_v|_p = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\big|_p$. To see that this makes sense, note that the latter is a linear map $C^\infty(\mathbb{R}^n) \to \mathbb{R}$ which satisfies the product rule. Such a map is called a *derivation at* p of $C^\infty(\mathbb{R}^n)$. It is straightforward to check that the sum and scalar multiple of derivations at p are also derivations at p, so these maps form a vector space, which we denote by $D_p(\mathbb{R}^n)$. We then have the following theorem:

Theorem 5.1. The map $\phi_p : T_p(\mathbb{R}^n) \to D_p(\mathbb{R}^n)$ defined by

$$\phi_p(v) = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\bigg|_p$$

is an isomorphism.

Proof. We leave it to the reader to verify that ϕ_p is linear. To show that ϕ_p is surjective, suppose that $v \in \ker \phi_p$. Then

$$\sum_{i=1}^n v^i \frac{\partial f}{\partial x^i}\bigg|_p = 0$$

for all $f \in C^\infty(\mathbb{R}^n)$. In particular, this must be true when $f = x^j$. Now $\partial x^j/\partial x^i = 1$ if i and j are the same, and 0 if i and j are different. We can write this compactly as $\partial x^j/\partial x^i = \delta_i^j$, where

$$\delta_i^j := \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases}$$

is called the *Kronecker δ-symbol.* We therefore have

$$0 = \phi_p(v)(x^j) = \sum_{i=1}^n v^i \frac{\partial x^j}{\partial x^i}\bigg|_p = \sum_{i=1}^n v^i \delta_i^j = v^j.$$

Since this is true for each $j = 1, \ldots, n$, this shows that $v = 0$. Hence, $\ker \phi_p = \{0\}$, so ϕ_p is injective.

The proof that ϕ_p is surjective uses Taylor's Theorem. This would require too much of a digression, so we omit the proof of surjectivity. □

Thus, by the isomorphism $T_p\mathbb{R}^n \cong D_p(\mathbb{R}^n)$ in Theorem 5.1, we can identify a tangent vector v at p with the operator which takes the directional derivative in the direction of v at p:

$$\boxed{v \longleftrightarrow \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\bigg|_p}$$

Recall that a vector field is then an assignment of a vector v_p to each point $p \in \mathbb{R}^n$; in other words, a vector field is a mapping $\mathbb{R}^n \to \bigcup_p T_p\mathbb{R}^n$. By the discussion above, we write a vector field as

$$v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}, \tag{5.1}$$

where

$$v_p = \sum_{i=1}^{n} v^i(p) \frac{\partial}{\partial x^i}\bigg|_p.$$

The vector fields $\partial/\partial x^i$ form a basis for $T_p\mathbb{R}^n$ at each point P. By definition, a vector field v is *smooth* if and only if each component function $v^i : \mathbb{R}^3 \to \mathbb{R}$ in Equation (5.1) is smooth. We denote the set of smooth vector fields on $\mathbb{R}^n$ by $\mathfrak{X}(\mathbb{R}^n)$. Since one can add two smooth vector fields to get another smooth vector field, and multiply a smooth vector field by a constant to get another vector field, $\mathfrak{X}(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. In fact, we can multiply vector fields not just by real scalars $c \in \mathbb{R}$, but by smooth functions $g \in C^\infty(\mathbb{R}^n)$ and all of the vector space axioms still hold. Since $C^\infty(\mathbb{R}^n)$ is only a ring (it is not a field since not every nonzero element has a multiplicative inverse), one says that this makes $\mathfrak{X}(\mathbb{R}^n)$ a $C^\infty(\mathbb{R}^n)$-*module*.

5.3 1-forms

Recall that the gradient ∇f of a scalar function f is a vector field. In rectangular coordinates, it is given by

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle.$$

However, the gradient does not take this same form in every coordinate system. For example, in spherical coordinates (ρ, θ, ϕ), it takes the form

$$\nabla f = \left\langle \frac{\partial f}{\partial r}, \frac{1}{r}\frac{\partial f}{\partial \theta}, \frac{1}{r\sin(\theta)}\frac{\partial f}{\partial \phi} \right\rangle.$$

Since the gradient is coordinate-dependent, it is not a valid vector field on a manifold. However, the gradient plays a fundamental role in calculus, so we need some object that does make intrinsic sense on a manifold to play the role of the gradient.

The key is to recall that the gradient ∇f of a function f keeps track of the directional derivatives of f at all points. That is, given any tangent vector v, we can use ∇f to define a mapping

$$\begin{aligned} T_p\mathbb{R}^n &\to \mathbb{R} \\ v &\mapsto v \cdot \nabla f|_p. \end{aligned}$$

What are the defining properties of this map? There is really just one: linearity. Such linear maps can be added and scalar multiplied to get other linear maps, so they form a vector space. This vector space is called the *cotangent space at* p and is an example of a *dual vector space*, which we now describe.

5.3.1 The dual space

Given a vector space V, denote by V^* the set of linear maps from $V \to \mathbb{R}$. Elements of V^* are called *linear functionals* or *covectors* on V.

Assume now that V is finite-dimensional, and let $\{e_1, \dots, e_n\}$ be a basis for V. Then every $v \in V$ is uniquely a linear combination $v = \sum_{i=1}^n v^i e_i$ with $v^i \in \mathbb{R}$. Define $\alpha^i \in V^*$ to be be the linear map that picks out the ith coordinate of v, $\alpha^i(v) = v^i$. It follows from linearity that that α^i is defined by

$$\alpha^i(e_j) = \delta^i_j = \begin{cases} 1 & \text{for } i = j, \\ 0 & \text{for } i \neq j. \end{cases}$$

Proposition 5.2. The set $\{\alpha^1, \dots, \alpha^n\}$ is a basis for V^*.

Proof. If $f \in V^*$ and $v = \sum_{i=1}^n v^i e_i \in V$, then

$$\begin{aligned} f(v) &= f\left(\sum_{i=1}^n v^i e_i\right) \\ &= \sum_{i=1}^n v^i f(e_i) \qquad \text{(since } f \text{ is linear)} \\ &= \sum_{i=1}^n \alpha^i(v) f(e_i) \qquad \text{(by definition of } \alpha^i) \\ &= \left(\sum_{i=1}^n f(e_i)\alpha^i\right)(v) \qquad \text{(since a linear combination of linear maps is linear).} \end{aligned}$$

Hence, $f = \sum_{i=1}^n f(e_i)\alpha^i$, which shows that $V^* = \text{Span}\{\alpha^1, \dots, \alpha^n\}$.

Now suppose $\sum_{i=1}^n c_i \alpha^i = 0$ for some $c_i \in \mathbb{R}$. This means that $\sum_{i=1}^n c_i \alpha^i(v) = 0$ for every $v \in V$. In particular, taking $v = e_j$, we have

$$\begin{aligned} 0 &= \sum_{i=1}^n c_i \alpha^i(e_j) \\ &= \sum_{i=1}^n c_i \delta^i_j \\ &= c_j. \end{aligned}$$

Since j was arbitrary, this shows that $c_j = 0$ for all $j = 1, \dots, n$, hence $\{\alpha^1, \dots, \alpha^n\}$ is linearly independent, and hence a basis for V^*. □

Definition 5.3 (Dual basis). The (ordered) basis $(\alpha^1, \dots, \alpha^n)$ for V^* is said to be *dual* to the (ordered) basis $(e_1, \dots, e_n)$ for V. In particular, this proves:

Corollary 5.4. The dual space V^* of a finite-dimensional vector space V has the same dimension as V.

5.3.2 1-forms

Definition 5.5 (Cotangent space). Define the *cotangent space to* $\mathbb{R}^n$ *at* p, denoted by $T_p^*\mathbb{R}^n$, to be the dual of the tangent space at p; that is

$$T_p^*\mathbb{R}^n := (T_p\mathbb{R}^n)^*.$$

Elements of $T_p^*\mathbb{R}^n$ are called *cotangent vectors*.

The we denote the basis for $T_p^*\mathbb{R}^n$ dual to the basis $(\partial/\partial x^1|_p, \ldots, \partial/\partial x^n|_p)$ for $T_p\mathbb{R}^n$ by $(dx^1|_p, \ldots, dx^n|_p)$. By definition of the dual basis,

$$dx^i|_p \left(\frac{\partial}{\partial x^j}\bigg|_p \right) = \delta^i_j.$$

Definition 5.6 (1-form). A *1-form* on $\mathbb{R}^n$ is an assignment of a cotangent vector to each point $p \in \mathbb{R}^n$; that is, a 1-form is a mapping $\mathbb{R}^n \to \bigcup_p T_p^*\mathbb{R}^n$. A 1-form ω on $\mathbb{R}^n$ can be written as

$$\omega = \sum_{i=1}^n \omega_i dx^i,$$

where

$$\omega_p = \sum_{i=1}^n \omega_i(p) dx^i|_p.$$

A 1-form ω is smooth if the functions $\omega_i : \mathbb{R}^3 \to \mathbb{R}$ are smooth. We denote the set of all smooth 1-forms on $\mathbb{R}^n$ by $\Omega^1(\mathbb{R}^n)$.

Example 5.7 (Differential of a function). Given $f \in C^\infty(\mathbb{R}^n)$, we can construct a 1-form df, called the *differential of* f, as follows. For $p \in \mathbb{R}^n$ and $v_p \in T_p\mathbb{R}^n$, define $(df)_p \in T_p^*\mathbb{R}^n$ by

$$(df)_p(v_p) = v_p f. \tag{5.2}$$

The 1-form df is then defined by $df : p \mapsto (df)_p$. The differential df of f turns out to be the desired analog of the gradient ∇f. To see this, expand $(df)_p$ in the dual basis $(dx^i|_p)$ as

$$(df)_p = \sum_{i=1}^n a_i(p) dx^i|_p,$$

for some real numbers $a_i(p)$. To solve for the $a^i(p)$'s, apply Equation (5.2) to $v_p = \frac{\partial}{\partial x^i}\big|_p$:

$$\begin{aligned}(df)_p(v_p) &= v_p f \\ \sum_{i=1}^n a_i(p) dx^i|_p \left(\frac{\partial}{\partial x^j}\bigg|_p\right) &= \frac{\partial f}{\partial x^j}\bigg|_p \\ \sum_{i=1}^n a_i(p)\delta^i_j &= \frac{\partial f}{\partial x^j}\bigg|_p \\ a_j(p) &= \frac{\partial f}{\partial x^j}\bigg|_p.\end{aligned}$$

This shows that at each point $p \in \mathbb{R}^n$ we have

$$(df)_p = \sum_{i=1}^n \frac{\partial f}{\partial x^i}\bigg|_p dx^i|_p,$$

and therefore the gradient is given in coordinates as

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i.$$

Since $f \in C^\infty(\mathbb{R}^n)$, this shows that $df \in \Omega^1(\mathbb{R}^n)$.

Given any point $p \in \mathbb{R}^n$ and tangent vector $v_p = \sum_{j=1}^n v^i(p)\frac{\partial}{\partial x^i}\big|_p$, we can use the differential of a function f to calculate the directional derivative

$$\begin{aligned}(df)_p(v_p) &= \sum_{i=1}^n \frac{\partial f}{\partial x^i}\bigg|_p dx^i|_p \left(\sum_{j=1}^n v^j(p)\frac{\partial}{\partial x^j}\bigg|_p\right) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x^i}\bigg|_p \sum_{j=1}^n v^j(p) dx^i|_p \left(\frac{\partial}{\partial x^j}\bigg|_p\right) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x^i}\bigg|_p \sum_{j=1}^n v^j(p)\delta^i_j \\ &= \sum_{i=1}^n v^i(p)\frac{\partial f}{\partial x^i}\bigg|_p \\ &= D_{v_p}|_p f.\end{aligned}$$

Unlike the gradient ∇f, Equation (5.2) is coordinate-independent, and therefore df has intrinsic meaning on a manifold.

In the special case $f = x^i$, then

$$df = \sum_{j=1}^n \frac{\partial x^i}{\partial x^j} dx^j = \sum_{j=1}^n \delta^i_j dx^j = dx^i.$$

Hence, the 1-forms dx^i are the differentials of the coordinate functions $f(x^1, \ldots, x^n) = x^i$ on $\mathbb{R}^n$, which is why we chose the notation '$dx^i|_p$' for the basis vector dual to $\partial/\partial x^i|_p$.

5.4 The Exterior Algebra of Alternating Tensors

5.4.1 Tensors on a Vector Space

Let V be a vector space. In addition to linear maps $V \to \mathbb{R}$, *multilinear* maps are also important.

Definition 5.8 (k-tensor). Let $V^k = V \times \cdots \times V$ be the Cartesian product of k copies of a real vector space V. This is a vector space with respect to the operations

$$(v_1, v_2, \ldots, v_n) + (w_1, w_2, \ldots, w_n) := (v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n)$$
$$c(v_1, v_2, \ldots, v_n) := (cv_1, cv_2, \ldots, cv_n).$$

A function $f : V^k \to \mathbb{R}$ is *k-linear* if it is linear in each of its k arguments, while holding the other variables constant:

$$f(\ldots, cv + w, \ldots) = cf(\ldots, v, \ldots) + f(\ldots, w, \ldots) \quad \text{for all } v, w \in V \text{ and } c \in \mathbb{R}.$$

A 1-linear map is the same as a linear map. A 2-linear map is also said to be *bilinear* and a 3-linear map is also called *trilinear*.

A multilinear map $f : V^k \to \mathbb{R}$ is also called a *k-tensor on V*. The non-negative integer k is called the *degree* of the tensor f.

Example 5.9 (Dot product on $\mathbb{R}^n$). The dot product $f : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, $f(v, w) = \sum_{i=1}^n v^i w^i$ is a 2-tensor on $\mathbb{R}^n$, since

$$f(v, cw_1 + w_2) = \sum_{i=1}^n v^i(cw_1^i + w_2^i) = c\sum_{i=1}^n v^i w_1^i + \sum_{i=1}^n v^i w_2^i = cf(v, w_1) + f(v, w_2),$$
$$f(cv_1 + v_2, w) = \sum_{i=1}^n (cv_1^i + v_2^i)w = c\sum_{i=1}^n v_1^i w^i + \sum_{i=1}^n v_2^i w^i = cf(v_1, w) + f(v_2, w).$$

Example 5.10 (Determinant on $\mathbb{R}^n$). The determinant $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $\det[v_1 \ \cdots \ v_n]$, viewed as a function of the n column vectors $v_1, \ldots, v_n$, is an n-tensor on $\mathbb{R}^n$.

The set of all k-tensors on V is denoted $L_k(V)$, which is a real vector space with respect to the usual addition and scalar multiplication of functions.

Definition 5.11 (Tensor product). Define the *tensor product* of a k-tensor and and ℓ-tensor $\otimes : L_k(V) \times L_\ell(V) \to L_{k+\ell}(V)$ by

$$(f \otimes g)(v_1, \ldots, v_{k+\ell}) := f(v_1, \ldots, v_k)g(v_{k+1}, \ldots, v_{k+\ell}).$$

That is, given a $k + \ell$-tuple of vectors $(v_1, \ldots, v_{k+\ell})$ in V, evaluate f on the first k vectors and g on the remaining ℓ vectors. The tensor product $f \otimes g$ is clearly multilinear, since f and g are.

Proposition 5.12. The tensor product is associative; that is, if $f \in L_k(V), g \in L_\ell(V)$, and $h \in L_m(V)$, then

$$(f \otimes g) \otimes h = f \otimes (g \otimes h).$$

Proof. If we evaluate both sides on $(v_1, \ldots, v_{k+\ell+m})$, then both sides are equal to

$$f(v_1, \ldots, v_k)g(v_{k+1}, \ldots, v_{k+\ell})h(v_{k+\ell+1}, \ldots, v_{k+\ell+m}).$$

□

5.4.2 Symmetric and Alternating Tensors

There are two important classes of k-tensors $f(v_1, \ldots, v_k)$, which are characterized by their behavior under a permutation (rearrangement) of the k arguments $v_1, \ldots, v_k$. Basic facts about permutations are collected in Appendix A. It would be a good idea to look over Appendix A now before reading the rest of this section, and to refer back to Appendix A as a reference, as needed.

Definition 5.13 (Symmetric and Alternating tensors). A k-tensor $f : V^k \to \mathbb{R}$ is *symmetric* if

$$f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) = f(v_1, \ldots, v_k)$$

for all permutations $\sigma \in S_k$; it is *alternating* if

$$f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) = (\operatorname{sgn}(\sigma))f(v_1, \ldots, v_k)$$

for all $\sigma \in S_k$.

Example 5.14 (Dot product on $\mathbb{R}^n$). Consider again the dot product $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $f(v_1, v_2) = v_1 \cdot v_2 = \sum_{i=1}^n v_1^i v_2^i$. There are two permutations of $\{1, 2\}$:

$$\sigma = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}, \ \tau = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Since $f(v_{\sigma(1)}, v_{\sigma(2)}) = f(v_1, v_2)$ and $f(v_{\tau(1)}, v_{\tau(2)}) = f(v_2, v_1) = v_2 \cdot v_1 = v_1 \cdot v_2 = f(v_1, v_2)$, we see that f is symmetric.

Example 5.15 (Determinant on $\mathbb{R}^n$). Consider again the determinant $f : \mathbb{R}^n \times \cdots \times \mathbb{R}^n$, $f(v_1, \ldots, v_n) = \det[v_1 \ \cdots \ v_n]$. A permutation $\sigma \in S_n$ rearranges the columns of the matrix $[v_1 \ \cdots \ v_n]$. A property of determinants says that swapping two columns (i.e., a transposition) multiplies the determinant by -1. Since every permutation σ can be decomposed into a product of transpositions, we see that a permutation $\sigma \in S_n$ of the columns multiplies the determinant by $\operatorname{sgn}(\sigma)$. Hence, the determinant is alternating.

We denote by $A_k(V)$ the set of alternating k-tensors on V. If $k = 1$, then the only permutation of $\{1\}$ is the identity permutation $\sigma : 1 \mapsto 1$, which has $\operatorname{sgn}(\sigma) = 1$, so if $f \in A_1(V)$, then $f(v_{\sigma(1)}) = f(v_1) = \operatorname{sgn}(\sigma)f(v_1)$. This shows that every linear functional $V \to \mathbb{R}$ is alternating, so $A_1(V) = V^*$. We define an alternating 0-tensor on V to be a constant, so $A_0(V) = \mathbb{R}$.

5.4.3 The Wedge Product

The tensor product $\otimes$ is a mapping $L_k(V) \times L_\ell(V) \to L_{k+\ell}(V)$. We would like to define a product which takes two alternating tensors to another alternating tensor: $A_k(V) \times A_\ell(V) \to A_{k+\ell}(V)$. The problem with using the tensor product, is that if f and g are alternating, then $f \otimes g$ is not always alternating. This motivates the definition of the *wedge product* (also called the *exterior product*) below.

First, define a mapping (it would be a good idea to look over Appendix B on the permutation action on tensors before reading the rest of this section)

$$\begin{aligned} \text{Alt} : L_k(V) &\to A_k(V) \\ \text{Alt}(f) &= \sum_{\sigma \in S_k} (\text{sgn}(\sigma))\sigma f. \end{aligned}$$

This means that if $v_1, \ldots, v_k$ are vectors in V, then

$$\text{Alt}(f)(v_1, \ldots, v_k) = \sum_{\sigma \in S_k} (\text{sgn}(\sigma)) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}).$$

Proposition 5.16. The k-tensor $\text{Alt}(f)$ is alternating.

Proof. Let $\tau \in S_k$. Then

$$\begin{aligned} \tau(\text{Alt} f) &= \sum_{\sigma \in S_k} (\text{sgn}(\sigma))\tau(\sigma f) \\ &= \sum_{\sigma \in S_k} (\text{sgn}(\sigma))(\tau\sigma f) \\ &= \sum_{\sigma \in S_k} (\text{sgn}(\tau))^2(\text{sgn}(\sigma))(\tau\sigma f) \qquad (\text{since } (\text{sgn}(\tau))^2 = 1) \\ &= (\text{sgn}(\tau)) \sum_{\sigma \in S_k} (\text{sgn}(\tau))(\text{sgn}(\sigma))(\tau\sigma) f \\ &= (\text{sgn}(\tau)) \sum_{\sigma \in S_k} (\text{sgn}(\tau\sigma))(\tau\sigma) f \\ &= (\text{sgn}(\tau)) \sum_{\tau\sigma \in S_k} (\text{sgn}(\tau\sigma))(\tau\sigma) f \qquad (\text{since } \tau\sigma \text{ runs over all of } S_k \text{ if } \sigma \text{ does}) \\ &= (\text{sgn}(\tau)) \sum_{\sigma' \in S_k} (\text{sgn}(\sigma'))(\sigma') f \qquad (\text{renaming the summation index}) \\ &= (\text{sgn}(\tau))\text{Alt}(f). \end{aligned}$$

□

Example 5.17. If $f \in L_3(V)$ and $v_1, v_2, v_3 \in V$, then

$$\text{Alt}(f)(v_1, v_2, v_3) = f(v_1, v_2, v_3) - f(v_1, v_3, v_2) + f(v_2, v_3, v_1) - f(v_2, v_1, v_3) + f(v_3, v_1, v_2) - f(v_3, v_2, v_1)$$

Definition 5.18 (Wedge product). The *wedge product* is defined to be the mapping $\wedge : A_k(V) \times A_\ell(V) \to A_{k+\ell}(V)$ defined by

$$f \wedge g = \frac{1}{k!\ell!}\mathrm{Alt}(f \otimes g).$$

Explicitly, if $v_1, \ldots, v_{k+\ell}$ are vectors in V, then

$$(f \wedge g)(v_1, \ldots, v_{k+\ell}) = \frac{1}{k!\ell!} \sum_{\sigma \in S_{k+\ell}} (\mathrm{sgn}(\sigma)) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+\ell)})$$

By Proposition 5.16, $f \wedge g$ is alternating. The factor $1/k!\ell!$ is conventional.

When $k = 0$, $f \in A_0(V)$ is simply a constant $c \in \mathbb{R}$. In this case, if $g \in A_\ell(V)$, then

$$\begin{aligned}
(c \wedge g)(v_1, \ldots, v_\ell) &= \frac{1}{\ell!} \sum_{\sigma \in S_\ell} (\mathrm{sgn}(\sigma)) c g(v_{\sigma(1)}, \ldots, v_{\sigma(\ell)}) \\
&= c\frac{1}{\ell!} \sum_{\sigma \in S_\ell} (\mathrm{sgn}(\sigma)) g(v_{\sigma(1)}, \ldots, v_{\sigma(\ell)}) \\
&= c\frac{1}{\ell!} \sum_{\sigma \in S_\ell} (\mathrm{sgn}(\sigma))^2 g(v_1, \ldots, v_\ell) \qquad \text{(since } g \text{ is alternating)} \\
&= c\frac{1}{\ell!} \sum_{\sigma \in S_\ell} g(v_1, \ldots, v_\ell) \qquad \text{(since } (\mathrm{sgn}(\sigma))^2 = 1\text{)} \\
&= c\frac{1}{\ell!} \ell! g(v_1, \ldots, v_\ell) \qquad \text{(since we are adding } g(v_1, \ldots, v_k)\ \ell! \text{ times)} \\
&= c g(v_1, \ldots, v_\ell),
\end{aligned}$$

which shows that the wedge product of a constant and an ℓ-tensor is just scalar multiplication.

Example 5.19. If $\alpha^1, \alpha^2 \in A_1(V) = V^*$ and $v_1, v_2 \in V$, then

$$\begin{aligned}
(\alpha^1 \wedge \alpha^2)(v_1, v_2) &= \frac{1}{1!1!} \sum_{\sigma \in S_2} (\mathrm{sgn}(\sigma)) f(v_{\sigma(1)}) g(v_{\sigma(2)}) \\
&= \alpha^1(v_1)\alpha^2(v_2) - \alpha^1(v_2)\alpha^2(v_1) \\
&= \det \begin{bmatrix} \alpha^1(v_1) & \alpha^1(v_2) \\ \alpha^2(v_1) & \alpha^2(v_2) \end{bmatrix}.
\end{aligned}$$

Proposition 5.20 (Properties of the wedge product). The wedge product has the following properties for all $f_1, f_2, f \in A_k(V)$ and $g_1, g_2, g \in A_\ell(V)$, and $c \in \mathbb{R}$.

(i) (Anticommutativity) $f \wedge g = (-1)^{k\ell} g \wedge f$.

(ii) (Bilinearity) $(cf_1 + f_2) \wedge g = c(f_1 \wedge g) + f_2 \wedge g$ and $f \wedge (cg_1 + g_2) = c(f \wedge g_1) + f \wedge g_2$.

(iii) (Associativity) $(f \wedge g) \wedge h = f \wedge (g \wedge h)$.

Example 5.21. The proof of Proposition 5.20 is rather tedious, so instead let's verify property (i) for the example of $f \in A_2(V)$ and $g \in A_1(V)$. If $v_1, v_2, v_3 \in V$, then

$$\begin{aligned}
&(f \wedge g)(v_1, v_2, v_3) \\
&= \frac{1}{2!1!} \sum_{\sigma \in S_3} (\operatorname{sgn}(\sigma)) f(v_{\sigma(1)}, v_{\sigma(2)}) g(v_{\sigma(3)}) \\
&= \frac{1}{2!1!} [f(v_1, v_2)g(v_3) - f(v_2, v_1)g(v_3) + f(v_2, v_3)g(v_1) - f(v_3, v_2)g(v_1) + f(v_3, v_1)g(v_2) - f(v_1, v_3)g(v \\
&= \frac{1}{2!1!} [g(v_1)f(v_2, v_3) - g(v_1)f(v_3, v_2) + g(v_2)f(v_3, v_1) - g(v_2)f(v_1, v_3) + g(v_3)f(v_1, v_2) - g(v_3)f(v_2, v \\
&= (-1)^2 \frac{1}{2!1!} \sum_{\sigma \in S_3} (\operatorname{sgn}(\sigma)) g(v_{\sigma(1)}) f(v_{\sigma(2)}, v_{\sigma(3)}) \\
&= (-1)^{2 \cdot 1} (g \wedge f)(v_1, v_2, v_3).
\end{aligned}$$

Example 5.22. Let's use Proposition 5.20(iii) to compute the wedge product of three covectors. If $\alpha^1, \alpha^2, \alpha^3 \in A_1(V) = V^*$ and $v_1, v_2, v_3 \in V$, then

$$\begin{aligned}
&(\alpha^1 \wedge \alpha^2 \wedge \alpha^3)(v_1, v_2, v_3) \\
&= \frac{1}{1!1!1!} \sum_{\sigma \in S_3} (\operatorname{sgn}(\sigma)) \alpha^1(v_{\sigma(1)}) \alpha^2(v_{\sigma(2)}) \alpha^3(v_{\sigma(3)}) \\
&= \alpha^1(v_1)\alpha^2(v_2)\alpha^3(v_3) - \alpha^1(v_1)\alpha^2(v_3)\alpha^3(v_2) + \alpha^1(v_2)\alpha^2(v_3)\alpha^3(v_1) - \alpha^1(v_2)\alpha^2(v_1)\alpha^3(v_3) \\
&+ \alpha^1(v_3)\alpha^2(v_1)\alpha^3(v_2) - \alpha^1(v_3)\alpha^2(v_2)\alpha^3(v_1) \\
&= \alpha^1(v_1)[\alpha^2(v_2)\alpha^3(v_3) - \alpha^2(v_3)\alpha^3(v_2)] - \alpha^1(v_2)[\alpha^2(v_1)\alpha^3(v_3) - \alpha^2(v_3)\alpha^3(v_1)] \\
&+ \alpha^1(v_3)[\alpha^2(v_3)\alpha^3(v_1) - \alpha^2(v_1)\alpha^3(v_3)] \\
&= \det \begin{bmatrix} \alpha^1(v_1) & \alpha^1(v_2) & \alpha^1(v_3) \\ \alpha^2(v_1) & \alpha^2(v_2) & \alpha^2(v_3) \\ \alpha^3(v_1) & \alpha^3(v_2) & \alpha^3(v_3) \end{bmatrix}.
\end{aligned}$$

More generally, if $\alpha^1, \ldots, \alpha^k \in A_1(V) = V^*$ and $v_1, \ldots, v_k$, then $(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) = \det[\alpha^i(v_j)]$.

5.4.4 Exterior Algebra

Definition 5.23 (Exterior algebra). For a finite-dimensional vector space V, with $\dim V = n$, define

$$A_\bullet(V) = \bigoplus_{k=0}^{n} A_k(V).$$

That is, an element f of $A_\bullet(V)$ can be written uniquely as a sum

$$f = f_0 + f_1 + f_2 \cdots f_n$$

with $f_i \in A_i(V)$. This is an algebra over $\mathbb{R}$, with product given by the wedge product. The algebra $A_\bullet(V)$ is called the *exterior algebra* or *Grassmann algebra*. Because of the direct sum structure and the fact that $\wedge : A_k(V) \times A_\ell(V) \to A_{k+\ell}(V)$, one says that the exterior algebra is a *graded algebra*, which in this case is graded by the degree k. Moreover, because of the property that $f \wedge g = (-1)^{k\ell} g \wedge f$, one says that the exterior algebra is an *anticommutative graded algebra*.

5.4.5 A basis for alternating k-tensors

Let $(e_1, \ldots, e_n)$ be an ordered basis for a finite-dimensional vector space V with $\dim V = n$, and let $(\alpha^1, \ldots, \alpha^n)$ be the dual basis for V^*.

Proposition 5.24. The set of all wedge products of the form $\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}$, where $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, form a basis for $A_k(V)$.

Proof. First we show linear independence. Suppose $\sum c_{i_1 \ldots i_k} \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} = 0$. This means that $\sum c_{i_1 \ldots i_k} \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}(v_1, \ldots, v_k) = 0$ for all k-tuples of vectors $(v_1, \ldots, v_k)$ in V^k. Then in particular, if $1 \leq j_1 < j_2 < \cdots < j_k \leq n$, we have

$$\begin{aligned} 0 &= \sum c_{i_1 \ldots i_k} \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}(e_{j_1}, \ldots, e_{j_k}) \\ &= \sum c_{i_1 \ldots i_k} \alpha^{i_1}(e_{j_1}) \cdots \alpha^{i_k}(e_{j_k}) \\ &= \sum c_{i_1 \ldots i_k} \delta^{i_1}_{j_1} \cdots \delta^{i_k}_{j_k} \\ &= c_{j_1 \cdots j_k}, \end{aligned}$$

since the product of Kronecker δ-functions is zero unless $i_1 = j_1, i_2 = j_2, \ldots, i_k = j_k$. Since $(j_1, \ldots, j_k)$ was an arbitrary strictly-ascending k-tuple of indices, this shows that all the $c_{i_1 \cdots i_k}$ with $i_1 < i_2 < \cdots < i_k$ are zero, which proves linear independence.

To show that the set of all $\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}$ with $(i_1, \ldots, i_k)$ a strictly-increasing k-tuple of indices spans $A_k(V)$, let $f \in A_k(V)$. We claim that

$$f = \sum f(e_{i_1}, \ldots, e_{i_k})(\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}).$$

Let $g = \sum f(e_{i_1}, \ldots, e_{i_k})(\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k})$. By k-linearity and the alternating property, if two k-tensors agree on $(e_{j_1}, \ldots, e_{j_k})$, where $j_1 < j_2 < \cdots < j_k$, then they are equal. We see that

$$\begin{aligned} g(e_{j_1}, \ldots, e_{j_k}) &= \sum f(e_{i_1}, \ldots, e_{i_k})(\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k})(e_{j_1}, \ldots, e_{j_k}) \\ &= \sum f(e_{i_1}, \ldots, e_{i_k}) \alpha^{i_1}(e_{j_1}) \cdots \alpha^{i_k}(e_{j_k}) \\ &= \sum f(e_{i_1}, \ldots, e_{i_k}) \delta^{i_1}_{j_1} \cdots \delta^{i_k}_{j_k} \\ &= f(e_{j_1}, \ldots, e_{j_k}). \end{aligned}$$

Hence, $f = g$. □

Corollary 5.25. If a finite-dimensional vector space V has dimension n, then the vector space $A_k(V)$ of k-tensors on V has dimension $\binom{n}{k}$.

Proof. A strictly ascending k-tuple of indices $(i_1, \ldots, i_k)$ is obtained by choosing a subset of k numbers from $1, \ldots, n$. This can be done in $\binom{n}{k}$ ways. □

Corollary 5.26. If $k > \dim(V)$, then $A_k(V) = 0$.

Proof. If $\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}$, at least two of the factors must be the same, say $\alpha^j = \alpha^\ell = \alpha$. Because $\alpha \in A_1(V) = V^*$, $\alpha \wedge \alpha = 0$ (since, by the antisymmetry property, $\alpha \wedge \alpha = (-1)^{1\cdot 1}\alpha \wedge \alpha = -\alpha \wedge \alpha$), from which it follows that $\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} = 0$. □

5.5 Differential k-forms

Definition 5.27 (Differential k-form). A *differential k-form* ω on $\mathbb{R}^n$ is an assignment of a k-tensor on the tangent space $T_p\mathbb{R}^n$ for each point $p \in \mathbb{R}^n$; that is, ω is a mapping $\mathbb{R}^n \to \bigcup_p A_k(T_p\mathbb{R}^n)$. Note that $A_1(T_p\mathbb{R}^n) = T_p^*\mathbb{R}^n$.

By Proposition 5.24, a basis for $A_k(T_p\mathbb{R}^n)$ is given by the set of all

$$dx_p^{i_1} \wedge \cdots \wedge dx_p^{i_k}, \quad 1 \le i_1 < i_2 < \cdots < i_k \le n.$$

Therefore, at each point $p \in \mathbb{R}^n$, ω_p is a linear combination

$$\omega_p = \sum a_{i_1 \cdots i_k}(p) dx_p^{i_1} \wedge \cdots \wedge dx_p^{i_k}.$$

We say that ω is smooth if and only if the coefficient functions $a_{i_1 \cdots i_k} : \mathbb{R}^n \to \mathbb{R}$ are smooth. We denote by $\Omega^k(\mathbb{R}^n)$ the vector space of smooth k-forms on $\mathbb{R}^n$. A 0-form assigns to each point p an element of $A_0(T_p\mathbb{R}^n) = \mathbb{R}$. Thus, a 0-form on $\mathbb{R}^n$ is simply a function on $\mathbb{R}^n$, and $\Omega^0(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$. By Corollary 5.26, there are no nonzero differential forms of degree $> n$ on an open subset of $\mathbb{R}^n$.

We define the wedge product of a k-form ω and an ℓ-form τ pointwise:

$$(\omega \wedge \tau)_p := \omega_p \wedge \tau_p.$$

In coordinates, if $\omega = \sum a_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ and $\tau = \sum b_{j_1 \cdots j_k} dx^{j_1} \wedge \cdots \wedge dx^{j_\ell}$, then

$$\begin{aligned} \omega \wedge \tau &= \left(\sum a_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}\right) \wedge \left(\sum b_{j_1 \cdots j_k} dx^{j_1} \wedge \cdots \wedge dx^{j_\ell}\right) \\ &= \sum\sum (a_{i_1 \cdots i_k} b_{i_1 \cdots i_k}) dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \qquad \text{(by bilinearity)}, \end{aligned}$$

which shows that the wedge product of two smooth forms is smooth. Hence, the wedge product is a bilinear map

$$\wedge : \Omega^k(\mathbb{R}^n) \times \Omega^\ell(\mathbb{R}^n) \to \Omega^{k+\ell}(\mathbb{R}^n).$$

By Proposition 5.20, the wedge product of differential forms is anticommutative and associative. In the case when one of the factors, say k, has degree 0, we wedge product

$$\wedge : \Omega^0(\mathbb{R}^n) \times \Omega^\ell(\mathbb{R}^n) \to \Omega^\ell(\mathbb{R}^n)$$

is the pointwise multiplication of a C^∞ ℓ-form by a C^∞ function:

$$(f \wedge \omega)_p = f(p) \wedge \omega_p = f(p)\omega_p,$$

since, as we noted before, the wedge product wtih a 0-tensor is scalar multiplication. Thus, if $f \in C^\infty(\mathbb{R}^3)$ and $\omega \in \Omega^\ell(\mathbb{R}^3)$, then $f \wedge \omega = f\omega$.

Example 5.28. Let x, y, z be the coordinates on $\mathbb{R}^3$. If $\omega \in \Omega^1(\mathbb{R}^3)$, then it is of the form

$$\omega = Pdx + Qdy + Rdz,$$

where $P, Q, R \in C^\infty(\mathbb{R}^3)$. If $\omega \in \Omega^2(\mathbb{R}^3)$, then it is of the form

$$Pdy \wedge dz + Qdx \wedge dz + Rdx \wedge dy,$$

where $P, Q, R \in C^\infty(\mathbb{R}^3)$. Finally, if $\omega \in \Omega^3(\mathbb{R}^3)$, then

$$\omega = fdx \wedge dy \wedge dz$$

with $f \in C^\infty(\mathbb{R}^3)$.

With the wedge product as multiplication as the degree of a form as the grading, the direct sum

$$\Omega^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^{n} \Omega^k(\mathbb{R}^n)$$

becomes an anticommutative graded algebra over $\mathbb{R}$. Since one can multiply C^∞ k-forms by C^∞ functions, the set $\Omega^k(\mathbb{R}^n)$ is both a vector space over $\mathbb{R}$ and a module over $C^\infty(\mathbb{R}^n)$, which means $\Omega^\bullet(\mathbb{R}^n)$ is also a $C^\infty(\mathbb{R}^n)$-module.

5.6 Looking ahead

All of this hard work will pay off in the next section when we see how differential k-forms unify and generalize vector calculus on $\mathbb{R}^3$. We will then rewrite Maxwell's equations in terms of differential forms, which will be of great help in understanding their properties and looking for solutions.

5.7 Exercises

1. Prove that $D_p(\mathbb{R}^n)$ is a subspace of the vector space of all maps $C^\infty(\mathbb{R}^n) \to \mathbb{R}$.

2. A smooth vector field $v \in \mathfrak{X}(\mathbb{R}^n)$ induces a mapping $C^\infty(\mathbb{R}^n) \to C^\infty(\mathbb{R}^n)$ defined by $f \mapsto vf$, where $(vf)(p) := v_p f$. Compute vf when $v = x\partial/\partial x + y\partial/\partial y \in \mathfrak{X}(\mathbb{R}^3)$ and $f(x, y, z) = x^2 + y^2 + z^2 \in C^\infty(\mathbb{R}^3)$.

3. Let $e_1, \ldots, e_n$ be a basis for a vector space V and $\alpha^1, \ldots, \alpha^n$ be the dual basis for V^*. Suppose $[g_{ij}] \in \mathbb{R}^{n\times n}$ is an $n \times n$ matrix. Define a bilinear function $f : V \times V \to \mathbb{R}$ by

$$f(v, w) = \sum_{1\le i,j\le n} g_{ij} v^i w^j$$

for $v = \sum_{i=1}^n v^i e_i$ and $w = \sum_{j=1}^n w^j e_j$. Describe f in terms of the tensor products of α^i and α^j, $1 \le i, j \le n$.

4. Prove that if f is an alternating k-linear function on a vector space V, then $\mathrm{Alt}(f) = k!f$.

5. Let V be a vector space. For $a, b \in \mathbb{R}$, $f \in A_k(V)$, and $g \in A_\ell(V)$, show that $af \wedge bg = (ab) f \wedge g$.

6. Prove that if $\alpha^1, \ldots, \alpha^k \in V^*$ and $v_1, \ldots, v_k \in V$, then $(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) = \det[\alpha^i(v_j)]$. (*Hint*: First convince yourself that $A = [a^i_j] \in \mathbb{R}^{n\times n}$ is an $n \times n$ matrix, then $\det A = \sum_{\sigma\in S_n} (\mathrm{sgn}(\sigma)) a^1_{\sigma(1)} \cdots a^n_{\sigma(n)}$, where the right-hand-side is the result of cofactor expansion across the first row of A.)

7. Suppose two sets of covectors on V, $\beta^1, \ldots, \beta^k$ and $\gamma^1, \ldots, \gamma^k$ are related by

$$\beta^i = \sum_{j=1}^k a^i_j \gamma^j, \quad i = 1, \ldots, k,$$

for a $k \times k$ matrix $A = [a^i_j]$. Show that

$$\beta^1 \wedge \cdots \wedge \beta^k = (\det A) \gamma^1 \wedge \cdots \wedge \gamma^k.$$

8. Let $f \in A_k(V)$. Suppose two sets of vectors $u_1, \ldots, u_k$ and $v_1, \ldots, v_k$ in V are related by

$$u_j = \sum_{i=1}^k a^i_j v_i, \quad j = 1, \ldots, k,$$

for a $k \times k$ matrix $A = [a^i_j]$. Show that

$$f(u_1, \ldots, u_k) = (\det A) f(v_1, \ldots, v_k).$$

9. Let $\alpha = a_1 dx + a_2 dy + a_3 dz \in \Omega^1(\mathbb{R}^3)$ and $\beta = b_1 dy \wedge dz + b_2 dz \wedge dx + b_3 dx \wedge dy \in \Omega^2(\mathbb{R}^3)$. Simplify the expression $\alpha \wedge \beta \in \Omega^3(\mathbb{R}^3)$ as much as possible.

10. If $x = r\cos(\theta)$ and $y = r\sin(\theta)$, calculate dx, dy, and $dx \wedge dy$ in terms of dr and $d\theta$.

6 Rewriting Maxwell's Equations

6.1 The Exterior Derivative

Recall that a differential form is a mapping from $\mathbb{R}^n$ (or an open subset thereof) to $A_k(T_p\mathbb{R}^n)$. Since a differential form is defined on an open set, and not just at a single point, there is a notion of differentiation for differential forms, called the *exterior derivative*, which we now define.

6.1.1 The Exterior Derivative

We first define the exterior derivative of a 0-form, which is just a smooth function f, to be the differential of f.

Definition 6.1 (Exterior derivative of a 0-form).

$$\begin{aligned} d : \Omega^0(\mathbb{R}^n) &\to \Omega^1(\mathbb{R}^n) \\ f &\mapsto df. \end{aligned}$$

As we saw last time, in terms of coordinates $(x^1, \ldots, x^n)$,

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i.$$

Definition 6.2 (Exterior derivative of a k-form). For $k \geq 1$, if $\omega = \sum a_{i_1 \ldots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$, where $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, then

$$\begin{aligned} d : \Omega^k(\mathbb{R}^n) &\to \Omega^{k+1}(\mathbb{R}^n) \\ d\omega &= \sum da_{i_1 \ldots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ &= \sum \left(\sum_{j=1}^{n} \frac{\partial a_{i_1 \ldots i_k}}{\partial x^j} dx^j \right) \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}. \end{aligned}$$

Example 6.3. Let $\omega = f dx + g dy \in \Omega^2(\mathbb{R}^2)$. Then

$$\begin{aligned} d\omega &= df \wedge dx + dg \wedge dy \\ &= \left(\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \right) \wedge dx + \left(\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy \right) \wedge dy \\ &= \frac{\partial f}{\partial x} \underbrace{dx \wedge dx}_{=0} + \frac{\partial f}{\partial y} \underbrace{dy \wedge dx}_{=-dx \wedge dy} + \frac{\partial g}{\partial x} dx \wedge dy + \frac{\partial g}{\partial y} \underbrace{dy \wedge dy}_{=0} \\ &= \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx \wedge dy. \end{aligned}$$

Proposition 6.4 (Properties of the exterior derivative). The exterior derivative

$$d : \Omega^\bullet(\mathbb{R}^n) \to \Omega^\bullet(\mathbb{R}^n)$$

has the following properties for all $\omega, \tau \in \Omega^\bullet(\mathbb{R}^n)$ and all $c \in \mathbb{R}$:

(i) $d(c\omega + \tau) = cd\omega + d\tau$.

(ii) $d(\omega \wedge \tau) = (d\omega) \wedge \tau + (-1)^{\deg \omega} \omega \wedge d\tau$.

(iii) $d^2 = 0$.

Proof. (i) Since d is linear on $\Omega^k(\mathbb{R}^n)$ for each k (this follows almost immediately from Definition 6.2) and the sum of linear maps is linear, this implies d is linear on $\Omega^\bullet(\mathbb{R}^n)$.

(ii) Since both sides are linear in ω and τ, it suffices to check this equation for $\omega \in \Omega^k(\mathbb{R}^n)$ and $\tau \in \Omega^\ell(\mathbb{R}^n)$. In coordinates $(x^1, \ldots, x^n)$,

$$\omega = \sum \omega_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k},$$
$$\tau = \sum \tau_{j_1 \cdots j_\ell} dx^{j_1} \wedge \cdots \wedge dx^{j_\ell},$$

where $(i_1, \ldots, i_k)$ and $(j_1, \ldots, j_\ell)$ are strictly-ascending tuples of indices,

$$\begin{aligned}
d(\omega \wedge \tau) &= d(\omega_{i_1 \cdots i_k} \tau_{j_1 \cdots j_\ell}) dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&= \left(\sum_{m=1}^n \frac{\partial (\omega_{i_1 \cdots i_k} \tau_{j_1 \cdots j_\ell})}{\partial x^m} dx^m \right) \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&= \left(\sum_{m=1}^n \left(\frac{\partial \omega_{i_1 \cdots i_k}}{\partial x^m} \tau_{j_1 \cdots j_\ell} + \omega_{i_1 \cdots i_k} \frac{\partial \tau_{j_1 \cdots j_\ell}}{\partial x^m} \right) dx^m \right) \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&= \left(\sum_{m=1}^n \frac{\partial \omega_{i_1 \cdots i_k}}{\partial x^m} \tau_{j_1 \cdots j_\ell} dx^m \right) \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&+ \left(\sum_{m=1}^n \omega_{i_1 \cdots i_k} \frac{\partial \tau_{j_1 \cdots j_\ell}}{\partial x^m} dx^m \right) \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&= \sum_{m=1}^n \frac{\partial \omega_{i_1 \cdots i_k}}{\partial x^m} \tau_{j_1 \cdots j_\ell} dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&+ \sum_{m=1}^n \omega_{i_1 \cdots i_k} \frac{\partial \tau_{j_1 \cdots j_\ell}}{\partial x^m} dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&= \sum_{m=1}^n \frac{\partial \omega_{i_1 \cdots i_k}}{\partial x^m} \tau_{j_1 \cdots j_\ell} dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&+ (-1)^k \omega_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge \sum_{m=1}^n \frac{\partial \tau_{j_1 \cdots j_\ell}}{\partial x^m} dx^m \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell} \\
&= (d\omega) \wedge \tau + (-1)^{\deg \omega} \omega \wedge d\tau.
\end{aligned}$$

(iii) Again, by linearity of d, it suffices to show that $d^2\omega = 0$ for $\omega \in \Omega^k(\mathbb{R}^n)$. Writing $\sum \omega_{i_1\cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ with $(i_1, \ldots, i_k)$ a strictly-increasing k-tuple,

$$\begin{aligned} d^2\omega &= d\left(\sum_{m=1}^{n} \frac{\partial \omega_{i_1\cdots i_k}}{\partial x^m} dx^m\right) \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ &= \sum_{m=1}^{n} d\left(\frac{\partial \omega_{i_1\cdots i_k}}{\partial x^m} dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}\right) \\ &= \sum_{m=1}^{n} d\left(\frac{\partial \omega_{i_1\cdots i_k}}{\partial x^m}\right) \wedge dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ &= \sum_{m=1}^{n} \left(\sum_{\ell=1}^{n} \frac{\partial^2 \omega_{i_1\cdots i_k}}{\partial x^\ell \partial x^m} dx^\ell\right) \wedge dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ &= \sum_{m=1}^{n} \sum_{\ell=1}^{n} \frac{\partial^2 \omega_{i_1\cdots i_k}}{\partial x^\ell \partial x^m} dx^\ell \wedge dx^m \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} \end{aligned}$$

In this sum, if $\ell = m$, then $dx^\ell \wedge dx^m = 0$; if $\ell \neq m$, then $\frac{\partial^2 \omega_{i_1\cdots i_k}}{\partial x^\ell \partial x^m}$ is symmetric in ℓ and m while $dx^\ell \wedge dx^m = -dx^m \wedge dx^\ell$ is alternating in ℓ and m, so the terms with $\ell \neq m$ cancel in pairs. Hence, $d^2\omega = 0$.

□

6.1.2 Generalization of Vector Calculus on $\mathbb{R}^3$

We are now ready to see how the theory of differential forms unifies many theorems in vector calculus on $\mathbb{R}^3$.

Let $\mathbf{F}(x, y, z) = P(x, y, z)\hat{\mathbf{i}} + Q(x, y, z)\hat{\mathbf{j}} + R(x, y, z)\hat{\mathbf{k}}$ be a vector field on $\mathbb{R}^3$. Recall the three operators gradient, curl, and divergence:

$$\{\text{scalar functions}\} \xrightarrow{\text{grad}} \{\text{vector fields}\} \xrightarrow{\text{curl}} \{\text{vector fields}\} \xrightarrow{\text{div}} \{\text{scalar functions}\}$$

$$\begin{aligned} \nabla f &= \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \\ \partial f/\partial z \end{bmatrix}, \\ \nabla \times \begin{bmatrix} P \\ Q \\ R \end{bmatrix} &= \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \times \begin{bmatrix} P \\ Q \\ R \end{bmatrix} = \begin{bmatrix} \partial R/\partial y - \partial Q/\partial z \\ \partial P/\partial z - \partial R/\partial x \\ \partial Q/\partial x - \partial P/\partial y \end{bmatrix}, \\ \nabla \cdot \begin{bmatrix} P \\ Q \\ R \end{bmatrix} &= \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \cdot \begin{bmatrix} P \\ Q \\ R \end{bmatrix} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}. \end{aligned}$$

Since every 1-form on $\mathbb{R}^3$ is a linear combination with function coefficients of dx, dy, and

dz, we can identify 1-forms with vector fields on $\mathbb{R}^3$ via

$$Pdx + Qdy + Rdz \longleftrightarrow \begin{bmatrix} P \\ Q \\ R \end{bmatrix}.$$

Similarly, 2-forms on $\mathbb{R}^3$ can also be identified with vector fields on $\mathbb{R}^3$:

$$Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy \longleftrightarrow \begin{bmatrix} P \\ Q \\ R \end{bmatrix},$$

and 3-forms on $\mathbb{R}^3$ can be identified with functions on $\mathbb{R}^3$:

$$fdx \wedge dy \wedge dz \longleftrightarrow f.$$

In terms of these identifications, the exterior derivative of a 0-form f is

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz \longleftrightarrow \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \\ \partial f/\partial z \end{bmatrix} = \nabla f;$$

the exterior derivative of a 1-form is

$$d(Pdx + Qdy + Rdz) = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) dy \wedge dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right) dz \wedge dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy,$$

which corresponds to

$$\nabla \times \begin{bmatrix} P \\ Q \\ R \end{bmatrix} = \begin{bmatrix} \partial R/\partial y - \partial Q/\partial z \\ \partial P/\partial z - \partial R/\partial x \\ \partial Q/\partial x - \partial P/\partial y \end{bmatrix};$$

the exterior derivative of a 2-form is

$$d(Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy) = \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx \wedge dy \wedge dz,$$

which corresponds to

$$\nabla \cdot \begin{bmatrix} P \\ Q \\ R \end{bmatrix} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Thus, after appropriate identifications, the exterior derivatives d on 0-forms, 1-forms, and 2-forms are simply the three operators grad, curl, and div. In summary, on an open subset U of $\mathbb{R}^3$, there are identifications=

Recall that $\nabla \times (\nabla f) = 0$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$. These equations express the property $d^2 = 0$ of the exterior derivative on open subsets of $\mathbb{R}^3$.

$$\begin{array}{ccccccc} \Omega^0(U) & \xrightarrow{d} & \Omega^1(U) & \xrightarrow{d} & \Omega^2(U) & \xrightarrow{d} & \Omega^3(U) \\ \simeq\downarrow & & \simeq\downarrow & & \simeq\downarrow & & \simeq\downarrow \\ C^\infty(U) & \xrightarrow[\text{grad}]{} & \mathfrak{X}(U) & \xrightarrow[\text{curl}]{} & \mathfrak{X}(U) & \xrightarrow[\text{div}]{} & C^\infty(U). \end{array}$$

6.2 Rewriting Maxwell's Equations

We are now ready to rewrite Maxwell's Equations

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0, \tag{6.1}$$

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j}. \tag{6.2}$$

in terms of differential forms.

6.2.1 Rewriting the homogeneous equations

Let's begin with the first two equations in the static case:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = 0. \tag{6.3}$$

By the correspondence of the previous section, the gradient becomes the exterior derivative of a 1-form, while the divergence becomes the exterior derivative of a 2-form. Thus, we will identify the electric field $\mathbf{E}$ as a 1-form

$$E = E_x dx + E_y dy + E_z dz \tag{6.4}$$

and the magnetic field $\mathbf{B}$ as a 2-form

$$B = B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy. \tag{6.5}$$

The two equations in (6.3) then become

$$dE = 0, \quad dB = 0.$$

Next we consider the general, time-dependent case. The component functions of E and B in (6.4) and (6.5), respectively, are now defined on spacetime $\mathbb{R}^4 = \{(x^0, x^1, x^2, x^3)\}$, where $(x^0, x^1, x^2, x^3) = (t, x, y, z)$. Note that $\dim \Omega^2(\mathbb{R}^4) = \binom{4}{2} = \frac{4!}{2!2!} = 6$, which is exactly the number of independent components of the electric and magnetic fields. We therefore combine these into a 2-form on $\mathbb{R}^4$. The magnetic field B is already a 2-form. The electric field E is only a 1-form, so we need wedge it with another 1-form to create a 2-form. If we wedge E with either dx, dy, or dz then we will lose the corresponding component; instead we obtain a 2-form by wedging E with dt. We therefore define the *electromagnetic field*, $F \in \Omega^2(\mathbb{R}^4)$, by

$$F = B + E \wedge dt.$$

The electromagnetic field F can be written in the basis $\{dt \wedge dx, dt \wedge dy, dt \wedge dz, dx \wedge dy, dx \wedge dz, dy \wedge dz\}$ for $\Omega^2(\mathbb{R}^4)$ as

$$\begin{aligned} F &= -E_x dt \wedge dx - E_y dt \wedge dy - E_z dt \wedge dz + B_z dx \wedge dy - B_y dx \wedge dz + B_x dy \wedge dz \\ &= -E_x dx^0 \wedge dx^1 - E_y dx^0 \wedge dx^2 - E_z dx^0 \wedge dx^3 + B_z dx^1 \wedge dx^2 - B_y dx^1 \wedge dx^3 + B_x dx^2 \wedge dx^3 \\ &= \sum_{0 \le \mu < \nu \le 3} F_{\mu\nu} dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} \sum_{\mu=0}^{3} \sum_{\nu=0}^{3} F_{\mu\nu} dx^\mu \wedge dx^\nu \end{aligned}$$

The coefficient functions $F_{\mu\nu}$ of the electromagnetic field are the entries of the antisymmetric matrix

$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}. \tag{6.6}$$

Lemma 6.5. If $\omega = \sum \omega_{\mu_1 \dots \mu_k} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} \in \Omega^k(\mathbb{R}^4)$ (where the sum is over all $0 \le \mu_1 < \mu_2 < \cdots < \mu_k \le 3$), then $d\omega \in \Omega^{k+1}(\mathbb{R}^4)$ can be written as the sum of timelike and spacelike parts as

$$d\omega = dt \wedge \frac{\partial \omega}{\partial t} + d_S \omega,$$

where the timelike part

$$dt \wedge \frac{\partial \omega}{\partial t} := dx^0 \wedge \sum \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^0} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k},$$

and the spacelike part

$$d_S \omega := \sum \sum_{j=1}^{3} \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^j} dx^j \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k}.$$

Proof.

$$\begin{aligned} d\omega &= \sum d\omega_{\mu_1 \dots \mu_k} \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} \\ &= \sum \sum_{\nu=0}^{3} \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^\nu} dx^\nu \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} \\ &= \sum \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^0} dx^0 \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} + \sum \sum_{j=1}^{3} \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^j} dx^j \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} \\ &= dx^0 \wedge \sum \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^0} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} + \sum \sum_{j=1}^{3} \frac{\partial \omega_{\mu_1 \dots \mu_k}}{\partial x^j} dx^j \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k} \\ &= dt \wedge \frac{\partial \omega}{\partial t} + d_S \omega. \end{aligned}$$

□

Proposition 6.6. The first two Maxwell equations in (6.1) are equivalent to the equation

$$dF = 0.$$

Proof. By Lemma 6.5,

$$\begin{aligned}
dF &= d(B + E \wedge dt) \\
&= dB + dE \wedge dt \\
&= dt \wedge \frac{\partial B}{\partial t} + d_S B + \left(dt \wedge \frac{\partial E}{\partial t} + d_S E \right) \wedge dt \\
&= dt \wedge \frac{\partial B}{\partial t} + d_S B + dt \wedge \frac{\partial E}{\partial t} \wedge dt + d_S E \wedge dt \\
&= dt \wedge \frac{\partial B}{\partial t} + d_S B - \frac{\partial E}{\partial t} \wedge \underbrace{dt \wedge dt}_{=0} + d_S E \wedge dt \\
&= dt \wedge \frac{\partial B}{\partial t} + d_S B + d_S E \wedge dt \\
&= dt \wedge \frac{\partial B}{\partial t} + d_S B + (-1)^2 dt \wedge d_S E \\
&= dt \wedge \frac{\partial B}{\partial t} + d_S B + dt \wedge d_S E \\
&= dt \wedge \left(\frac{\partial B}{\partial t} + d_S E \right) + d_S B \\
&= \left[\frac{\partial B_x}{\partial t} + \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \right] dt \wedge dy \wedge dz + \left[\frac{\partial B_y}{\partial t} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \right] dt \wedge dx \wedge dz \\
&+ \left[\frac{\partial B_z}{\partial t} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \right] dt \wedge dx \wedge dy + \frac{\partial B_x}{\partial x} dx \wedge dy \wedge dz + \frac{\partial B_y}{\partial y} dy \wedge dz \wedge dx \\
&+ \frac{\partial B_z}{\partial z} dz \wedge dx \wedge dy \\
&= \left[\frac{\partial B_x}{\partial t} + \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \right] dt \wedge dy \wedge dz + \left[\frac{\partial B_y}{\partial t} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \right] dt \wedge dx \wedge dz \\
&+ \left[\frac{\partial B_z}{\partial t} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \right] dt \wedge dx \wedge dy + \frac{\partial B_x}{\partial x} dx \wedge dy \wedge dz + \frac{\partial B_y}{\partial y} dx \wedge dy \wedge dz \\
&+ \frac{\partial B_z}{\partial z} dx \wedge dy \wedge dz \\
&= \left[\frac{\partial B_x}{\partial t} + \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \right] dt \wedge dy \wedge dz + \left[\frac{\partial B_y}{\partial t} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \right] dt \wedge dx \wedge dz \\
&+ \left[\frac{\partial B_z}{\partial t} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \right] dt \wedge dx \wedge dy + \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) dx \wedge dy \wedge dz
\end{aligned}$$

First suppose that $dF = 0$. Then, since $\{dt \wedge dx \wedge dy, dt \wedge dx \wedge dz, dt \wedge dy \wedge dz, dx \wedge dy \wedge dz\}$ is a basis for $\Omega^3(\mathbb{R}^4)$ and therefore linearly independent, each coefficient must be zero,

which implies

$$\begin{aligned}
\frac{\partial B_x}{\partial t} + \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) &= 0 \\
\frac{\partial B_y}{\partial t} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) &= 0 \\
\frac{\partial B_z}{\partial t} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right) &= 0 \\
\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} &= 0
\end{aligned}$$

The first three equations are equivalent to the vector equation

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0,$$

while the last equation is

$$\nabla \cdot \mathbf{B} = 0.$$

Hence, $dF = 0$ implies the two Maxwell equations in (6.1). Conversely, if the equations (6.1) hold, then all four coefficient functions of dF are zero, and therefore $dF = 0$. □

6.2.2 Rewriting the inhomogeneous equations

To rewrite the first pair of Maxwell equations

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0$$

in differential form, we identified $\mathbf{E}$ with a 1-form and $\mathbf{B}$ with a 2-form. The second pair of equations

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j}.$$

look much like the first pair with the roles of $\mathbf{E}$ and $\mathbf{B}$ reversed, so it would seem that we need to now think of $\mathbf{E}$ as a 2-form and $\mathbf{B}$ as a 1-form! What comes to the rescue is the existence of a duality on an n-dimensional manifold between k-forms and $n-k$ forms. Explaining this here would involve a significant digression, so I won't attempt to do so, but it takes the following simple form on $\mathbb{R}^3$:

Definition 6.7 (Hodge star operator on $\mathbb{R}^3$). The *Hodge star operator on* $\mathbb{R}^3$ is a linear map

$$* : \Omega^k(\mathbb{R}^3) \to \Omega^{3-k}(\mathbb{R}^3)$$

satisfying $*^2 = 1$ (that is, it is its own inverse) which sends

$$*dx = dy \wedge dz, \quad *dy = dz \wedge dx, \quad *dz = dx \wedge dy, \quad *1 = dx \wedge dy \wedge dz.$$

Example 6.8. Let $\omega, \tau \in \Omega^1(\mathbb{R}^3)$, and write these in terms of coordinates as

$$\begin{aligned}\omega &= \omega_x dx + \omega_y dy + \omega_z dz,\\ \tau &= \tau_x dx + \tau_y dy + \tau_z dz.\end{aligned}$$

(a)

$$\begin{aligned}*(\omega \wedge \tau) &= *[(\omega_x dx + \omega_y dy + \omega_z dz) \wedge (\tau_x dx + \tau_y dy + \tau_z dz)]\\ &= *[(\omega_y\tau_z - \omega_z\tau_y)dy \wedge dz + (\omega_z\tau_x - \omega_x\tau_z)dz \wedge dx + (\omega_x\tau_y - \omega_y\tau_x)dx \wedge dy]\\ &= (\omega_y\tau_z - \omega_z\tau_y) * dy \wedge dz + (\omega_z\tau_x - \omega_x\tau_z) * dz \wedge dx + (\omega_x\tau_y - \omega_y\tau_x) * dx \wedge dy\\ &= (\omega_y\tau_z - \omega_z\tau_y)dx + (\omega_z\tau_x - \omega_x\tau_z)dy + (\omega_x\tau_y - \omega_y\tau_x)dz,\end{aligned}$$

which corresponds to $\omega \times \tau$.

(b)

$$\begin{aligned}*d\omega &= * \left[\left(\frac{\partial \omega_z}{\partial y} - \frac{\partial \omega_y}{\partial z} \right) dy \wedge dz + \left(\frac{\partial \omega_x}{\partial z} - \frac{\partial \omega_z}{\partial x} \right) dz \wedge dx + \left(\frac{\partial \omega_y}{\partial x} - \frac{\partial \omega_x}{\partial y} \right) dx \wedge dy \right]\\ &= \left(\frac{\partial \omega_z}{\partial y} - \frac{\partial \omega_y}{\partial z} \right) * dy \wedge dz + \left(\frac{\partial \omega_x}{\partial z} - \frac{\partial \omega_z}{\partial x} \right) * dz \wedge dx + \left(\frac{\partial \omega_y}{\partial x} - \frac{\partial \omega_x}{\partial y} \right) * dx \wedge dy\\ &= \left(\frac{\partial \omega_z}{\partial y} - \frac{\partial \omega_y}{\partial z} \right) dx + \left(\frac{\partial \omega_x}{\partial z} - \frac{\partial \omega_z}{\partial x} \right) dy + \left(\frac{\partial \omega_y}{\partial x} - \frac{\partial \omega_x}{\partial y} \right) dz\end{aligned}$$

which corresponds to $\nabla \times \omega$.

(c)

$$\begin{aligned}*d * \omega &= *d(\omega_x * dx + \omega_y * dy + \omega_z * dz)\\ &= *d(\omega_x dy \wedge dx + \omega_y dz \wedge dx + \omega_z dx \wedge dy)\\ &= * \left(\frac{\partial \omega_x}{\partial x} dz \wedge dy \wedge dx + \frac{\partial \omega_y}{\partial y} dy \wedge dz \wedge dx + \frac{\partial \omega_z}{\partial x} dz \wedge dx \wedge dy \right)\\ &= * \left(\frac{\partial \omega_x}{\partial x} + \frac{\partial \omega_y}{\partial y} + \frac{\partial \omega_z}{\partial z} \right) dx \wedge dy \wedge dz\\ &= \left(\frac{\partial \omega_x}{\partial x} + \frac{\partial \omega_y}{\partial y} + \frac{\partial \omega_z}{\partial z} \right) * dx \wedge dy \wedge dz\\ &= \left(\frac{\partial \omega_x}{\partial x} + \frac{\partial \omega_y}{\partial y} + \frac{\partial \omega_z}{\partial z} \right) 1\\ &= \frac{\partial \omega_x}{\partial x} + \frac{\partial \omega_y}{\partial y} + \frac{\partial \omega_z}{\partial z}\end{aligned}$$

which corresponds to $\nabla \cdot \omega$.

To rewrite the second pair of Maxwell equations, we need the Hodge star operator on differential forms on $\mathbb{R}^4$.

Definition 6.9 (Hodge star operator on $\mathbb{R}^4$). The *Hodge star operator on* $\mathbb{R}^3$ is a linear map

$$* : \Omega^k(\mathbb{R}^4) \to \Omega^{4-k}(\mathbb{R}^4)$$

satisfying $*^2 = (-1)^{k(4-k)+1}$ which sends

$$*dt = -dx \wedge dy \wedge dz, \quad *dx = -dt \wedge dy \wedge dz, \quad *dy = dt \wedge dx \wedge dz, \quad *dz = -dt \wedge dx \wedge dy,$$
$$*dt \wedge dx = -dy \wedge dz, \quad *dt \wedge dy = -dz \wedge dx, \quad *dt \wedge dz = -dx \wedge dy, \quad *dx \wedge dy = dt \wedge dz,$$
$$*dx \wedge dz = -dt \wedge dy, \quad *dy \wedge dz = dt \wedge dx.$$

The dual of the electromagnetic field is

$$\begin{aligned} *F &= *(-E_x dt \wedge dx - E_y dt \wedge dy - E_z dt \wedge dz + B_z dx \wedge dy - B_y dx \wedge dz + B_x dy \wedge dz) \\ &= -E_x * dt \wedge dx - E_y * dt \wedge dy - E_z * dt \wedge dz + B_z * dx \wedge dy - B_y * dx \wedge dz + B_x * dy \wedge dz \\ &= E_x dy \wedge dz - E_y dx \wedge dz + E_z dx \wedge dy + B_z dy \wedge dz + B_y dt \wedge dy + B_x dt \wedge dx \end{aligned}$$

Writing $*F = \frac{1}{2}\sum_{\mu=0}^{3}\sum_{\nu=0}^{3}(*F)_{\mu\nu}dx^\mu \wedge dx^\nu$, the component functions $*F_{\mu\nu}$ are given by

$$*F_{\mu\nu} = \begin{bmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & E_z & -E_y \\ -B_y & -E_z & 0 & E_x \\ -B_z & E_y & -E_x & 0 \end{bmatrix} \tag{6.7}$$

Comparing the matrix in (6.7) to the one in (6.6), we see that taking the dual of f amounts to doing the replacements

$$E_i \mapsto -B_i, \quad B_i \mapsto E_i.$$

Now combine the charge and current densities into the *4-current*

$$J = j - \rho dt \in \Omega^1(\mathbb{R}^4).$$

Proposition 6.10. The second pair of Maxwell equations

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j}$$

can be written as

$$*d * F = J.$$

Proof. Left as an exercise. □

6.3 Summary

In the language of differential forms, Maxwell's Equations can be written as

$$dF = 0,$$
$$*d * F = J,$$

where $E \in \Omega^1(\mathbb{R}^4)$, $B \in \Omega^2(\mathbb{R}^4)$, $F = B + E \wedge dt \in \Omega^2(\mathbb{R}^4)$, $j \in \Omega^1(\mathbb{R}^4)$, $\rho \in \Omega^0(\mathbb{R}^4) = C^\infty(\mathbb{R}^4)$, $J = j - \rho dt \in \Omega^1(\mathbb{R}^4)$.

6.4 Exercises

1. Prove Proposition 6.10.

2. Prove that the continuity equation $\nabla \cdot \mathbf{j} + \frac{\partial \rho}{\partial t} = 0$ can be written as

$$d * J = 0.$$

3. In the case $J = 0$ (the vacuum), Maxwell's equations become

$$dF = 0, \quad d * F = 0.$$

These are preserved by duality: $F \mapsto *F$. Show that this is equivalent to $\mathbf{B} \mapsto \mathbf{E}$, $\mathbf{E} \mapsto -\mathbf{B}$.

4. In $\mathbb{R}^4$, the dual of a 2-form is a 2-form, so $* : \Omega^2(\mathbb{R}^4) \to \Omega^2(\mathbb{R}^4)$ with $*^2 = -1$.

   (a) Suppose we allow complex-valued differential forms on $\mathbb{R}^4$. Prove that $* : \Omega^2(\mathbb{R}^4) \to \Omega^2(\mathbb{R}^4)$ has eigenvalues $\pm i$. We say that $F \in \Omega^2(\mathbb{R}^4)$ is *self-dual* if $*F = iF$, and *anti-self-dual* if $*F = -iF$.

   (b) Show that if a self-dual or anti-self-dual 2-form F satisfies $dF = 0$, then it automatically satisfies $d * F = 0$.

   (c) Show that we can write any $F \in \Omega^2(\mathbb{R}^4)$ as the sum of self-dual and anti-self-dual parts:

$$F = F_+ + F_-, \quad *F_\pm = \pm i F_\pm.$$

   (d) Write out the components of F_+ and F_- in terms of $\mathbf{E}$ and $\mathbf{B}$.

   (e) Consider the following monochromatic plane wave propagating in the z-direction:

$$\mathbf{E} = E_0 \cos(kz - \omega t)\hat{\mathbf{i}}$$
$$\mathbf{B} = E_0 \cos(kz - \omega t)\hat{\mathbf{j}},$$

   where $\omega k = 1$ (in units with $c = 1$). Form self-dual electric and magnetic fields from these, and show that the direction in which the electric field points rotates at a constant rate in the xy-plane as the wave propagates. Such a wave is said to be *circularly polarized*. (*Hint*: It is helpful to write $\cos(kz - \omega t) = \mathrm{Re}\, e^{i(kz-\omega t)}$).

7 de Rham Cohomology

7.1 Closed and Exact 1-forms

Definition 7.1 (Closed and Exact 1-forms). Let $\omega \in \Omega^k(\mathbb{R}^n)$.

(a) ω is said to be *closed* if $d\omega = 0$.

(b) ω is said to be *exact* if there exists $\tau \in \Omega^{k-1}(\mathbb{R}^n)$ such that $\omega = d\tau$.

Proposition 7.2. Every exact form is closed.

Proof. Since $d^2 = 0$, if $\omega = d\tau$, then $d\omega = d^2\tau = 0$. □

Recall that we found the two homogeneous Maxwell equations can be written as

$$dF = 0,$$

which, in the language of Definition 7.1 says that the electromagnetic field $F \in \Omega^2(\mathbb{R}^4)$ is closed. In the static case, this says that $E \in \Omega^1(\mathbb{R}^3)$ and $B \in \Omega^2(\mathbb{R}^3)$ are also closed:

$$dE = 0, \quad dB = 0.$$

Let's try to solve these equations. Proposition 7.2 suggests a way to find lots of solutions: if $F = dA$ for some $A \in \Omega^1(\mathbb{R}^4)$, then $dF = d^2A = 0$. Similarly for E and B in the static case. This simple observation is the basis of a surprisingly large amount of mathematics and physics. It leads to a very interesting question: can one get *all* the solutions of the first pair of Maxwell equations this way? Or, in the language of Definition 7.1, is every closed form exact?

Consider the case of a closed 1-form $dE = 0$. Can we cook up a smooth function ϕ, called a *scalar potential*, such that $E = d\phi$? Let's try and see what, if anything, prevents us. We will attempt to find such a function ϕ by integrating the 1-form E along a smooth curve $\mathbf{r}(t)$. Since $\mathbf{r}'(t)$ is a tangent vector to the curve at $\mathbf{r}(t)$, we can apply the cotangent vector $E_{\mathbf{r}(t)}$ to $\mathbf{r}'(t)$ to get a number $E_{\mathbf{r}(t)}(\mathbf{r}'(t))$, which we integrate from $t = 0$ to $t = 1$. Our plan will be to define

$$\Phi(q) = \int_C E = \int_0^T E_{\mathbf{r}(t)}(\mathbf{r}'(t))dt,$$

where C is a path from some fixed point p to q, parametrized by $\mathbf{r}(t)$, $0 \leq t \leq 1$.

There are a number of potential problems with this plan:

(1) If the domain consists of disconnected components, with p in one and q in another, then there will not exist any path from p to q. A domain for which there exists a path between any two points is said to be *path-connected*. From now on, we will assume the domain is path connected. (Otherwise, we could just apply our strategy in each path-connected component.)

(2) The next problem, which is more serious, is that $\int_C E$ will in general depend on the details of the path C, not just its endpoints $\mathbf{r}(0) = p$ and $\mathbf{r}(1) = q$.

Let's see what conditions are necessary for the integral $\int_C E$ to be independent of path. First, let's see how the integral changes when we smoothly vary the path C. Consider two smooth paths $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ from $\mathbf{r}_1(0) = \mathbf{r}_2(0) = p$ to $\mathbf{r}_1(T) = \mathbf{r}_2(1) = q$. We say that $\mathbf{r}_1$ can be *smoothly deformed* into $\mathbf{r}_2$ if there exists a function $\mathbf{H}(s,t)$ defined on $[0,1] \times [0,1]$ such that

$$\begin{aligned}
\mathbf{H}(s,0) &= p \text{ for all } s,\\
\mathbf{H}(s,1) &= q \text{ for all } s,\\
\mathbf{H}(0,t) &= \mathbf{r}_1(t) \text{ for all } t,\\
\mathbf{H}(1,t) &= \mathbf{r}_2(t) \text{ for all } t.
\end{aligned}$$

The function $\mathbf{H}$ is said to be a *smooth homotopy* from $\mathbf{r}_1$ to $\mathbf{r}_2$, and we say that the two paths are *smoothly homotopic*. If we think of s as time, then the path smoothly "morphs" from $\mathbf{r}_1$ into $\mathbf{r}_2$ as time goes from $s = 0$ to $s = 1$.

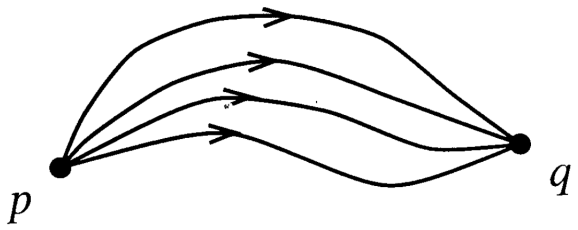


Figure 24: A smoothly varying family of paths from p to q.

Example 7.3 (Straight-line homotopy). A subset B of $\mathbb{R}^n$ is *convex* if any two points in B can be connected by a straight line which lies in B. If $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ are two smooth paths from p to q in a convex subset B of $\mathbb{R}^n$, then

$$\mathbf{H}(s,t) = s\mathbf{r}_2(t) + (1-s)\mathbf{r}_1(t)$$

is a smooth homotopy from $\mathbf{r}_1$ to $\mathbf{r}_2$, called a *straight-line homotopy*.

To see how

$$I_s = \sum_{i=1}^{3} \int_0^1 E_i(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) dt$$

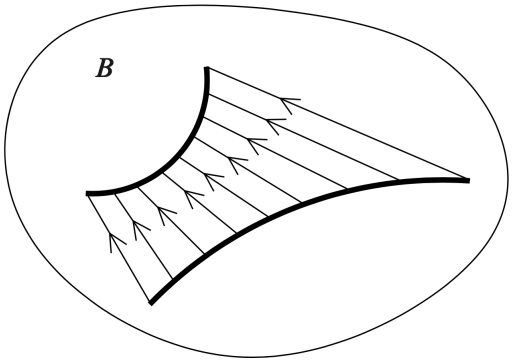


Figure 25: A straight-line homotopy.

depends on s, we differentiate it with respect to s:

$$
\begin{aligned}
&\frac{\partial I_s}{\partial s} \\
&= \frac{\partial}{\partial s} \sum_{i=1}^{3} \int_0^1 E_i(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) dt \\
&= \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) + E_i(\mathbf{H}(s,t)) \left(\frac{\partial^2 \mathbf{H}^i(s,t)}{\partial s \partial t} \right) \right] dt \\
&= \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) + E_i(\mathbf{H}(s,t)) \left(\frac{\partial^2 \mathbf{H}^i(s,t)}{\partial t \partial s} \right) \right] dt \\
&= \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) + \frac{\partial}{\partial t} \left(E_i(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) \right) - \frac{\partial E_i}{\partial t}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt \\
&= \left[\sum_{i=1}^{3} E_i(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right]_0^1 + \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) - \frac{\partial E_i}{\partial t}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt \\
&= \sum_{i=1}^{3} E_i(\mathbf{H}(s,1)) \left(\frac{\partial \mathbf{H}^i(s,1)}{\partial s} \right) - \sum_{i=1}^{3} E_i(\mathbf{H}(s,0)) \left(\frac{\partial \mathbf{H}^i(s,0)}{\partial s} \right) \\
&+ \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) - \frac{\partial E_i}{\partial t}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt \\
&= \sum_{i=1}^{3} E_i(q) \underbrace{\left(\frac{\partial q}{\partial s} \right)}_{=0} - \sum_{i=1}^{3} E_i(p) \underbrace{\left(\frac{\partial p}{\partial s} \right)}_{=0} \\
&+ \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) - \frac{\partial E_i}{\partial t}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt \\
&= 0 + \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) - \frac{\partial E_i}{\partial t}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt \\
&= \sum_{i=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial s}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) - \frac{\partial E_i}{\partial t}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt \\
&= \sum_{i=1}^{3} \sum_{j=1}^{3} \int_0^1 \left[\frac{\partial E_i}{\partial x^j}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^j(s,t)}{\partial s} \right) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t} \right) - \frac{\partial E_i}{\partial x^j}(\mathbf{H}(s,t)) \left(\frac{\partial \mathbf{H}^j(s,t)}{\partial t} \right) \left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s} \right) \right] dt.
\end{aligned}
$$

Re-indexing the sums in the second term by exchanging $i \leftrightarrow j$, we have

$$\begin{aligned}
&\frac{\partial I_s}{\partial s} \\
&= \sum_{i=1}^{3}\sum_{j=1}^{3}\int_0^1 \left[\frac{\partial E_i}{\partial x^j}(\mathbf{H}(s,t))\left(\frac{\partial \mathbf{H}^j(s,t)}{\partial s}\right)\left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t}\right) - \frac{\partial E_j}{\partial x^i}(\mathbf{H}(s,t))\left(\frac{\partial \mathbf{H}^i(s,t)}{\partial t}\right)\left(\frac{\partial \mathbf{H}^j(s,t)}{\partial s}\right)\right] dt \\
&= \sum_{i=1}^{3}\sum_{j=1}^{3}\int_0^1 \left(\frac{\partial E_i}{\partial x^j}(\mathbf{H}(s,t)) - \frac{\partial E_j}{\partial x^i}(\mathbf{H}(s,t))\right)\left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s}\right)\left(\frac{\partial \mathbf{H}^j(s,t)}{\partial t}\right) dt \\
&= \sum_{i=1}^{3}\sum_{j=1}^{3}\int_0^1 2(dE)_{ij}(\mathbf{H}(s,t))\left(\frac{\partial \mathbf{H}^i(s,t)}{\partial s}\right)\left(\frac{\partial \mathbf{H}^j(s,t)}{\partial t}\right) dt,
\end{aligned}$$

where $(dE)_{ij}$ is the ij-th component of $dE = \frac{1}{2}\sum_{i=1}^{3}\sum_{j=1}^{3}\left(\frac{\partial E_i}{\partial x_j} - \frac{\partial E_j}{\partial x^i}\right) dx^i \wedge dx^j \in \Omega^2(\mathbb{R}^3)$. From the expression above, see that I_s is independent of s when $dE = 0$. This shows that I_s will be the same for any two smoothly homotopic paths. Thus, defining

$$\phi(q) = \int_C E$$

will make sense if all paths from p to q are smoothly homotopic. Whether or not this is true depends on the domain of E. For example, suppose E is defined on $\mathbb{R}^2$ with the origin removed. In this case, the two paths γ_0 and γ_1 from $(-1,0)$ to $(1,0)$ shown in Figure 26 below are not homotopic:

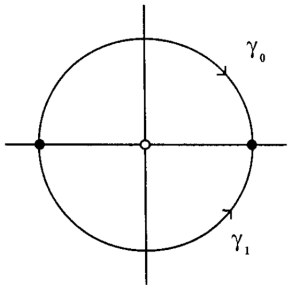


Figure 26: Two paths that are not homotopic in $\mathbb{R}^2 \setminus \{0\}$.

It is pretty obvious that there is no way to smoothly deform the path γ_0 to the path γ_1 without getting caught on the hole at the origin. Indeed, consider the 1-form

$$E = \frac{-y}{x^2+y^2}dx + \frac{x}{x^2+y^2}dy. \tag{7.1}$$

Parametrize the curves γ_0, γ_1 as

$$\begin{aligned}
\gamma_0(t) &= \langle -\cos(t), \sin(t)\rangle,\ 0 \le t \le \pi, \\
\gamma_1(t) &= \langle -\cos(t), -\sin(t)\rangle,\ 0 \le t \le \pi.
\end{aligned}$$

Then

$$\int_{\gamma_0} E = \int_0^\pi E_{\gamma_0(t)}\gamma_0'(t)dt = -\int_0^\pi (\sin^2(t)+\cos^2(t))dt = -\int_0^\pi dt = -\pi,$$

while

$$\int_{\gamma_1} E = \int_0^\pi E_{\gamma_1(t)}\gamma_1'(t)dt = \int_0^\pi (\sin^2(t)+\cos^2(t))dt = \int_0^\pi dt = \pi.$$

This means that we cannot use $\phi(q) = \int_\gamma E$ to define ϕ in a path-independent manner.

Given a path-connected domain S, we say that S is *simply-connected* if any two paths between any two points p and q are homotopic. If S is simply-connected, we can carry out our plan and define ϕ unambiguously when E is closed. In particular, by Example 7.3 things are fine on $\mathbb{R}^n$, since there is always a straight-line homotopy between any two paths between any two points p and q in $\mathbb{R}^n$.

Proposition 7.4. If S is simply-connected, E a closed 1-form on S, and

$$\phi(q) = \int_\gamma E,$$

where γ is any smooth path from a fixed point p to q, then $E = d\phi$.

Proof. We need to show that, if q is any point in S and $v_q \in T_qS$, then

$$E_q(v_q) = d\phi(v_q) = v_q\phi.$$

Let $\gamma : [0,1] \to S$ be a smooth path with $\gamma(0) = p$, $\gamma(1) = q$, and $\gamma'(1) = v_q$. [6] Then we have

$$\begin{aligned} E_q(v_q) &= E_{\gamma(1)}(\gamma'(1)) \\ &= \frac{d}{ds}\int_0^s E_{\gamma(t)}(\gamma'(t))dt\Big|_{s=1} \\ &= \frac{d}{ds}\phi(\gamma(s))\Big|_{s=1} \\ &= v_q(\phi_q), \end{aligned}$$

where we have used the fact that the derivative of $\phi(\gamma(s))$ at q with respect to s is the same as the derivative of ϕ at q in the direction $(\gamma'(s))_q = v_q$. □

To summarize, we have shown that on a simply-connected space, every closed 1-form is exact. In this case, we can always find a scalar potential for the electric field.

[6] It follows from the existence and uniqueness of solutions to first-order initial value problems that such a curve always exists.

7.2 de Rham Cohomology

7.2.1 Definition

Rather than studying homotopies of paths, there is an alternative, algebraic way of studying the question of whether all closed 1-forms are exact. Consider the collection of vector spaces $\{\Omega^k(M)\}_{k=0}^{\infty}$ of k-forms on an n-dimensional manifold M. If $k > n$, we set $\Omega^k(M) = \{0\}$. The exterior derivative $d_k : \Omega^k(M) \to \Omega^{k+1}(M)$ gives a sequence of linear maps [7]

$$0 \to \Omega^0(M) \xrightarrow{d_0} \Omega^1(M) \xrightarrow{d_1} \Omega^2(M) \to \dots$$

where we have extended the sequence of vector spaces so that the first linear mapping $0 \to \Omega^0(M)$ denotes the inclusion map of the zero vector space $\{0\}$ into $\Omega^0(M) = C^\infty(M)$. As we have seen before, applying the exterior derivative twice gives zero: $d_k \circ d_{k-1} = 0$. A sequence of vector spaces $\{V_k\}$ and linear mappings $d_k : V_k \to V_{k+1}$ such that $d_k \circ d_{k-1} = 0$ is said to be a *cochain complex*. The cochain complex with $V_k = \Omega^k(M)$ and $d_k : \Omega^k(M) \to \Omega^{k+1}(M)$ the exterior derivative on k-forms is called the *de Rham complex* of M.

$$\dots \to \Omega^{k-1}(M) \xrightarrow{d_{k-1}} \Omega^k(M) \xrightarrow{d_k} \Omega^{k+1}(M) \to \dots$$

From this point of view, closed k-forms are precisely the elements of $\ker d_k$, while exact k-forms are elements of the $\operatorname{im} d_{k-1}$, both of which are subspaces of $\Omega^k(M)$. Since $d_k \circ d_{k-1} = 0$ (that is, since all exact forms are closed), $\operatorname{im} d_{k-1}$ is a subset of $\ker d_k$. Since d_k is linear, any linear combination of exact forms is also closed. Hence, $\operatorname{im} d_{k-1}$ is a subspace of $\ker d_k$, so we can form the quotient vector space $\ker d_k / \operatorname{im} d_{k-1}$. Following the standard notation, we denote the subspaces $\ker d_k$ and $\operatorname{im} d_{k-1}$ by

$$Z^k(M) := \ker d_k = \{\text{closed } k\text{-forms on } M\},$$
$$B^k(M) := \operatorname{im} d_k = \{\text{exact } k\text{-forms on } M\}.$$

The quotient vector space

$$H^k(M) := \ker d_k / \operatorname{im} d_{k-1} = \frac{\{\text{closed } k\text{-forms on } M\}}{\{\text{exact } k\text{-forms on } M\}}$$

is called the *kth de Rham cohomology group of* M. [8] As we have seen, the quotient vector space construction introduces an equivalence relation on $Z^k(M)$:

$$\omega' \sim \omega \text{ in } Z^k(M) \text{ if and only if } \omega' - \omega \in B^k(M).$$

[7] We usually just denote all of the d_k's as d, but for the moment it will be instructive to keep track of the domain on which d_k acts.

[8] While $H^k(M)$ is a vector space, in what follows we will not need the scalar multiplication operation on $H^k(M)$, so we are just viewing it as an abelian group.

That is, two closed forms are equivalent if their difference is an exact form: $\omega' = \omega + d\tau$. In this case we say that the two closed forms ω and ω' are *cohomologous*. The equivalence class $[\omega] = \omega + B^k(M)$ of a closed form ω is called its *cohomology class*.

Notice that if every closed k-form is exact, then $B^k(M) = Z^k(M)$, and we have

$$H^k(M) = Z^k(M)/Z^k(M) \cong \{0\},$$

so $H^k(M)$ is trivial. On the other hand, if no closed k-form is exact (except for 0, which is automatically exact), then

$$H^k(M) = Z^k(M)/\{0\} \cong Z^k(M).$$

Hence, $H^k(M)$ **provides a measure of the extent to which closed k-forms on M fail to be exact**. It follows that every static electric field on a manifold M is conservative if and only if $H^1(M) = \{0\}$.

7.2.2 Examples

Example 7.5 ($H^0(M)$). The simplest case of these definitions is $H^0(M)$. A 0-form on an n-dimensional manifold M is just a smooth function f. In local coordinates,

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i.$$

This is 0 if and only if $\partial f/\partial x^i = 0$ for all $i = 1, \ldots, n$. Such a function is *locally constant*. A function can be locally constant but not constant if M is not connected. In this case, M is a disjoint union $M = \cup_j M_j$, where each M_j is called a *connected component*. (For example, if M is the union of disjoint intervals of real numbers, then each M_j is a disjoint interval.) The most general locally constant function is one that takes the constant value c_j on the jth component.

When is a 0-form exact? When it is d of a (-1)-form, presumably, but there are no (-1)-forms! In order for $H^0(M)$ to be defined, then, we define $\Omega^{-1}(M) := \{0\}$. We therefore have

$$H^0(M) = Z^0(M)/B^0(M) = Z^0(M)/\{0\} \cong Z^0(M).$$

In other words, the 0th de Rham cohomology group of M is isomorphic to the space of locally constant functions on M. If M has r connected components, then a locally constant function $f \in Z^0(M)$ is specified by the r numbers

$$f(x) = \begin{cases} c_1, & \text{if } x \in M_1, \\ c_2, & \text{if } x \in M_2, \\ \vdots & \\ c_r & \text{if } x \in M_r. \end{cases}$$

It is straightforward to check that the map

$$\begin{aligned}\psi : Z^0(M) &\to \mathbb{R}^r \\ \psi(f) &= \langle c_1, c_2, \ldots, c_r \rangle\end{aligned}$$

is an isomorphism. Therefore,

$$H^0(M) \cong \mathbb{R}^r,$$

where r is the number of connected components of M.

Thus, $H^0(M)$ conveys some very basic information about the topology of M. In particular, $H^0(M) = \{0\}$ if and only if M is connected.

Example 7.6 ($H^k(M) = \{0\}$ for $k >$ dim M). Let M be an n-dimensional manifold. At any point $p \in M$, the tangent space T_pM to M at p is a vector space of dimension n. If $\omega \in \Omega^k(M)$, then $\omega_p \in A_k(T_pM)$. We have seen that if $k > n$, then $A_k(T_pM) = \{0\}$. Hence, for $k > n$, the only k-form on M is the zero form. Hence

$$H^k(M) = Z^k(M)/B^k(M) = \{0\}/\{0\} \cong \{0\}.$$

Example 7.7 ($H^k(\mathbb{R})$). Since $\mathbb{R}$ is connected, by Example 7.2

$$H^0(\mathbb{R}) = \mathbb{R}.$$

Since $\dim \mathbb{R} = 1$, there are no nonzero 2-forms on $\mathbb{R}$. This implies that every 1-form on $\mathbb{R}$ is closed: $Z^1(\mathbb{R}) = \Omega^1(\mathbb{R})$. A 1-form $f(x)dx$ on $\mathbb{R}$ is exact if and only if there exists a C^∞ function $g(x)$ on $\mathbb{R}$ such that

$$f(x)dx = dg = \frac{dg}{dx}(x)dx.$$

Such a function $g(x)$ is an antiderivative of f. Since f is, in particular, continuous, the Fundamental Theorem of Calculus says that

$$g(x) = \int_0^x f(t)dt$$

is an antiderivative for f. Since f is smooth, so is g, hence $g \in \Omega^0(\mathbb{R}) = C^\infty(\mathbb{R})$. This proves that every closed 1-form on $\mathbb{R}$ is exact, therefore

$$H^1(\mathbb{R}) = \{0\}.$$

Since $\dim \mathbb{R} = 1$, by Example 7.6 $H^k(\mathbb{R}) = \{0\}$ for $k > 1$. We therefore have

$$H^k(\mathbb{R}) = \begin{cases} \mathbb{R}, & \text{for } k = 0, \\ 0 & \text{for } k \geq 1. \end{cases}$$

Example 7.8 ($H^k(S^1)$). Let S^1 be the unit circle in the xy-plane. By Example 7.5, because S^1 is connected, $H^0(S^1) = \mathbb{R}$. By Example 7.6, because $\dim S^1 = 1$, $H^k(S^1) = \{0\}$ for all $k \geq 2$. All that remains is to compute $H^1(S^1)$. Since $\dim S^1 = 1$, there are non non-zero 2 forms on S^1, hence all 1-forms on S^1 are closed. Therefore, $Z^1(S^1) = \Omega^1(S^1)$. The integration of 1-forms on S^1 defines a linear map

$$\begin{aligned} \varphi : Z^1(S^1) = \Omega^1(S^1) &\to \mathbb{R} \\ \varphi(\alpha) &= \int_{S^1} \alpha. \end{aligned}$$

To see that φ is a surjection, parametrize S^1 by $\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$, $0 \leq t \leq 2\pi$. Recall the 1-form

$$\omega = -ydx + xdy$$

from Equation (7.1), restricted to S^1 (so that $x^2 + y^2 = 1$). We have

$$\varphi(\omega) = \int_{S^1} \omega = \int_0^{2\pi} \omega_{\mathbf{r}(t)}(\mathbf{r}'(t))dt = \int_0^{2\pi} dt = 2\pi \neq 0,$$

which shows that $\dim \operatorname{im} \varphi = 1 = \dim \mathbb{R}$, hence φ is surjective.

After quite a bit of work (there is a much better way to do this after developing some tools from homological algebra, so we skip this calculation) one finds that $B^1(S^1) = \ker \varphi$. By the First Isomorphism Theorem, the surjective linear map $\varphi : Z^1(S^1) \to \mathbb{R}$ induces an isomorphism

$$Z^1(S^1)/\ker \varphi \cong \operatorname{im} \varphi.$$

Since $\ker \varphi = B^1(S^1)$ and $\operatorname{im} \varphi = \mathbb{R}$, this says that

$$H^1(S^1) \cong Z^1(S^1)/B^1(S^1) \cong \mathbb{R}.$$

Therefore

$$H^k(S^1) = \begin{cases} \mathbb{R}, & k = 0, 1 \\ 0, & k \geq 2. \end{cases}$$

Note that $\dim H^1(S^1) = \dim \mathbb{R} = 1$, which is exactly the number of '1-dimensional holes' in S^1. More generally, $\dim H^k(M)$ counts the number of 'k-dimensional holes' in M.

Example 7.9 ($H^k(T^2)$). Let $M = T^2$ denote the 2-torus, show in Figure 27 below:

It can be shown that

$$H^k(T^2) = \begin{cases} \mathbb{R}, & k = 0, \\ \mathbb{R}^2, & k = 1 \\ \mathbb{R}, & k = 2, \\ 0, & k \geq 3. \end{cases}$$

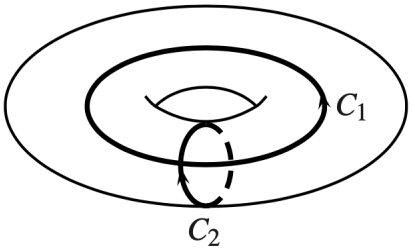


Figure 27: The 2-torus T^2.

By the comment above, $\dim H^2(T^2) = 2$ is consistent with the fact that there are two independent '1-dimensional holes' in T^2 (which prevent the paths C_1 and C_2 in Figure 27 from shrinking to a point), and $\dim H^2(T^2) = 1$ is consistent with the fact that there is one '2-dimensional hole' (which, if one maps a 2-sphere into T^2 in such a way that it contains the hole in the center, then this hole prevents the 2-sphere from shrinking to a point).

7.3 Exercises

The ring structure on de Rham cohomology

We have seen that the wedge product on the vector space

$$\Omega^\bullet(M) = \bigoplus_{k=0}^{n} \Omega^k(M)$$

is associative an distributes over addition. The operations of vector addition and the wedge product therefore make $\Omega^\bullet(M)$ into a *ring*. Since $\wedge : \Omega^k(M) \times \Omega^\ell(M) \to \Omega^{k+\ell}(M)$, this is a *graded ring*, graded by the degree k. Since $\alpha \wedge \beta = (-1)^{k\ell}\beta \wedge \alpha$, the ring is *anti-commutative*. Thus, $\Omega^\bullet(M)$ is an *anticommutative graded ring*.

We now show that the these operations are well-defined in cohomology, which makes

$$H^\bullet(M) := \bigoplus_{k=0}^{n} H^k(M)$$

an anticommutative graded ring as well, called the *cohomology ring* of M.

1. If $[\omega] \in H^k(M)$ and $[\tau] \in H^\ell(M)$, define

$$[\omega] + [\tau] := [\omega + \tau],$$
$$[\omega] \wedge [\tau] := [\omega \wedge \tau].$$

Check that these operations are well-defined by checking the following:

(a) $\omega + \tau$ and $\omega \wedge \tau$ are closed.

(b) If $\omega' \sim \omega$ and $\tau' \sim \tau$, then $\omega' + \tau' \sim \omega + \tau$ and $\omega' \wedge \tau' \sim \omega \wedge \tau$.

Categories and Functors

Many problems in mathematics share common features. This has given rise to the theory of *categories* and *functors*, which tries to clarify the structural similarities among different areas of mathematics.

A *category* C is a collection of *objects*, $\mathrm{Obj}(\mathsf{C})$ and a set of *morphisms* $\mathrm{Mor}(A, B)$ between each pair of objects $A, B \in \mathrm{Obj}(\mathsf{C})$ such that

- Composition of morphisms is defined and is associative.
- There is an identity morphism id_A for each object A, such that $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$ for every morphism $f : A \to B$.

A morphism $f : A \to B$ is called an *isomorphism* if there exists a two-sided inverse for f; that is, a map $g : B \to A$ such that $f \circ g = \mathrm{id}_B$ and $g \circ f = \mathrm{id}_A$.

The simplest example is the category Set, whose objects are sets and whose morphisms are functions. If $f : A \to B$ and $g : B \to C$ are functions, then $g \circ f : A \to C$ is defined by $(g \circ f)(x) := g(f(x))$ for all $x \in A$, which is an associative operation. The identity map on a set A is defined by $\mathrm{id}_A(x) = x$ for all $x \in A$. Isomorphisms in Set are bijections.

Another example is the category $\mathsf{SmoothMan}$, which has smooth manifolds as the objects and smooth mappings of manifolds as morphisms. The isomorphisms in $\mathsf{SmoothMan}$ are *diffeomorphisms*, which are bijective smooth maps with smooth inverses.

2. Show that there is a category $\mathsf{Vect}_{\mathbb{R}}$, whose objects are real vector spaces and whose morphisms are linear maps. The isomorphisms in $\mathsf{Vect}_{\mathbb{R}}$ are bijective linear maps. (*Hint*: Since vector spaces are sets and linear maps are functions, all that needs to be checked is that composition of linear maps is linear and that the identity map is linear.)

The general philosophy of category theory, motivating the very definition of a category, is that any mathematical object should be considered together with its accompanying notion of structure-preserving morphism. Categories are themselves mathematical objects, and morphisms between categories are called *functors*.

Let C and D be categories. A *functor* from C to D, denoted

$$\mathsf{C} \xrightarrow{F} \mathsf{D}$$

consists of the following data:

- A function between the classes of objects: $F : \mathrm{Obj}(\mathsf{C}) \to \mathrm{Obj}(\mathsf{D})$
- For each pair $X, Y \in \mathrm{Obj}(\mathsf{C})$ a function

$$F : \mathrm{Mor}_{\mathsf{C}}(X, Y) \to \mathrm{Mor}_{\mathsf{D}}(F(X), F(Y))$$

These data are required to satisfy the following two *functorality axioms*:

(i) For each object $X \in \mathrm{Obj}(\mathsf{C})$, the identity is preserved:

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$$

(ii) For each triple $X_1, X_2, X_3 \in \mathrm{Obj}(\mathsf{C})$, composition is preserved: given

$$X_1 \xrightarrow{f} X_2 \xrightarrow{g} X_3$$

we have

$$F(g \circ f) = F(g) \circ F(f).$$

In other words, a functor consists of a mapping on objects and a mapping on morphisms that preserves all of the structure of a category, namely domains and codomains, compositions, and identities. If the target category of a functor is simpler than the original category, then the functor provides a way to simplify problems in the first category.

Here is a relevant example. Given a function $F : U \to V$, where U is an open subset of $\mathbb{R}^n$ and V is an open subset of $\mathbb{R}^m$, define the *differential of F at p* in U by

$$\begin{aligned} F_{*,p} : T_pU &\to T_{F(p)}V \\ v_p &\mapsto dF_p(v_p), \end{aligned}$$

where $F_{*,p}(v_p) \in T_{F(p)}V$ is defined by

$$F_{*,p}(v_p)f = v_p(f \circ F)$$

for $f \in C^\infty(V)$. If U and V are both open subsets of $\mathbb{R}$, then denoting by x the coordinate on U and $y = F(x)$ the coordinate on V, then since $\{d/dx|_p\}$ is a basis for T_pU and $\{d/dy|_{F(p)}\}$ is a basis for $T_{F(p)}V$, in particular we have

$$F_{*,p}\left(\left.\frac{d}{dx}\right|_p\right) = a\left.\frac{d}{dy}\right|_{F(p)}$$

for some $a \in \mathbb{R}$. To solve for a, apply both sides to the function $f(y) = y$:

$$\begin{aligned} F_{*,p}\left(\left.\frac{d}{dx}\right|_p\right) y &= a\left.\frac{d}{dy}\right|_{F(p)} y \\ \left.\frac{d}{dx}\right|_p (y \circ F) &= a\frac{dy}{dy}(F(p)) \\ \frac{dy}{dx}(F(p)) &= a, \end{aligned}$$

which shows that

$$F_{*,p}\left(\left.\frac{d}{dx}\right|_p\right) = dF_p\left(\left.\frac{d}{dx}\right|_p\right) = \frac{dy}{dx}(F(p))\left.\frac{d}{dy}\right|_{F(p)}.$$

A similar calculation shows that, if U is an open subset of $\mathbb{R}^n$ with coordinates $(x^1, \ldots, x^n)$ and V is an open subset of $\mathbb{R}^m$ with coordinates $(y^1, \ldots, y^m)$, then

$$F_{*,p}\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \sum_{j=1}^{m} \frac{\partial y^j}{\partial x^i}\left.\frac{\partial}{\partial y^j}\right|_{F(p)}.$$

The matrix of $F_{*,p}$ with respect to the bases $\{\partial/\partial x^i|_p\}$ and $\{\partial/\partial y^j|_{F(p)}\}$ is therefore $[\frac{\partial y^j}{\partial x^i}(F(p))]$, which is the *Jacobian matrix* of the change of coordinates $(x^1, \ldots, x^n) \mapsto (y^1, \ldots, y^m) = (F^1(x^1, \ldots, x^n), \ldots, F^m(x^1, \ldots, x^n))$.

To see that the differential is functiorial, note that

(i) If $\mathrm{id}_U : U \to U$ is the identity map on U, then for any $v_p \in T_pU$ and $f \in C^\infty(U)$,

$$(\mathrm{id}_U)_{*,p}(v_p)f = v_p(f \circ \mathrm{id}_U) = v_p,$$

which shows that $(\mathrm{id}_U)_{*,p}(v_p) = v_p$ and therefore

$$(\mathrm{id}_U)_{*,p} = \mathrm{id}_{T_pU}.$$

(ii) If $F : U \to V$ and $G : V \to W$ and $p \in U$, then for any $v_p \in T_pU$ and $f \in C^\infty(W)$,

$$\begin{aligned}(G \circ F)_{*,p}(v_p)f &= v_p(f \circ (G \circ F)) \\ &= v_p((f \circ G) \circ F) \\ &= F_{*,p}(v_p)(f \circ G) \\ &= G_{*,F(p)}(F_{*,p}(v_p))f \\ &= (G_{*,F(p)} \circ F_{*,p})(v_p)f,\end{aligned}$$

which shows that

$$(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p}.$$

This shows that $F : U \to V \implies F_{*,p} : T_pU \to T_{F(p)}V$ is a functor from $\mathsf{SmoothMan} \to \mathsf{Vect}_{\mathbb{R}}$.

An important fact about functors is that if $F : \mathsf{C} \to \mathsf{D}$ is a functor, and $f : A \to B$ an isomorphism in C, then $F(f) : F(A) \to F(B)$ is an isomorphism in D. This can be seen by applying F to both sides of

$$g \circ f = \mathrm{id}_A,$$

which gives

$$\begin{aligned}F(g \circ f) &= F(\mathrm{id}_A) \\ F(g) \circ F(f) &= \mathrm{id}_{F(A)}\end{aligned}$$

and similarly applying F to both sides of

$$f \circ g = \mathrm{id}_B$$

gives

$$F(f \circ g) = F(\mathrm{id}_B)$$
$$F(f) \circ F(g) = \mathrm{id}_{F(B)},$$

which shows that $F(g)$ is a 2-sided inverse for $F(f)$, and therefore $F(f)$ is an isomorphism.

By our previous example, this implies that if $F : N \to M$ is a diffeomorphism of smooth manifolds, then $F_{*,p} : T_pN \to T_{F(p)}M$ is an isomorphism of vector spaces.

The next problem gives an example of the usefulness of functors. Recall that an n-dimensional manifold M is a space that locally looks like $\mathbb{R}^n$. More precisely, this means that we can cover M with open sets U_α and we can find a homeomorphism $\varphi_\alpha : U_\alpha \to \varphi_\alpha(U) \subset \mathbb{R}^n$ for each α. Two smooth manifolds M and N are *diffeomorphic* if there is a *diffeomorphism* $F : M \to N$ (i.e., a smooth bijection with a smooth inverse). If M and N are diffeomorphic (denoted by $M \cong N$), then they are equivalent objects in the category SmoothMan. For the dimension of a manifold to be well-defined, it must therefore be a diffeomorphism invariant. That is, if N is an n-dimensional manifold and M is an m-dimensional manifold, then $M \cong N$ implies $n = m$. This boils down to the statement that if $U \subset \mathbb{R}^n$ is diffeomorphic to $V \subset \mathbb{R}^m$, then $n = m$. This turns out to be quite difficult to prove directly in the category SmoothMan. However, as the next problem shows, the observation above about the functor $F : U \to V \implies F_{*,p} : T_pU \to T_{F(p)}V$ turns this into an easy question in the category $\mathsf{Vect}_{\mathbb{R}}$!

3. Use previous observation to prove the following statement: If an open set $U \subset \mathbb{R}^n$ is diffeomorphic to an open set $V \subset \mathbb{R}^m$, then $n = m$. This result is called the *smooth invariance of dimension*.

The de Rham cohomology functor

Let U be an open subset of $\mathbb{R}^n$ and V an open subset of $\mathbb{R}^m$. Given a smooth map $F : U \to V$, define the *pullback map*

$$F^* : \Omega^k(V) \to \Omega^k(U)$$
$$\omega \mapsto F^*\omega$$

for $k \geq 1$ by setting, for $\omega \in \Omega^k(V)$ and $v_1, \ldots, v_k$ vector fields on V,

$$F^*\omega(v_1, \ldots, v_k) = \omega(F_*(v_1), \ldots, F_*(v_k)).$$

If $k = 0$, then $f \in \Omega^0(V) = C^\infty(V)$ and we define $F^*f = f \circ F$.

4. Show that $\mathrm{id}_U^* = \mathrm{id}_{\Omega^k(U)}$ and $(G \circ F)^* = F^* \circ G^*$. This resembles the defining properties of a functor, but the order of the composition is reversed in the composition. Functors which reverse the order of compositions are said to be *contravariant*, while functors which preserve the order of compositions are said to be *covariant*.

A straightforward but somewhat lengthy calculation shows that the exterior derivative commutes with pullbacks:

$$dF^*\omega = f^*d\omega.$$

5. Show that $F^* : \Omega^k(V) \to \Omega^k(V)$ takes closed forms to closed forms and exact forms to exact forms.

6. Define the *pullback map in cohomology* by

$$F^\sharp : H^k(V) \to H^k(U)$$
$$F^\sharp([\omega]) = [F^*\omega].$$

   Show that $F^\sharp$ is well-defined.

7. Check that $F^\sharp$ has the same functorial properties as a F^*; namely,

$$\mathrm{id}_U^\sharp = \mathrm{id}_{H^k(U)}, \quad (G \circ F)^\sharp = F^\sharp \circ G^\sharp$$

   It follows that $F : M \to N \implies F^\sharp : H^k(N) \to H^k(M)$ is a contravariant functor from SmoothMan to the category of anticommutative graded rings. By the above, if $F : M \to N$ is a diffeomorphism, then $F^\sharp : H^\bullet(N) \to H^\bullet(M)$ is an isomporphism of anticommutative graded rings. In this way, de Rham cohomology becomes a very important diffeomorphism invariant of smooth manifolds.

While we have used the smooth structure of M to define $H^k(M)$, it turns out that the groups $H^k(M)$ are actually a *topological* invariants of M. That is, if M and N are *homeomorphic* as topological manifolds, then their de Rham cohomology groups are isomorphic. While we will not explore the implications of this fact, it is worth pointing out that one can use the topological invariance of de Rham cohomology groups to prove a topological version of problem 3: if an open set $U \subset \mathbb{R}^n$ is homeomorphic to an open set $V \subset \mathbb{R}^m$, then $n = m$.

8 Applications in Electromagnetism

8.1 Gauge Transformations

Last time we considered the equations of electrostatics

$$dE = 0, \quad *d*E = \rho,$$

where the electric field $E \in \Omega^1(M)$ and the charge density $\rho \in C^\infty(M)$, and saw that the first of these two equations was automatically satisfied if E is exact; that is, if there exists a scalar potential $\phi \in C^\infty(M)$ such that $E = d\phi$.

Now consider the equations of magnetostatics

$$dB = 0, \quad *d*B = j,$$

where the magnetic field $B \in \Omega^2(M)$ and the current density $j \in \Omega^1(M)$. If B is exact, then

$$B = dA,$$

the first equation is automatically true, and the second one reduces to the following equation for the 1-form A:

$$*d*dA = j.$$

The 1-form A is called a *vector potential* for B. Note that A is not uniquely determined, since we can add any closed 1-form to A without changing B. That is, if we change $A \mapsto A + df$, then

$$d(A + df) = dA + d^2f = dA + 0 = dA = B.$$

The transformation $A \mapsto A + df$ is called a *gauge transformation*, and one says the equations

$$B = dA, \quad *d*dA = j$$

are *gauge-invariant*.

The same remarks hold more generally for the electromagnetic field $F \in \Omega^2(\mathbb{R}^4)$. Maxwell's equations say that

$$dF = 0, \quad *d*F = J.$$

We say that A is a vector potential for F if $dA = F$. In this case, Maxwell's equations reduce to

$$dA = F, \quad *d*dA = J.$$

These equations are invariant under a gauge transformation $A \mapsto A + df$, where $f \in C^\infty(M)$.

8.2 The Aharonov-Bohm Effect

The following application illustrates the importance of the vector potential in electromagnetism, especially in the context of quantum mechanics, and also shows that funny things can happen in regions of space that are not simply connected. It also has technological applications.

8.2.1 Magnetic Field of a Cylindrical Wire

Let's first solve a problem in magnetostatics that we will need later. Consider the magnetic field produced by a current running through an infinitely long cylindrical wire. Choose coordinates so that the wire runs along the z-axis.

Because of the rotational symmetry about the z-axis, we will use *cylindrical coordinates* (r, θ, z) for $\mathbb{R}^3$, where r and θ are the usual polar coordinates for the xy-plane, and z is the height of a point above the xy-plane.

If the current is cylindrically symmetric and flows in the z-direction, then

$$j = f(r)dz,$$

which is smooth away from the z-axis ($r = 0$). Then, away from the z-axis,

$$\begin{aligned}
*j &= f(r) * dz \\
&= f(r)dx \wedge dy \\
&= f(r)d(r\cos(\theta)) \wedge d(r\sin(\theta)) \\
&= f(r)(\cos(\theta)dr - r\sin(\theta)d\theta) \wedge (\sin(\theta)dr + r\cos(\theta)d\theta) \\
&= f(r)(r\cos^2(\theta)dr \wedge d\theta - r\sin^2(\theta)d\theta \wedge dr) \\
&= f(r)(r\cos^2(\theta)dr \wedge d\theta + r\sin^2(\theta)dr \wedge d\theta) \\
&= f(r)r(\sin^2(\theta) + \cos^2(\theta))dr \wedge d\theta \\
&= f(r)rdr \wedge d\theta.
\end{aligned}$$

One can check that, in rectangular coordinates,

$$d\theta = -\frac{y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy. \tag{8.1}$$

We have seen that the 1-form in Equation 8.1 is not exact, so the notation '$d\theta$', while standard, is unfortunately misleading.

Ampére's law says that the curl of the magnetic field is the current, so B must look something like Figure 28 below:

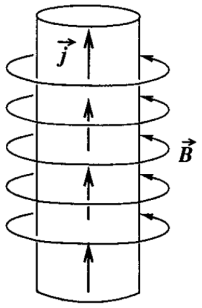


Figure 28: Magnetic field produced by a current running through a wire.

We therefore assume that B has the form

$$*B = g(r)d\theta$$

or

$$\begin{aligned}
B &= g(r) * d\theta \\
&= g(r)\left[-\frac{y}{x^2+y^2} * dx + \frac{x}{x^2+y^2} * dy\right] \\
&= g(r)\left[-\frac{\sin(\theta)}{r} * dx + \frac{\cos(\theta)}{r} * dy\right] \\
&= g(r)\left[-\frac{\sin(\theta)}{r} dy \wedge dz + \frac{\cos(\theta)}{r} dz \wedge dx\right] \\
&= g(r)\left[-\frac{\sin(\theta)}{r} d(r\sin(\theta)) \wedge dz + \frac{\cos(\theta)}{r} dz \wedge d(r\cos(\theta))\right] \\
&= g(r)\left[-\frac{\sin(\theta)}{r} (\sin(\theta)dr + r\cos(\theta)d\theta) \wedge dz + \frac{\cos(\theta)}{r} dz \wedge (\cos(\theta)dr - r\sin(\theta)d\theta)\right] \\
&= g(r)\left[-\frac{\sin^2(\theta)}{r} dr \wedge dz - \sin(\theta)\cos(\theta)d\theta \wedge dz + \frac{\cos^2(\theta)}{r} dz \wedge dr - \sin(\theta)\cos(\theta)dz \wedge d\theta\right] \\
&= g(r)\left[\frac{\sin^2(\theta)+\cos^2(\theta)}{r} dz \wedge dr + (\sin(\theta)\cos(\theta) - \sin(\theta)\cos(\theta))dz \wedge d\theta\right] \\
&= g(r)\left[\frac{1}{r} dz \wedge dr + 0\right] \\
&= \frac{g(r)}{r} dz \wedge dr.
\end{aligned}$$

Let's now use the equations of magnetostatics to solve for g. The first equation $dB = 0$ holds for any $g(r)$, since

$$dB = \frac{\partial}{\partial r}\left(\frac{g(r)}{r}\right) dr \wedge dz \wedge dr = -\frac{\partial}{\partial r}\left(\frac{g(r)}{r}\right) dz \wedge \underbrace{dr \wedge dr}_{=0} = 0$$

automatically, and therefore does not put any condition on $g(r)$. The second equation

$$\begin{aligned} d * B &= *j \\ d(g(r)) \wedge d\theta &= f(r) r dr \wedge d\theta \\ g'(r) dr \wedge d\theta &= f(r) r dr \wedge d\theta \end{aligned}$$

implies $g'(r) = f(r)r$. Integrating, we obtain

$$g(r) = g(0) + \int_0^r s f(s) ds,$$

so

$$B = \frac{1}{r} \left[g(0) + \int_0^r s f(s) ds \right] dz \wedge dr.$$

We set $g(0) = 0$ so that $\lim_{r \to 0} B$ is finite. Suppose the wire has radius R. Since $j = 0$ outside the wire, this implies $f(r) = 0$ for $r > R$. Letting

$$I = 2\pi \int_0^s f(s) ds,$$

we have

$$*B = \frac{I}{2\pi} d\theta, \quad B = \frac{I}{2\pi r} dz \wedge dr.$$

8.2.2 Magnetic Field of a Solenoid

We now solve a different magnetostatic problem, which turns out to be mathematically similar to the last one. A wire would around a cylinder in a tight spiral is called a *solenoid*. Let's choose coordinates so that the axis of the cylinder is the z-axis. In the limit where the solenoid is infinitely long and the wire is very thin, then one obtains a constant magnetic field inside the solenoid, and a zero magnetic field outside. That is,

$$*B = f(r) dz, \quad B = f(r) r dr \wedge d\theta,$$

where f is a constant for $r < R$ and zero for $r > R$. Note that $*B$ is exactly the same 1-form as j of the wire in the previous problem, which satisfied the equations

$$dB = 0, \quad d * B = *j.$$

It follows from the second equation that a vector potential A for the B of the solenoid is given by $*B$ of the wire. That is,

$$A = g(r) d\theta$$

where

$$g(r) = \int_0^r s f(s) ds.$$

In particular, outside the solenoid we have

$$A = \frac{\Phi}{2\pi} d\theta,$$

where

$$\Phi = 2\pi \int_0^R s f(s) ds$$

is the magnetic flux through the solenoid; that is, the integral of B over the disc in any plane of constant z.

8.2.3 The Aharonov-Bohm Effect

The Aharonov-Bohm effect occurs when a charged particle passes around a solenoid. It is a purely quantum-mechanical effect, so we need to explain a small amount of quantum mechanics.

In quantum mechanics, the state of a physical system is described by a unit vector in a *Hilbert space* $\mathscr{H}$, which is a kind of complex vector space with an *inner product* $\langle \cdot, \cdot \rangle : \mathscr{H} \times \mathscr{H} \to \mathbb{C}$. If one prepares the system in a state $\psi \in \mathscr{H}$, and immediately does an experiment to see if it is in the state $\phi \in \mathscr{H}$, then the probability then the probability that the answer is 'yes' is

$$|\langle \phi, \psi \rangle|^2.$$

Such a quantity is called a *transition probability*, while the inner product

$$\langle \phi, \psi \rangle$$

is called the *transition amplitude*. *Observables* (i.e., measurable quantities) are represented by *self-adjoint* operators $A : \mathscr{H} \to \mathscr{H}$. The *expected value* (average measured value) of an observable A in the state ψ is given by

$$\langle \psi, A\psi \rangle.$$

While we represent states as unit vectors, it is important to note that if two states ψ and ψ' differ by a phase, that is, if

$$\psi' = e^{i\theta} \psi$$

for some real number θ, then they describe the same state. The reason is that no transition probabilities are affected by using ψ' instead of ψ:

$$|\langle \phi, \psi' \rangle|^2 = |\langle \phi, \psi \rangle|^2$$

for all $\phi \in \mathscr{H}$.

Magnetism has a remarkable relationship to the phase in quantum mechanics, which is roughly as follows. Suppose we have a particle in $\mathbb{R}^3$ with electric charge q in the state

described by the vector ψ moving in a magnetic field B with vector potential A. If the particle travels around a loop γ, then

$$\psi \mapsto e^{-\frac{i}{\hbar} q \int_\gamma A} \psi,$$

where $\hbar$ is Planck's constant. In particular, if D is an oriented 2-disk in $\mathbb{R}^3$ such that γ runs counterclockwise around the boundary of D, then by Stokes' theorem we have

$$\int_\gamma A = \int_D B,$$

which is the magnetic flux through D.

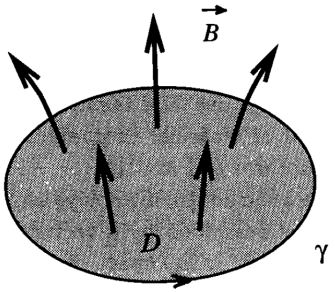


Figure 29: Magnetic flux through a 2-disk D.

Then

$$\psi \mapsto e^{-\frac{i}{\hbar} q \int_D B} \psi.$$

It turns out that the same formulas hold for a loop in spacetime, except that B must be replaced by the electromagnetic field F.

The discussion above is an oversimplification, as in quantum mechanics a particle does not really follow a well-defined trajectory. Moreover, we just said above that multiplying ψ by a phase does not change any transition probabilities, so what physical effect could this phase possible have?

To answer this question, we need to briefly mention the path integral approach to quantum mechanics. In classical mechanics, a particle moves along some path γ in $\mathbb{R}^3$. There is an important quantity called the *Lagrangian*, which is the kinetic energy minus the potential energy, and is a function of time that can be calculated from the particle's position and velocity:

$$L = L(\gamma(t), \gamma'(t)).$$

The exact formula for the Lagrangian depends on the forces acting on the particle. If we consider the particle's path only from time 0 to time T, integrating the Lagrangian over this interval of time gives the *action*

$$S(\gamma) = \int_0^T L dt.$$

The significance of the action in classical mechanics is that a particle moving from point p at time 0 to point q at time T will always follow the path γ which *extremizes* (usually minimizes) the action.

In quantum mechanics the action also plays an important role. Let the Hilbert space of states $\mathscr{H} = L^2(\mathbb{R}^3)$, the space of all complex functions on $\mathbb{R}^3$ such that

$$\int |\psi(x)|^2 d^3x < \infty.$$

The inner product on $\mathscr{H}$ is

$$\langle \phi, \psi \rangle = \int \overline{\phi}(x)\psi(x)d^3x.$$

A state vector $\psi \in \mathscr{H}$ is called a *wavefunction*. The square magnitude $|\psi(x)|^2$ is a probability density, meaning that the probability for the particle to be located in some set $U \subset \mathbb{R}^3$ is given by

$$\int_U |\psi(x)|^2 d^3x.$$

In the path integral formulation of quantum mechanics, if a particle is in the state ψ at $t = 0$, then its state ϕ at some later time T is given by the integral over all paths

$$\mathcal{P} = \{\gamma : [0, T] \to \mathbb{R}^3 : \gamma(0) = a, \gamma(T) = b\}$$

as

$$\phi(b) = \int_{\mathcal{P}} e^{\frac{i}{\hbar}S(\gamma)}\psi(a)D\gamma,$$

where $D\gamma$ is some sort of measure on the space $\mathcal{P}$. We are supposed to think of the particle as taking all paths from a to b, with each path weighted by the factor

$$e^{\frac{i}{\hbar}S(\gamma)},$$

where $S(\gamma)$ is the classical action. As $\hbar \to 0$ (the "classical limit"), this phase factor oscillates very rapidly, except near paths that are critical points of the action, cancelling out in such a way that only the classical path contributes.

Suppose now that, in addition to whatever forces were already acting on the particle, there is also a magnetic field B on $\mathbb{R}^3$ with vector potential A. For simplicity, let's consider the case where A and B are both independent of time. Then the path integral formula for the state ϕ at time T gets modified as

$$\phi(b) = \int_{\mathcal{P}} e^{\frac{i}{\hbar}\left(S(\gamma) - q\int_\gamma A\right)}\psi(a)D\gamma.$$

In other words, the phase factor is multiplied by the additional phase

$$e^{-\frac{i}{\hbar}q\int_\gamma A}.$$

Since $S = \int L dt$, we can alternatively say that the Lagrangian L is replaced by the Lagrangian

$$L - qA(\gamma'(t)).$$

In particular, if $a = b$, so that the path γ is a loop based at a, the extra phase factor is just

$$e^{-\frac{i}{\hbar}\oint_\gamma A},$$

or if γ bounds the disk D, then

$$e^{-\frac{i}{\hbar}\int_D B}.$$

This phase factor *does* have physical effects, since it can differ for different loops γ, producing constructive or destructive interference in the path integral.

Everything above applies equally well to space which is any manifold M, not just $\mathbb{R}^3$. The Aharonov-Bohm effect is an interesting phenomenon that occurs when M is not simply connected. For example, consider a cylindrical solenoid of radius R centered on the z-axis, as shown in Figure 30 below:

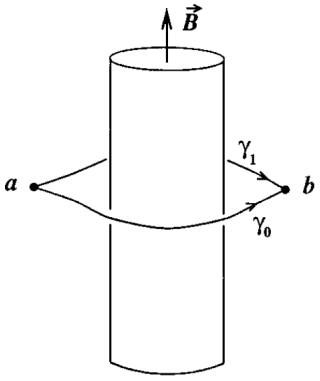


Figure 30: The Aharonov-Bohm effect: two paths from a to b.

Suppose an electron moves from the point $a = (-2R, 0, 0)$ to the point $b = (2R, 0, 0)$ in the space $M = \mathbb{R}^3 - \{r < R\}$, which is not simply-connected. While the magnetic field B is zero on M, in the last section we found that the vector potential

$$A = \frac{\Phi}{2\pi} d\theta$$

is nonzero on M. The electron takes all possible paths from a to b, each one giving a different phase

$$e^{-\frac{i}{\hbar}q\int_\gamma A},$$

producing interference in the path integral. This is the Aharonov-Bohm effect. In short, in quantum mechanics the vector potential can affect the wavefuntion in significant ways even in regions where the magnetic field is zero!

For the two paths shown in Figure 30, the phase factor is $e^{-\frac{iq\Phi}{2\hbar}}$ for the path γ_0, and is $e^{\frac{iq\Phi}{2\hbar}}$ for the path γ_1. By adjusting Φ to the appropriate value, we can arrange for this phase factor to be i for the path γ_0 and $-i$ for γ_1. Similarly, by symmetry, every path from a to b has a reflected version for which the phase factor has the opposite sign. On the other hand, by symmetry, the standard action S will be the same for these two reflected paths. Thus

$$\int_{\mathcal{P}} e^{\frac{i}{\hbar}(S(\gamma)-q\int_\gamma A)}\psi(a)D\gamma = 0.$$

In other words, for the right value of Φ there is *complete destructive interference*: an electron starting at a will never go to b! This effect has been observed and is the basis for the technology of SQUIDs - superconducting quantum interference devices - which are used to accurately measure magnetic flux.

It is crucial here that the space M is not simply connected: this is what allows integrals of a closed form along different paths from a to b to give different answers.

8.3 Magnetic monopoles

Magnetic monopoles were first seriously studied by Paul Dirac. No magnetic monopole has ever been experimentally detected, but certain 'grand unified theories' predict their existence. A very intriguing argument about monopoles is Dirac's argument that if a single monopole existed in the universe, then it would imply that all particles must be an integral multiple of a certain fundamental unit. All free particles have charges that are integer multiples of the electron charge. Quarks, which according to the standard model are confined inside of atomic nuclei, have charges that are multiples of $1/3$ of the electron charge. Nonetheless, it appears that all particles do have charges that are integer multiples of *some* basic charge. It would be nice to find some reason for this fact.

The electric field E of a point charge q at the origin in $\mathbb{R}^3$ is given in spherical coordinates by the 1-form on $\mathbb{R}^3 - \{0\}$

$$E = \frac{qdr}{4\pi r^2}.$$

Since

$$dE = -\frac{q}{2r^3}\underbrace{dr\wedge dr}_{=0} + \underbrace{\frac{\partial}{\partial\theta}\left(\frac{q}{4\pi r^2}\right)}_{=0} d\theta\wedge dr + \underbrace{\frac{\partial}{\partial\phi}\left(\frac{q}{4\pi r^2}\right)}_{=0} d\phi\wedge dr = 0,$$

E is closed. Intuitively, $\mathbb{R}^3 - \{0\}$ is simply-connected. Unlike in $\mathbb{R}^2 - \{0\}$, a closed loop circling the origin in the xy-plane can avoid the hole at the origin and shrink to a point by first moving in the z-direction. Indeed, a calculation in algebraic topology shows that $H^1(\mathbb{R}^3 - \{0\}) = \{0\}$, so every closed 1-form is exact. Indeed, it is straightforward to check that $E = d\phi$, where $\phi = -\frac{q}{4\pi r}$.

By electric-magnetic duality, a magnetic monopole of magnetic charge m at the origin in $\mathbb{R}^3$ would be given by

$$*B = \frac{mdr}{4\pi r^2}$$

which implies

$$B = \frac{m}{4\pi r^2} * dr.$$

In spherical coordinates, $r^2 = x^2 + y^2 + z^2$, so taking the differential of both sides gives

$$\begin{aligned} 2rdr &= 2xdx + 2ydy + 2zdz \\ dr &= \frac{xdx + ydy + zdz}{r}, \end{aligned}$$

so

$$\begin{aligned} B &= \frac{m}{4\pi r^3}(x * dx + y * dy + z * dz) \\ &= \frac{m}{4\pi r^3}(xdy \wedge dz + ydz \wedge dx + zdx \wedge dy) \\ &= \frac{m}{4\pi} \sin(\phi) d\theta \wedge d\phi. \end{aligned}$$

We see that B is closed, since

$$dB = \frac{m}{4\pi}\frac{\partial}{\partial r}(\sin(\phi))dr \wedge d\theta \wedge d\phi = 0.$$

However, if we integrate B over S^2, then

$$\int_{S^2} B = \frac{m}{4\pi}\int_0^{2\pi}\int_0^{\pi} \sin(\phi) d\theta d\phi = \frac{m}{4\pi}(4\pi) = m,$$

unlike in $\mathbb{R}^3$, where the equation $dB = 0$ implies that B is exact, since

$$\begin{aligned} \int_{S^2} B &= \int_{S^2} dA \\ &= \int_{\partial S^2} A \quad \text{(by Stokes's Theorem)} \\ &= 0 \quad \text{(since } \partial S^2 = \emptyset\text{)}. \end{aligned}$$

This shows that B is closed but not exact, which implies that $H^2(\mathbb{R}^3 - \{0\})$ is non-trivial. This is comfirmed by another calculation in algebraic topology, which shows that dim $H^2(\mathbb{R}^3 - \{0\}) = 1$, reflecting the fact that $\mathbb{R}^3 - \{0\}$ has a single '2-dimensional hole' at the origin.

Recall now the relationship between magnetic fields and the phase in the path integral from the previous section. If γ is the boundary of the 2-dimensional surface D, then a particle of electric charge q circling γ contributes a phase

$$e^{-\frac{i}{\hbar}q\int_D B}$$

to the path integral. Note that, if we choose γ to be the equator of a two-sphere S^2 enclosing the monopole at the origin, then this phase can be computed in two different ways. We can use the northern hemisphere, D_1, or the southern hemisphere, D_2.

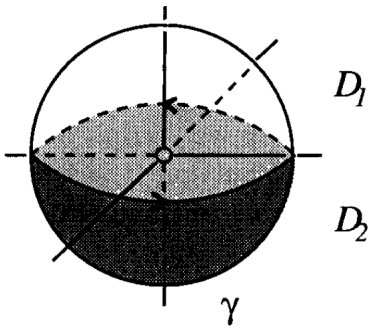


Figure 31: Path of an electrically charged particle around a magnetic monopole.

We have to be careful about orientations here. Suppose γ is traversed in counterclockwise around the disk at the equator, as shown in Figure 31. Then the standard orientation of the northern hemisphere is compatible with the orientation of γ. Then

$$\int_{D_1} B = \frac{m}{4\pi}\int_0^{\pi/2}\int_0^{2\pi}\sin(\phi)d\theta d\phi = \frac{m}{2},$$

giving a phase of

$$e^{-\frac{iqm}{2\hbar}}.$$

On the other hand, we need to use the opposite orientation on the southern hemisphere, so we get

$$\int_{D_2} B = -\frac{m}{4\pi}\int_0^{\pi/2}\int_0^{2\pi}\sin(\phi)d\theta d\phi = -\frac{m}{2},$$

giving a phase of

$$e^{\frac{iqm}{2\hbar}}.$$

For these to be equal, we need

$$e^{-\frac{iqm}{2\hbar}} = e^{\frac{iqm}{2\hbar}}$$

or

$$e^{\frac{iqm}{\hbar}} = 1,$$

which implies

$$\frac{qm}{\hbar} = 2\pi N$$

or

$$q = N\left(\frac{2\pi\hbar}{m}\right)$$

for some integer N. Thus, a non-zero magnetic charge forces quantization of electric charge.

A Permutations

Here we summarize some facts about permutations.

- A *permutation* of the set $A = \{1, 2, \ldots, k\}$ is a bijection $\sigma : A \to A$, which can be thought of as a reordering of the list $1, 2, \ldots, k$ to a new order $\sigma(1), \sigma(2), \ldots, \sigma(k)$. A convenient way to describe a permutation σ is by a matrix

$$\begin{bmatrix} 1 & 2 & \cdots & k \\ \sigma(1) & \sigma(2) & \cdots & \sigma(k) \end{bmatrix}$$

  where the inputs form the first row and the corresponding outputs form the second row. For example the permutation $1 \mapsto 3, 2 \mapsto 1, 3 \mapsto 2$ is represented by the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

  Note that nothing is multiplied by this matrix; it just keeps track of the inputs/outputs.

- Permutations of A form a *group* under composition (which is an associative operation), denoted by S_k. [9] For $\sigma, \tau \in S_k$, we write $\tau\sigma$ for the composition $\tau \circ \sigma$. The identity element of the group is the trivial permutation $\begin{bmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{bmatrix}$. We denote the inverse of a permutation σ by σ^{-1}.

- A *cyclic permutation* or *r-cycle*, denoted $(a_1\ a_2\ \cdots\ a_r)$, where the a_i are distinct, is the permutation σ such that $\sigma(a_i) = a_{i+1}$ for $i = 1, \ldots, r$, $\sigma(a_r) = a_1$, and $\sigma(a_\ell) = a_\ell$ for $\ell > r$. A 2-cycle is also called a *transposition*.

- Cycles $(a_1\ \cdots\ a_r)$ and $(b_1\ \cdots\ b_s)$ are *disjoint* if the sets $\{a_1, \ldots, a_r\}$ and $\{b_1, \ldots, b_r\}$ have no elements in common.

- Any permutation σ can be written as a product of disjoint transpositions. We say that σ is *even* or *odd* depending on whether it is the product of an even or odd number of transpositions.

- We define the *sign* of a permutation σ, denoted by $\operatorname{sgn}(\sigma)$, to be $+1$ or -1 depending on whether the permutation is even or odd. This satisfies $\operatorname{sgn}(\tau\sigma) = \operatorname{sgn}(\tau)\operatorname{sgn}(\sigma)$.

Example A.1. Consider the permutation

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{bmatrix} \in S_5.$$

To write σ as a product of disjoint cycles, start with any element of $\{1, 2, 3, 4, 5\}$, say 1, and apply σ to it repeatedly until we return to the initial element; this gives a cycle:

[9] A *group* is a set G with an associative binary operation, such that G has an identity element, and every element of G has an inverse.

$1 \mapsto 2 \mapsto 4 \mapsto 1$, or (124). Next, repeat this procedure beginning with any of the remaining elements, say 3, to get a second cycle: $3 \mapsto 5 \mapsto 3$, or (35). Since all elements of $\{1,2,3,4,5\}$ are now accounted for, $\sigma = (124)(35)$. Noting that the product $(14)(12)$ also sends $1 \mapsto 2 \mapsto 4 \mapsto 1$, we have $(124) = (14)(12)$, and therefore $\sigma = (14)(12)(35)$, so we see that $\operatorname{sgn}(\sigma) = -1$.

Example A.2. More generally, the decomposition

$$(a_1 a_2 \cdots a_r) = (a_1 a_r)(a_1 a_{r-1}) \cdots (a_1 a_3)(a_1 a_2).$$

shows that an r-cycle is an even permutation if and only if r is odd, and an odd permutation if and only if r is even. Thus, a faster way to compute the sign of a permutation is to decompose it into a product of cycles and to count the number of cycles of even length. For example, the permutation $\sigma = (124)(35)$ in Example A.1 is odd because (124) is even and (35) is odd.

B Permutation Action on Tensors

Definition B.1 (Group action). Let G be a group and X be a set. A *left group action* is a mapping

$$\begin{aligned} G \times X &\to X \\ (g, x) &\mapsto gx \end{aligned}$$

which satisfies the following two properties:

(i) $ex = x$, where e is the identity element of G and $x \in X$.

(ii) $h(gx) = (hg)x$ for all $g, h \in G$ and $x \in X$.

A right group action is defined similarly.

Definition B.2 (Orbit). If there is a left action of G on X, then the set

$$\mathcal{O}_x := \{gx \in X : g \in G\}$$

is called the *orbit* of the element $x \in X$.

We can define a left action of S_k on $L_k(V)$ as

$$\begin{aligned} S_k \times L_k(V) &\to L_k(V) \\ (\sigma, f) &= \sigma f, \end{aligned}$$

where

$$(\sigma f)(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}).$$

Note this is indeed a left action of S_k on $L_k(V)$, since

(i) If σ is the trivial permutation, then

$$(\sigma f)(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) = f(v_1, \ldots, v_k),$$

hence $\sigma f = f$.

(ii) If $\sigma, \tau \in S_n$, then

$$\begin{aligned}
\tau(\sigma f)(v_1, \ldots, v_k) &= (\sigma f)(v_{\tau(1)}, \ldots, v_{\tau(k)}) \\
&= (\sigma f)(w_1, \ldots, w_k) \qquad (\text{letting } w_i = v_{\tau(i)}) \\
&= f(w_{\sigma(1)}, \ldots, w_{\sigma(k)}) \\
&= f(v_{\tau(\sigma(1))}, \ldots, v_{\tau(\sigma(k))}) \\
&= f(v_{\tau\sigma(1)}, \ldots, v_{\tau\sigma(k)}) \\
&= (\tau\sigma) f(v_1, \ldots, v_k),
\end{aligned}$$

hence $\tau(\sigma f) = (\tau\sigma f)$.

Note that $f \in L_k(V)$ is symmetric if and only if $\sigma f = f$ for all $\sigma \in S_k$ and f is alternating if and only if $\sigma f = (\operatorname{sgn}(\sigma)) f$ for all $\sigma \in S_k$.

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